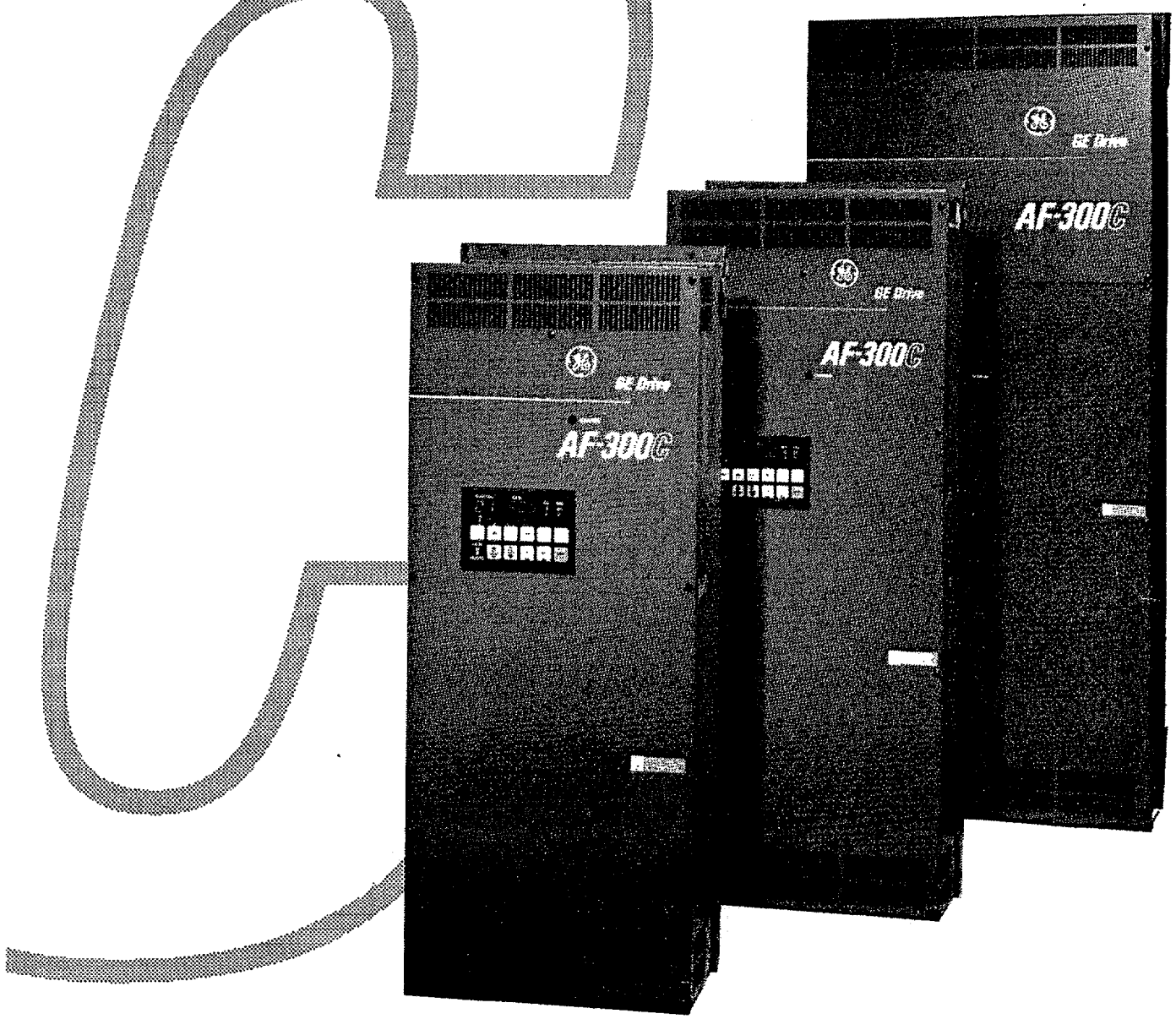
GEK-85716A

GE Drives

AF-300C™

40-300 HP

Instructions



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These instructions do not purport to coverall details or variations in equipment to provide for every possible contingency lo be met in connection with installation operation maintenance. Should further information bo desired or should particular probloms arise which are not covered sufficiently for the purchasers purposes, the matter should be referred to General Electric Comwnv

SUPPLEMENT #1

FOR INSTRUCTION MANUAL GEK-85716A

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particularproblems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia U.S.A.

This supplement contains additional information needed when installing and operating the AF-300C series of inverters. Refer to the information of this supplement in addition to that contained in GEK-85716A for the figures, tables, and paragraphs listed in this supplement.

Figure 4-15. Basic Connection Diagram
Figure S-12. Control Block Diagram

The 3-phase AC circuit breaker shown in these figures is incorrectly illustrated as three breakers. It is actually a single, 3-pole breaker.

Table 5: Function Code Data
Table 6: Function Code Descriptions

Function Code lb, Electronic Thermal Overload Protection, that is built in to the AF-300C series of inverters is factory set as Operational. If the user changes this to Non-Operational, conventional overload protection for the motor and wiring must then be provided by the user. Typical wiring for an overload relay is shown in figure 8-12 of GEK-85716A. All inverter installations must meet the applicable national (NEC, CEC) and local electrical codes. It is the system engineer's responsibility to be certain that the inverter installation complies with all applicable electrical codes. (If a system engineer is not involved, it is the user's responsibility.)

(10/14/94)

AF-300C Inverter

40 to 300 HP

380/400-460 VAC, 50/60 Hz

GEK-85716A

Issue Date: August 1994

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia, U.S.A.

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1. SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions on them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.

WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.

CAUTION labels will be amber in color with black letterina.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.

NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - MECHANICAL MOTION HAZARD: inverter systems cause mechanical motion. it is the responsibility of the user to insure that any such motion does not result in an unsafe condition. Factory provided Interlocks and operating limits should not be bypassed or modified.

WARNING - ELECTRICAL SHOCK AND BURN HAZARD: When using instruments such as oscilloscopes to work on live equipment, the oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See instrument manufacturers instruction book for proper operatlon and adjustments to the instrument.

WARNING - STRAIN HAZARD: improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

WARNING - FIRE AND EXPLOSION HAZARD: Fires or explosions might result from mounting inverters in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Inverters should be Installed away from hazardous areas, even if used with motors suitable for use in these locations.

WARNING - ELECTRICAL SHOCK HAZARD: All motor bases and equipment enclosure houslngs should be grounded in accordance with the National Electric Code or other required standards.

WARNING - MOTOR OVERSPEED HAZARD: Because it is possible to configure the inverter for operation up to 400 Hz maximum output frequency, the inverter is capable of running the motor at up to 8 - 7 times its base speed. Never operate the motor above its top mechanical speed or a catastrophic failure may occur. Never configure the inverter for a top frequency above the safe mechanical speed for the motor. Consult factory for any application exceeding 120 Hz.

WARNING - PERSONAL INJURY HAZARD: Carry the inverter only by the body, not by the terminals or top cover. Equipment movement should only be performed by trained personnel.

WARNING - BURN AND SHOCK HAZARD: Shut off the AC power supply before making the output connections. Voltage may be present between the output terminals even though the inverter Is stopped. Make sure that the "CHARGE LED indicator has gone out before making any outputconnections.

WARNING - BURN AND SHOCK HAZARD: Do not conduct any inspections until after disconnecting the power supply and after the "CHARGE" LED indicator on the inverter has gone out.

WARNING - MECHANICAL MOTION HAZARD: When the frequency reference is zero, Function Code 01 displays zero. If Function Code 13 or 15 contains a non-zero number, the motor will run when a RUN comand Is glven even If the frequency reference is zero.

WARNING - MECHANICAL MOTION HAZARD: When running from the Keypad Panel with the 'Local/Remote" switch in the "Local" position, pressing the FWD or REV key will cause the motor to rotate. Any RUN permissives that may be wired to the FWD or REV terminals are ignored.

WARNING - PERSONAL INJURY HAZARD: To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained persons should modify any programmable inverter.

WARNING - FIRE HAZARD: Use only dynamic braking resistors that are provided by GE Drive Systems. These are special resistors that have a thermal fuse built in that opens if the resistor gets too hot. This fuse should only open If the DB transistor In the Inverter or DB unit fails short circuited. The resistor will need to be replaced if this happens.

CAUTION: When removinn or replacing wires, handle them by the connector onlv and not bv the wire itself. Be careful not to mix wires when reconnecting them.

CAUTION: Do not connect power supply voltage that exceeds the standard specification voltage fluctuation permissible. If excessive voltaae is applied to the invetter, damage to the internal components will result.

CAUTION: Do not connect power supply to the output terminals (U, V, W) Connect power supply onlv to the power terminal; (L1, L2, L3).

CAUTION: Do not connect power supply to the brakina resistor connection terminals (P, DB). Never short-circuit between P-N or P-DB terminals, and do not connect any resistance with a resistance value less than the standard application brakina resistor.

CAUTION: Do not connect power supply to the control circuit terminals (except 30A, B, C).

CAUTION: For RUN and STOP, use the FWD-CM (forward) and REV-CM (reverse) terminals. Do not use a contactor (for ON/OFFoperation) on the line side of the inverter for RUN and STOP.

CAUTION: Do not use a contactor on the output side of the inverter for ON/OFF operation.

CAUTION: The AC Supply Power Transformer KVA ratina should be greater than 1.5 times the Horsepower ratina of the invetter, but not areater than 500 KVA. If an AC Supply Power Transformer greater than 500 KVA is to be used, install a line reactor (optional; must be pruchased seperately) on the line side of the inverter. A Line Reactor is also required if an SCR Type controller and/or Power Factor Correction Capacitors are used on the same power lines.

CAUTION: Do not connect filter capacitors on the output side of the invetter.

CAUTION: Do not operate the inverter without the ground wire connected.

CAUTION: If the inverter's fault alarm is activated, consult the TROUBLESHOOTING section of this instruction book, and after correcting the problem, resume operation. Do not reset the fault alarm automatically bv external sequence, etc.

CAUTION: Do not perform a meaaer test between the inverter terminals or on the control circuit terminals.

CAUTION: Motor Thermal Overload protection must be provided, either bv motor thermoswitch, external motor overload relav, or inverter electronic thermal overload.

CAUTION: Because the ambient temperature greatly affects inverter life and reliability, do not install the inverter in anv location that exceeds the allowable temperature. Leave the ventilation cover attached for temperatures of 40 degrees C or below, and remove the cover for temperatures between 40 and 50 dearees **C. If the cover needs to be removed, another type of enclosure may be required for safety purposes. The inverter should not be operated when the ambient temperature is above 50 dearees C.**

CAUTION: Do not connect a phase advance condensor in the output wires. The condensor and inverter will overheat and be damaged due to harmonics.

CAUTION: The control circuit terminal wirinn should be kept as far awav as possible from the main power wiring to prevent operational error due to noise interference. Never install both types of wiring in the same duct or conduit. (A separation distance of 10 centimeters 14 inches1 or more is recommended.) If the control circuit wiring must cross the main power wirinn, it should cross at a right angle (see Fiaure 4-8).

CAUTION: Use shielded or twisted wire for the control circuit wirina (wiring should be as short as possible, i.e. 20 meters [65 feet1 or less). (Connect outer covering of the shielded wires to the inverter around terminal and leave the other end unconnected, but taped.)

CAUTION: Install a suppressor in parallel with any relavs or solenoid type coils, etc. that mav be close to the inverter or used with the inverter.

NOTE: *Always read the complete instructions prior to applyingpoweror troubleshooting the equipmentand follow al/procedures step by step.*

NOTE: *The motor chassis should be grounded to earth ground through a separate ground lead from all other equipment ground leads to prevent noise coupling.*

NOTE: *Readandheedall WARNING, CAUTION, and NOTE labels posted on the equipment.*

NOTES;

NOTES:

2. DESCRIPTION, COMPONENT IDENTIFICATION, AND SPECIFICATIONS,

GENERAL DESCRIPTION

The AF-300C™ inverter is available in 40 to 300 HP versions (380-460 VAC @ 50Hz; 400-460 VAC @ 60 Hz). All of the inverters use two 16-bit "Central Processing Units' (CPUs) for multifunction processing to provide high performance in all applications.

General data and specifications for each individual inverter are listed on the nameplate attached to the inverter (see Figure 2-1 for nameplate example). See TABLE 1 for a complete listing of all AF-300C inverter specifications.

Inverter operation and Function Code setting can be performed from the "Keypad Panel" that also features an LED Data Monitor Display. Eleven Touch Keys and the LOCAL/REMOTE switch arefurnished on the Keypad Panel for operation and Function Code adjustment. Operation Monitoring, Function Code Setting, Fault Monitoring, Data Protection, Reset, and Mode Changeover can all be performed from the Touch Panel. Operation Commands and Frequency Setting Commands can also be entered from the Keypad Panel. (See Figure 2-4 for Keypad Panel component identification and function.)

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INSPECTION ITEMS UPON DELIVERY

Upon receipt of the inverter, inspect equipment for the following items:

1. Check the nameplate to insure that the inverter specifications correspond to those ordered.
2. Inspect the unit for any damage that may have occurred during shipment.
   -- If there are any problems or questions regarding the above, contact the distributor that the inverter was purchased from.

MOVEMENT AND STORAGE

WARNING - PERSONAL INJURY HAZARD: Carry the inverter only by the body, not by the terminals or top cover. Equipment movement should only be performed by trained personnel.

For temporary storage after delivery, the following guidelines should be followed:

1. Ambienttemperature range in the storage area should be between -25 to +65 degrees C.
2. Leave inverter in its packing carton to avoid exposure to dust and potential damage.
3. Avoid placing the inverter directly on a concrete floor or in a location exposed to direct sunlight. (Store on a stand or shelf.)

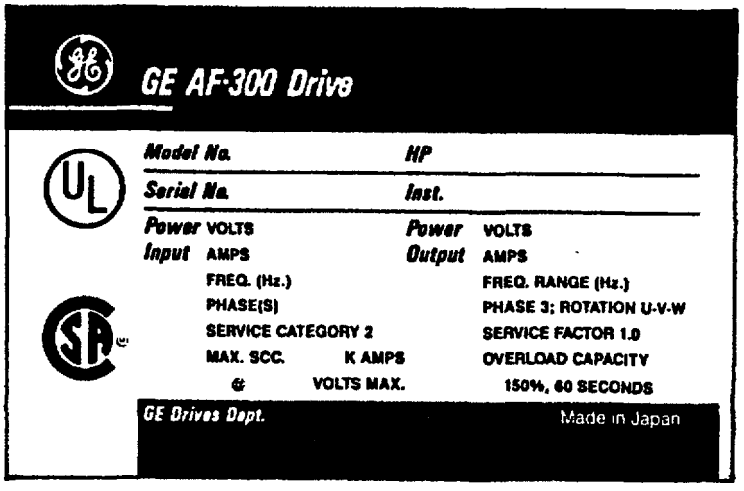


Figure 2-1. NAMEPLATE DATA IDENTIFICATION

4. Avoid storage in humid environments or in atmospheres that contain corrosive gases such as sulferized gas, ammonia gas, or chlorine gas.

NOTE: *If the inverter is stored or installed in a location where construction work is still being done, keep the inverter prorecred from exposure to water and dust until it is to be placed in service.*

INVERTER COMPONENTS AND COOLING

Refer to Figure 2-2 for identification and location of inverter components. There are two types of cooling methods depending on the installation method, "cooled inside control enclosure' and "cooled outside control enclosure' (see Figure 2-3).

Inverters rated 200,250, and 300 HP can be mounted "Cooled Inside Control Enclosure". However, a mounting pedastal will need to be constructed that can support the weight of the inverter. This pedastal should mount to the bottom of the inverter.

NOTE: Inverters are shipped with the adapter mounted for 'cooling inside the enclosure'. If "cooling outside the enclosure" is required, the adapter must be repositioned (see Figure 2-3).

The mounting adapter, positioned at either the top or bottom of the enclosure, must be located in accordance with the installation/cooling method.

With the "cooled outside the enclosure" method, the cooling fan is mounted externally and approximately 60% of the total heat generated by the inverter is discharged outside the unit.

NOTE: *When the cooling fan is installed outside the control enclosure, it must be periodically cleaned (especially in dusty environments).*

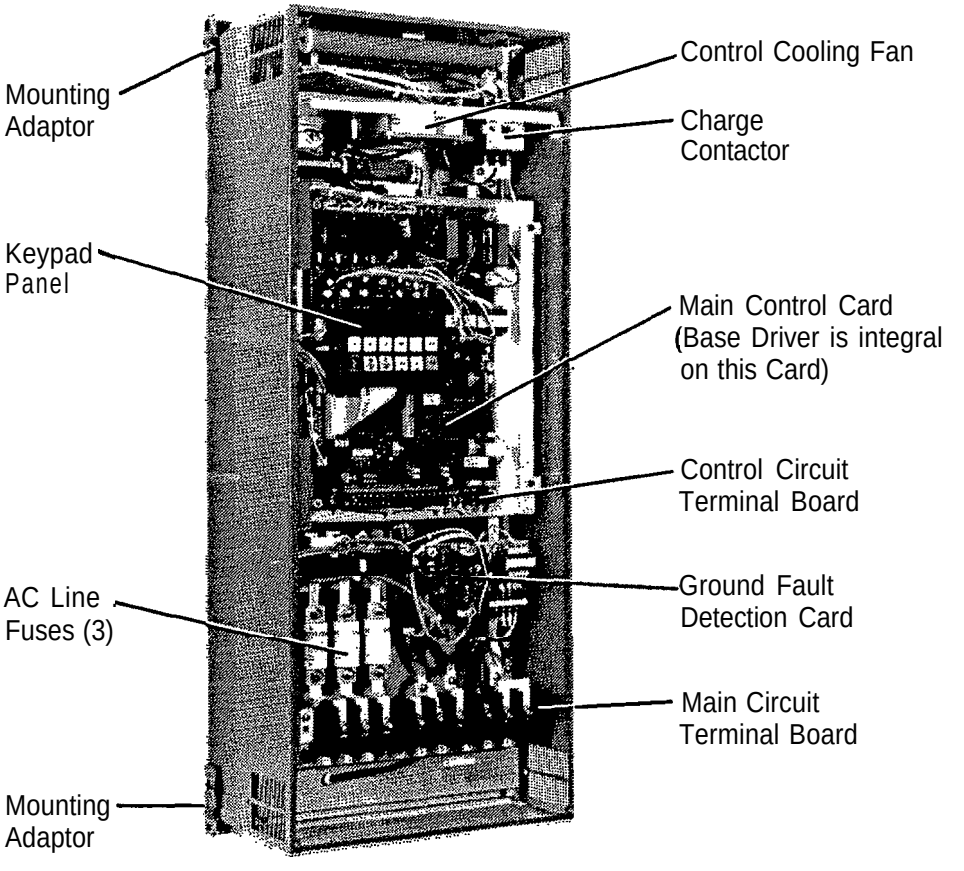


Figure 2-2. TYPICAL INVERTER COMPONENTS

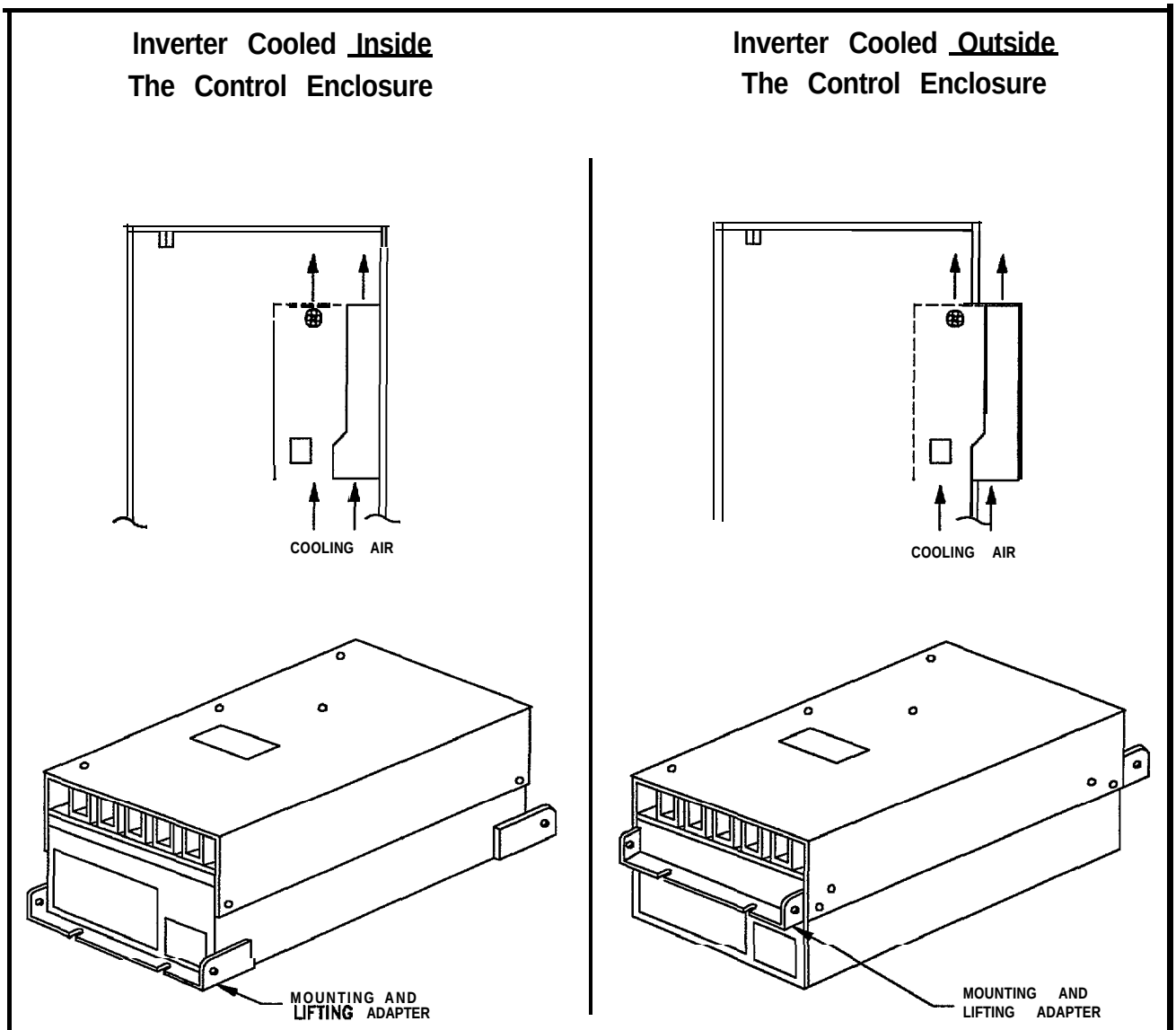


Figure 2-3. INVERTER COOLING METHODS

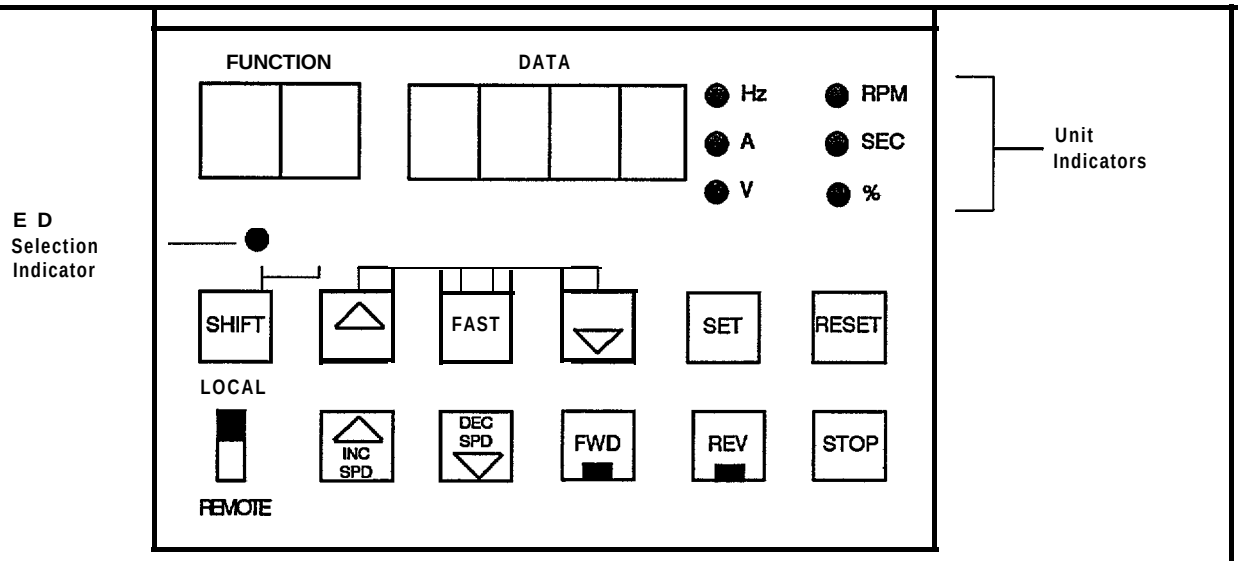


Figure 2-4. KEYPAD PANEL COMPONENT IDENTIFICATION

LED Indicator -The LED Indicator is made up of three sections as follows:

Function Display - Displays in two digits the selected Function Code number.

Data Display - Displays in four digits the operation data, Function Code setting data, and the fault codes.

Unit Indicator - The LED on the left of each unit symbol corresponds to the contents of the data display.

NOTE: All letters *are displayed in upper case except "b", "d", and "r".*

Touch Keys - There are eleven Touch Keys and the LOCAL/REMOTE switch that perform the following functions: (See Section 5 for a description of each individual key.)

Function		Description
Operation Monitoring		Monitor the operation status of the inverter.
Function Code Setting:	Basic Function Codes	Set data required for operation.
	Auxiliary Function Codes	Set data required for control.
	Calibration Function Codes	Adjust output signals to match with instruments externally installed.
Fault Monitoring		Display and retrieval of the Fault Code and the operating conditions at the time of the fault.
Set Data Protection		Protect set data against accidental modification.
Reset		Data reset at Function Code setting, reset to returr to operation monitoring mode after completing the setting, fault reset, and Error Display Reset.
Operation Mode Selection		Switch for changing between "Local" and "Remote' modes of operation.
Operation Commands		Forward, Reverse, and Stop commands can **be** entered in the "Local" mode.
Frequency Setting Commands		Increase and decrease reference frequency commands can be entered in the "Local' mode.

TABLE 1: AF-300C INVERTER STANDARD SPECIFICATIONS

ITEM	SPECIFICATION										
inverter HP (400460 VAC)	40	50	60	75	100	125	150	175	200	250	300
Standard Applicable Motors-460V (KW)	30	37	45	55	75	90	110	132	166	200	220
OUTPUT:											
Rated Capacity-460V (KVA) (*1)	46	57	69	65	114	134	160	193	232	287	316
Rated Output Current-460V (Amps)	60	75	91	112	150	176	210	253	304	377	415
Rated Output Voltage (*2)	3-Phase, 3-Wire Type, 400-460 VAC										
Rated Output Frequency (*3)	50 - 120 Hertz Base Frequency										
Overload Current Rating	150% for 1 minute duration (inverse time characteristic)										
POWER SUPPLY:											
Rated input AC Voltage	3-Phase, 3-Wire Type, 400-460 VAC @ 60 Hertz, 380-460 VAC @ 50 Hertz										
Allowable Fluctuations	Voltage: +10%, -15%; Voltage Unbalance: Within 3%; Frequency: +/-5%										
Control System	Sinusoidal Wave PWM With Flux Control										
Output Frequency Range (*3)	0.5 - 120 Hz (0.5 to 5.0 Hz start frequency; 50 to 120 Hz base frequency)										
CONTROL:											
Frequency/Temperature Fluctuation (Stability)	Analog Setting - Maximum frequency +/-0.2% (at 25°C, +/-10°C) Digital Setting - Maximum frequency +/-0.01% (at -10°C to +50°C)										
Frequency Setting Resolution	Analog Setting - 0.1% (0.05Hz/50Hz) of maximum frequency Digital Setting - 0.1 Hz (0.01 Hz/50Hz)										
Voltage/Frequency Characteristics (V/F) (*3)	Voltage - 320-460 VAC (Adjustable independently for both Base Frequency - 50 to 120 Hz voltage and frequency.)										
Torque Boost	21 selectable patterns plus energy conservation mode.										
Acceleration/Deceleration Chacteristics	0.2 to 3600 seconds, four Accel/Decel time settings available. Three patterns available (Linear, "S" Curve, Logarithmic Curve)										
Control Torque	Standard: Regenerative Brake, (DC Link Pump-Up) DC injection Brake, Starting Frequency of O-60 Hz at 0-10% Voltage for O-l 0 Seconds Time										
	Optional: Dynamic Brake, 100% (5% Duty Cycle).										

(*1) Indicates rated capacity when rated output voltage is 440 VAC.
(*2) It is not possible for output voltage to exceed power supply voltage.
(*3) Consult distributor for operation above 120 **Hertz.**

TABLE 1: AF-300C INVERTER STANDARD SPECIFICATIONS (continued)

ITEM	SPECIFICATION										
inverter HP (400-460V)	40	50	60	75	100	125	150	175	200	250	300
CONTROL: (continued), Torque Limit Control	Output current can be limited within a range of approximately 30% to 150% (1% step).										
Automatic Acceleration and Deceration Operation	Can be used to avoid DC Link overvoltage trips caused by motor regeneration during deceleration.										
Slip Compensation Control	Maintains motor at constant speed with load fluctuations.										
Multi-Step Frequency Setting	Seven adjustable frequency presets can be selected via three external contacts.										
Up-Down Control	External momentary contacts can be used to control frequency increase and decrease.										
Momentary Power Failure Restart	After momentary power failure, automatic restart is possible.										
inverter Bypass Logic	Allows switching the motor between the inverter and the power line.										
Bias Frequency Setting (*4)	Magnitude of the bias added to the frequency reference signal can be set from 0 to 100% (1% step).										
Frequency Jump	Up to three resonant frequencies can be skipped. Width adjustable from 0 - 5 Hertz (1Hz step).										
Reversing Operation By Frequency Setting Signal Characteristics	Makes bipolar frequency reference input possible. (Reverse motor rotation by changing frequency reference voltage polarity, rather than by using REV-CM contact.)										
Upper and Lower Limiters of Output Frequency	Output frequency can be limited to a specified range (overiding frequency reference).										
PFiOTECTiON/DETECTiON:	Stall Prevention, Overcurrent, Overvoltage, Undervoltage, instantaneous Power Failure, Inverter Overload, Motor Overload (electronic thermal action), External Failure (external thermal action), CPU Abnormal, Output Short Circuit, Ground Fault (for inverter protection only), and Input Surge.										
OPERATION: Frequency Reference Signals	Frequency Reference (Voltage Input): 0 to +/-10 VDC (0 to +/-5VDC). Current input: 4 to 20 Ma DC.										
input Signals	Forward Command, Reverse Command, Self-Holding Selection, Current Input Selection, Multi-Frequency Setting, Up-Down Control, Accel/Decel Tim Selection, Free-Run Command, Inverter Bypass Command, Output Switch Interlock Selection, External Alarm input, Alarm Reset Input, and Ground Protectionnput.										
Output Signals	Contact Outputs: Power-Side (or Motor-Side) electromagnetic contacto command, Fault Relay Output..										
	Open Collector outputs: See Section 6, Function Code 45.										

(*4) Bias does not affect operation when frequency command comes from the Keypad Panel.

TABLE 1: AF-300C INVERTER STANDARD SPECIFICATIONS (continued)

ITEM		SPECIFICATION										
Inverter HP (400-460 VAC)		40	50	60	75	100	125	150	175	200	250	300
INDICATION: Frequency Meter Output		Analog - 0 to +10 VDC, Pulse Frequency - (6 to 100) x Output Frequency.										
Touch Panel LED	Operation:	Output Frequency, Set Frequency, Synchronous Speed, Output Current, Output Voltage, Load Revolution Speed, and Presence of Input and Output Signals.										
	Setting:	Indication of Function Code and Set Data (Refer to Function Code List).										
	Fault:	OC1 - Overcurrent During Acceleration, OC2 - Overcurrent During Deceleration, OC3 - Overcurrent During Constant Speed, OV - Overvoltage, LV - Undervoltage, OL1 - Inverter Overload, OH1 - Inverter Overheat, OL2 - Motor Overload, OH2 - External Failure, Err0 - CPU Abnormal or Failure, and Failure History (three previous failure indications).										
LED Indication (charge light)		Lights up when DC Link capacitors are charged (DC Link voltage is present),										
ENVIRONMENT: Installation Location		Indoors, altitude of 3281 ft (1000 meters) or less, free of direct sunlight, corrosive gas, and dust. Motor wiring should be less than 66 feet, otherwise wire separation of 4 inches may be required.										
Ambient Temperature		-10 to +50° C										
Humidity		20 to 90% Relative Humidity (free of condensation).										
Vibration		0.5G or less.										
Temperature During Transportation		-25 to +65° C										
APPROXIMATE WEIGHTS												
Pounds		66	77	95	95	221	221	221	276	331	419	419
Kilograms		30	35	43	43	100	100	100	125	150	190	190

TABLE 1: AF-300C INVERTER STANDARD SPECIFICATIONS (continued)

ITEM	SPECIFICATION										
Inverter HP (400460 VAC)	40	50	60	75	100	125	150	175	200	250	300
Efficiencv and Loss:											
Loss (W)	1280	1520	1800	2050	2750	3150	3530	4330	5470	6650	7440
Efficiencv (%) Constant Torque Load:											
Base Speed Full Load	95.7	95.9	96.0	96.3	96.3	96.5	96.8	96.7	96.6	96.7	96.6
0.75 Base Speed Full Load	94.6	94.7	94.9	95.2	95.3	95.5	95.9	95.8	95.6	95.7	95.7
0.5 Base Speed Full Load	92.2	92.4	92.7	93.1	93.1	93.5	94.0	94.0	93.6	93.8	93.7
0.25 Base Speed Full Load	85.1	85.4	86.0	86.6	86.6	87.6	88.5	88.5	87.7	88.1	87.9
Efficiencv (%) Pump or Fan:											
0.8 Base Speed 64% Load	94.6	94.8	94.9	95.3	95.2	95.5	95.9	95.8	95.7	95.8	95.8
0.6 Base Speed 36% Load	93.5	94.1	94.1	94.2	94.4	94.9	95.4	95.3	95.3	95.5	95.8
0.4 Base Speed 16% Load	73.0	74.2	175.0	76.5	74.8	77.7	79.4	79.6	78.4	79.3	79.9

AFSOOC Catalog Numbers (460 VAC, 3-Phase, 50/60 Hertz)			
HP Rating	Catalog No.	HP Rating	Catalog No.
40	6VAF343040C-__(1)	125	6VAF343125C-__(1)
50	6VAF343050C-__(1)	150	6VAF343150C-__(1)
60	6VAF343060C-__(1)	200	6VAF343200C-__(1)
75	6VAF343075C-__(1)	250	6VAF343250C-__(1)
100	6VAF343100C-__(1)	300	6VAF343300C-__(1)

(1) The two character **positions** following a dash in the catalog number designate the following:

No dash or characters - inverter is not UL listed or CSA approved.

AI - Inverter is UL listed only.

A2 - Inverter is UL listed and CSA approved.

INVERTER OPTIONS

Various options are available for the AF-300C Inverter. These options can be ordered from the supplier of the inverter per the GE Catalog Number shown on the chart within this paragraph. The General Electric Company "6V" catalog numbers are coded as shown in the following Examples:

NOTE: *Option charts are divided into horsepower related and non-horsepower related items. Check the horsepower of the inverter before ordering options to be sure the correct option is ordered.*

Example: (Non-HP Related Items)

6VKAAA3AANN*

6V - GE Drives Products Operation
K - Option Kit
AAA - Applicable Series (1, 2, or 3 characters)
3 - AF-300 Family of inverters
AA - Option Abreviation
NN - Option Size/Length (if applicable)

Example: (HP Related Items)

6VKA3NAANN*

6V - GE Drives Products Operation
K - Option Kit
AAA - Applicable Series
3 - AF-300 Family of inverters
N - Voltage
AA - Option Abreviation
NN - Horsepower Rating (2 or 3 characters)

*A = *alphabetical character, N = Numerical char.*

AF-300C HORSEPOWER RELATED OPTIONS - Braking Units and Braking Resistors (See Section 4 for recommended wiring.)

Horsepower	40	50	60	75	100	125	150	175	200	250	300
Braking Unit 6VKC34BU- - -	050	050	075	075	150	150	150	175	150	150	150
Quantity Req.	1	1	1	1	1	1	1	1	2	2	2
Braking Resistor 6VKC34BR- - -	040	050	060	075	050	060	075	060	050	060	075
Quantity Req.	1	1	1	1	2	2	2	3	4	4	4
Capacity (KW)	3.6	4.8	6.0	7.2	9.6*	12.0*	14.4*	18.0*	19.2*	24.0"	28.8*
Resistance (Ohms)	15	12	10	7.5	6.0*	5.0*	3.75*	3.33*	3.0*	2.5*	1.88"

NOTE: *If the braking unit or resistor overheats, the thermal switch opens which turns off the braking unit transistors and trips the inverter on an "OH2" fault (provided thermal interlocks are wired correctly). These recommended braking unit and resistor combinations will provide up to 100% braking torque with a maximum 5% duty cycle (maximum continuous ON time of braking unit is 60 seconds).*

**Total value, not for one resistor.*

AF-300C NON-HORSEPOWER RELATED OPTIONS

Description	GE Catalog #	Description	GE Catalog #
RS-422/485 Card	6VKC3RS	Relay Unit	6VKC3RY
BCD Card	6VKC3BC	Keypad Cable - 2M	6VKC3CB02
Amps Signal Card	6VKC3AS	Speed Potentiometer	6VKC3SP
Output Signal Card	6VKC3AC	RFI Reactor	6VKABC3RF
Pulse Tach Feedback Card	6VKC3TF		

CAUTION: The AC Power Supply Transformer KVA rating should be greater than 1.5 times the HP rating of the Inverter, but not greater than 10 times the HP rating of the Inverter or 500 KVA (whichever is greater). If an AC Power Supply Transformer greater than this is to be used, install a Line Reactor (must be purchased seperately) on the line side of the Inverter. A Line Reactor is also required if an SCR type controller and/or power factor correction capacitors are used on the same power lines.

AC LINE REACTORS FOR AF-300C INVERTERS (See Caution)

HP	Volts*	KVA	Input Amps**	Part # 104X220	Inductance (in uH)
40	460	46	56	FA003	240
50	460	57	66	FA003	240
60	460	69	80	FA003	240
75	460	85	104	FA004	120
100	460	114	121	FA004***	120
125	460	134	145	FA004***	120
150	460	160	176	FA005***	75
200	*460*	*232*	*295*	FA005***	75
250	*460*	*287*	*370*	FA006***	50
300	*460*	*316*	*400*	FA006***	50

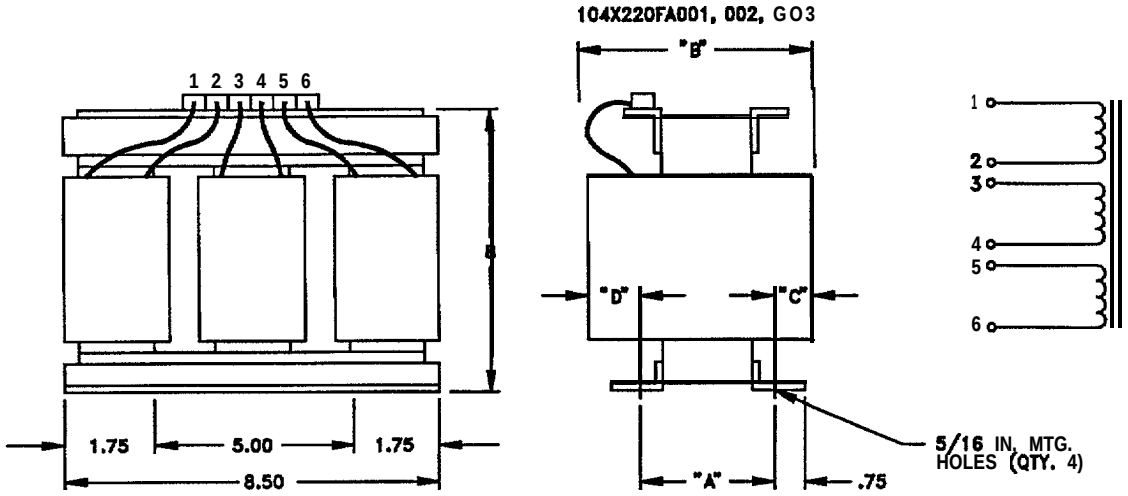
* *For input voltages other than 460, Contact General Electric Company, Drives Products Operation.*

** *Amps shown are at 460 VAC input voltage.*

***AC Line Reactors may also be required for inverters rated 100 UP and above even though a DC Link reacfor is provided with the inverter.*

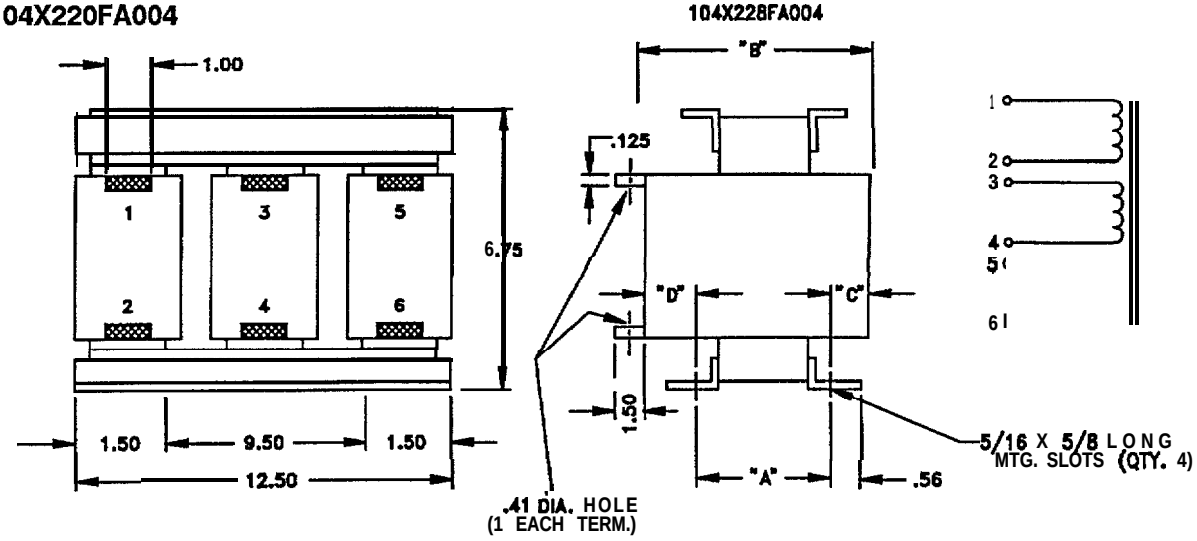
NOTE: If reactors are supplied by others, be sure that the peak (non-saturating) current rating of the reactor is greater than 2.8 times the input current rating of the inverter.

104X220FA003



NOTE: AU DIMENSIONS ARE IN INCHES.

104X220FA004



Reactor	Dlmensions (inches) A	B	C*	D	Terminal Board	Weight (pounds)
104X220FA003	3.125	5.563	0.875	1.750	16-6 AWG**	37
104X220FA004	3.625	7.250	0.625	1.50	----	67

* *Toierece of +/- 0.125 inches*

** *If* **4AWG** *is required, use* ***#4*** *solid or two* ***#10*** *stranded (****#4*** *stranded will not fit in connector).*

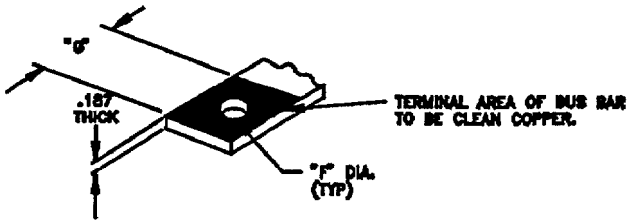
Figure 2-5. 104X220FA003, 004 LINE REACTOR OUTLINE AND SCHEMATIC

NOTES:

All dimensions in inches (not drawn to scale).

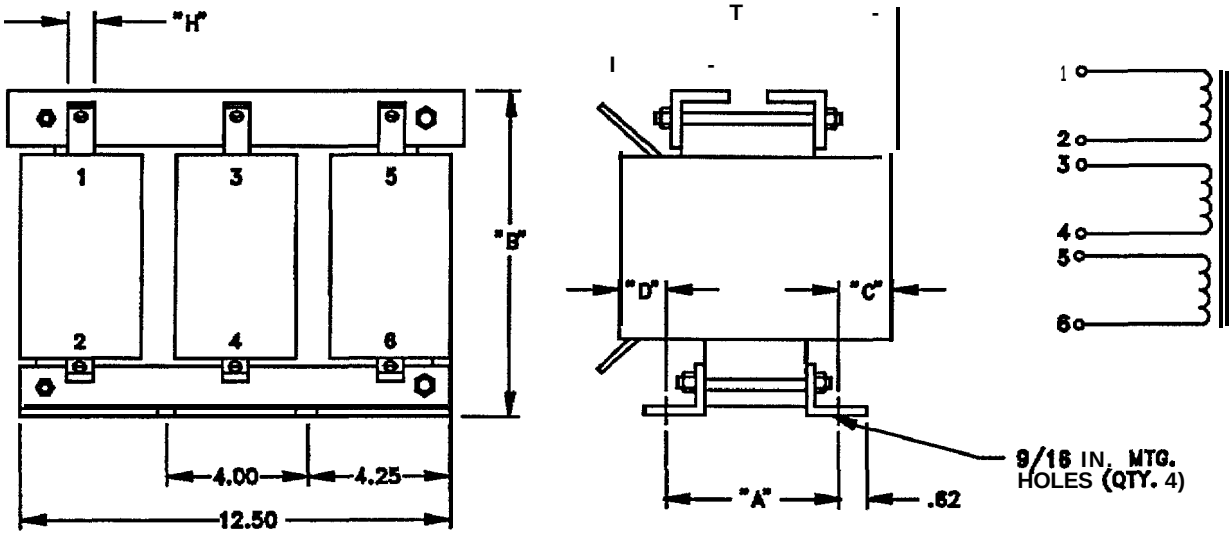
Dimension tolerance of +/-0.09 unless otherwise indicated.

Reactor may be slightly different than shown.



TYPICAL TERMINAL CONNECTION
(QUANTITY: 6)

104X220FA005, 006 (OUTLINE)



Reactor	DImensions (inches) A	B	C (max)	D	E	F	G (min)	H	Weight (lbs)
104X220FA005	5.00	8.875	1.00	1 .125	7.250	0.406	1.00	1.00	90
104X220FA006	5.50	12.50	.875	1.00	8.50	0.531	1.250	1.750	151

Figure 2-6.104X220FA005, 006 LINE REACTOR OUTLINE AND SCHEMATIC

3. INSTALLATION GUIDELINES

INSTALLATION ENVIRONMENT

Install the inverter in a location that meets the following requirements:

-- The ambient temperature is between -lCPC and +50°C (+14°F to +122°F).

-- The relative humidity is between 20% and 90%. Avoid any location subject to dew condensation, freezing, or where the inverterwould come in contact with water.

-- Do not install in any location subject to direct sunlight, dust, corrosive gas, inflammable gas, water vapor, or oil mist.

-- The inverter should be installed at an elevation below 1000 meters (3281 feet) and vibration should be less than 0.5G.

-- Motor wiring should be less than 66 feet. Otherwise, wire seperation of 4 inches may be required.

OPEN INSTALLATION MOUNTING CLEARANCE

CAUTION: Because the ambient temperature greatly affects inverter life and reliability, do not install the invetter in any location that exceeds the allowable temperatures.

-- Install the inverter perpendicular to the ground and with the lettering right side up. (If the inverter is installed up-side-down or horizontally, heat build-up will occur.)

-- Install at a sufficient distance from other equipment, walls, or wiring ducts as shown in Figure 3-l (these clearances are required to allow for proper cooling air circulation).

CAUTION: During operation, the temperature of the cooling fins of the inverter rise to approximatelv 90 degrees C (194 dearees F). For this reason, the mounting wall must be of heat resistant material.

NOTE: *When installing two or more* ***inverters*** *in close proximity, allow sufficient* **space as** *shown in Figure 3-1 and install them in a horizontal row. If they must be installed in a* ***vertical*** *row,* ***at least 19.7 inches (50cm)*** *spacemust be provided between each one to prevent the ambient temperature from rising.*

-- See the appropriate view in Figure 33 for the location of mounting holes.

-- Install the rubber bushings supplied with the inverter into the cable openings in the wiring lead-in plate to prevent cable damage and to minimize dust entry.

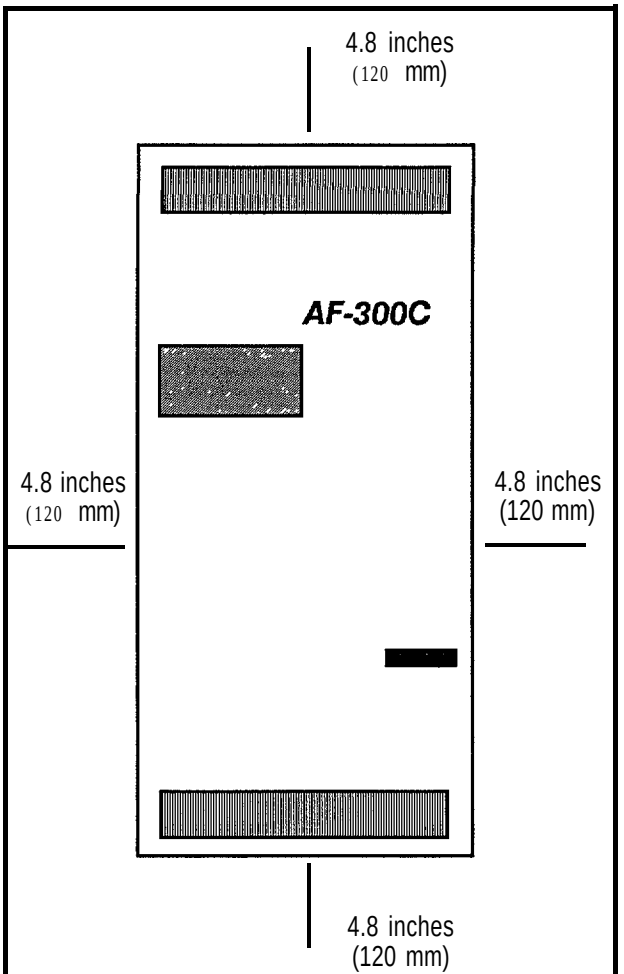


Figure 3-1. INVERTER MOUNTING CLEARANCE (Open Installation)

ENCLOSED MOUNTING INSTALLATION GUIDELINES

- When mounting inverter(s) inside of an enclosure, the temperature inside of the enclosure should not exceed 50°C (122°F).
- Do not install the inverter in a small sealed enclosure or mount other heat-generating components near the inverter within the enclosure.
- Cooling fans are to be installed in such a way that the cooling air passes through the heat-generating components.

NOTE: *Improper installation positions of the Inverter or Cooling Fan mayprevent the air temperature within the enclosure from being reduced to the specified value, even though the cooling fan is of the proper capacity.*

- When installing more than one inverter within the same enclosure, arrange them horizontally as shown in Figure 3-2. If the invetters must be arranged vertically, provide a partition board between the inverters to negate the heat generated by the lower unit from affecting the upper one (see Figure 3-2).

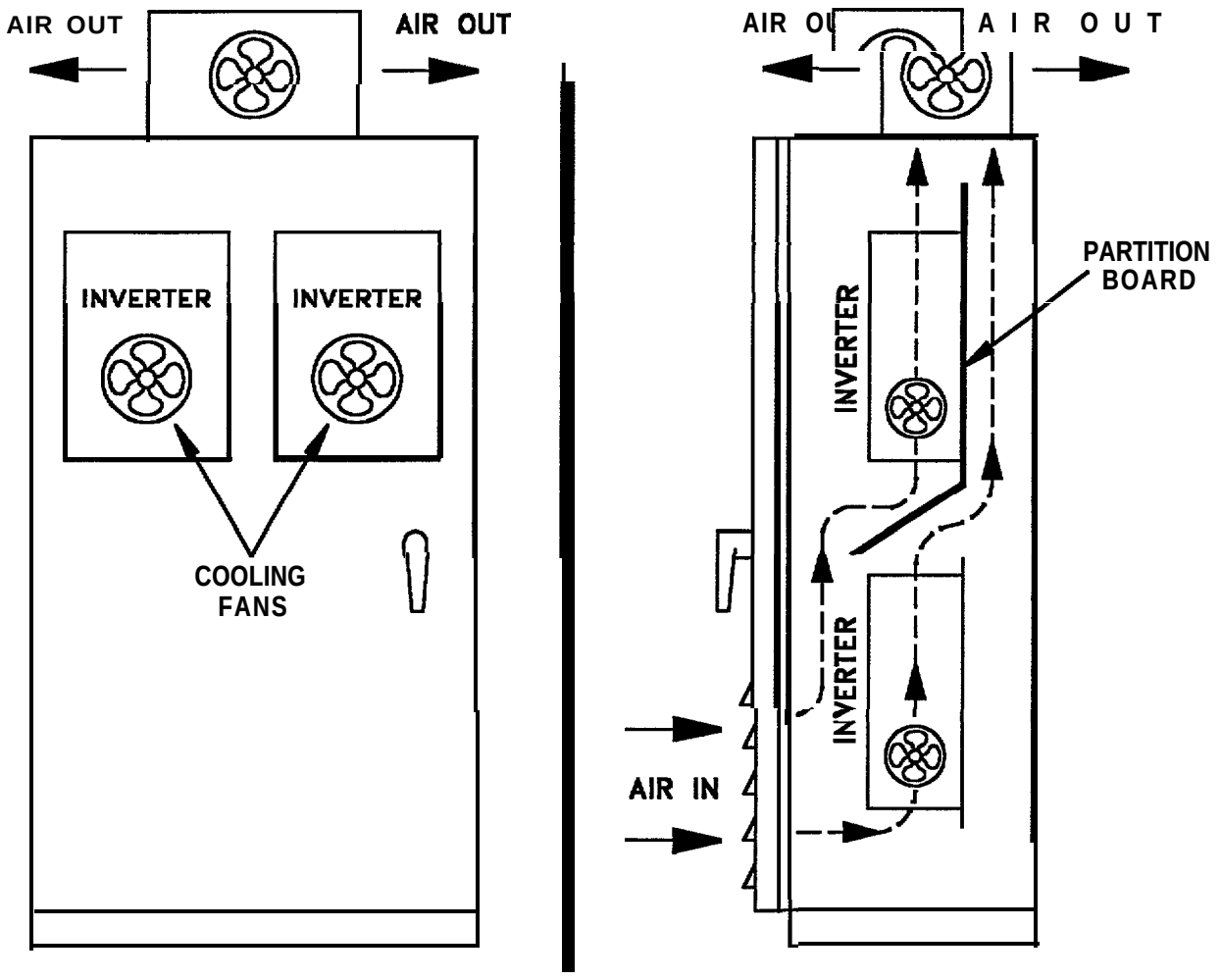


Figure 3-2. INVERTER ENCLOSURE MOUNTING ARRANGEMENTS

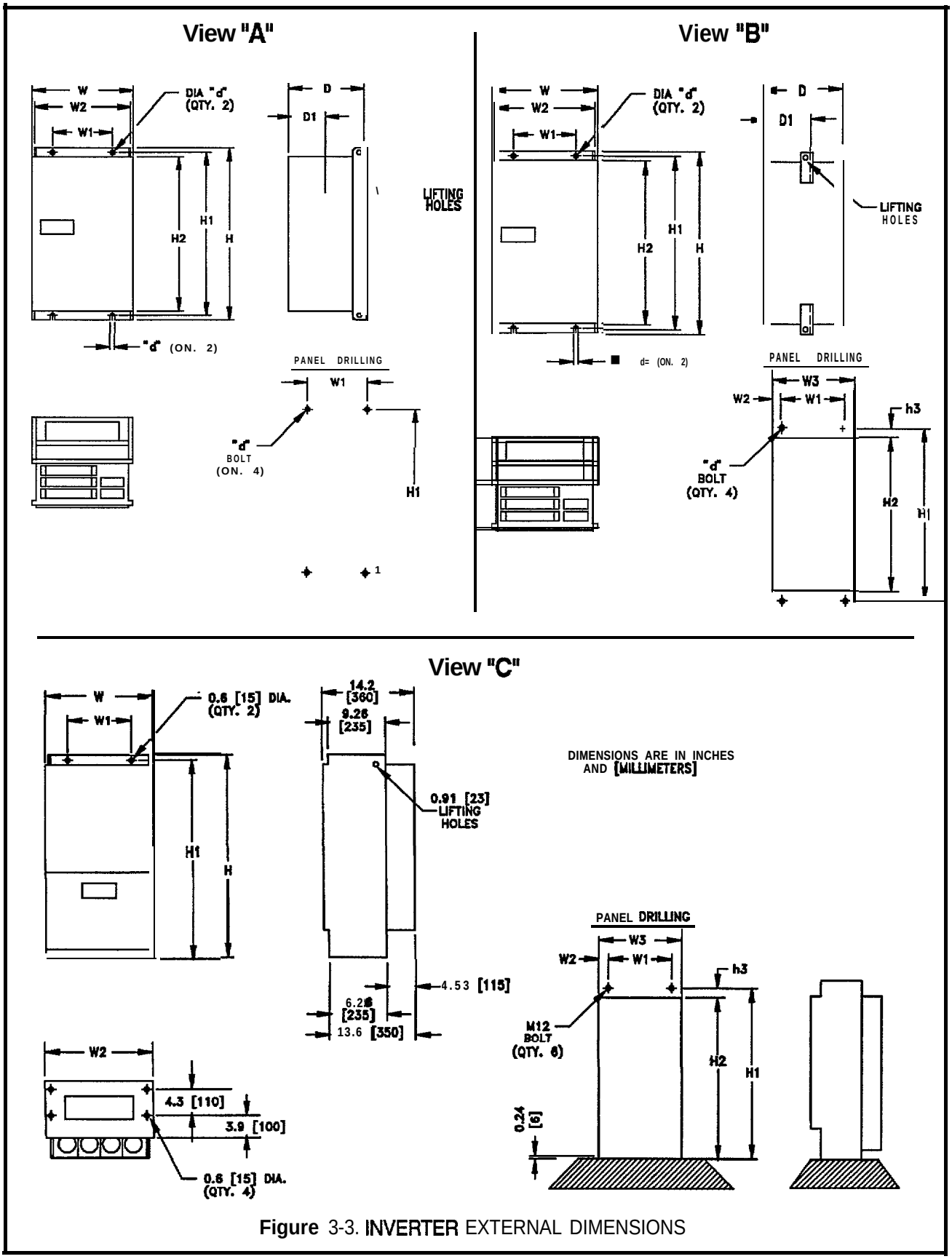


Figure 3-3. **INVERTER** EXTERNAL DIMENSIONS

AF-300C Dimensions (Refer to Figure 3-3)

Inverter HP	Catalog Number	View	Dimension in Inches and (mm) H	H1	H2	H3	H4	hl	h2	h3
40	6VAF343040C	A, B	35.25 (895)	84.45 (875)	33.45 (850)	33.75 (857)	11.26 (286)	.50 (13)	1.0 (25)	.35 (9)
50	6VAF343050C	A, B	39.20 (996)	88.45 (977)	37.40 (950)	37.75 (959)	14.02 (356)	.50 (13)	1.0 (25)	.35 (9)
60	6VAF343060C	A, B	43.90 (1115)	43.15 (1096)	42.15 (1071)	42.44 (1078)	16.70 (424)	.50 (13)	1.0 (25)	.35 (9)
75	6VAF343075C	A, B	48.90 (1115)	43.15 (1096)	42.15 (1071)	42.44 (1078)	16.70 (424)	.50 (13)	1.0 (25)	.35 (9)
100	6VAF243100C	A, B	51.20 (1300)	50.0 (1270)	48.60 (1234)	49.0 (1245)	20.25 (514)	.70 (18)	1.4 (36)	.50 (13)
125	6VAF343125C	A, B	51.20 (1300)	50.0 (1270)	48.60 (1234)	49.0 (1245)	20.25 (514)	.70 (18)	1.4 (36)	.50 (13)
150	6VAF343150C	A, B	51.20 (1300)	50.0 (1270)	48.60 (1234)	49.0 (1245)	20.25 (514)	.70 (18)	1.4 (36)	.50 (13)
200	6VAF343200C	C	64.60 (1641)	63.60 (1615)	63.20 (1605)	--- ---	32.40 (823)	--- ---	--- ---	--- ---
250	6VAF343250C	C	64.60 (1641)	63.60 (1615)	63.20 (1605)	--- ---	32.40 (823)	--- ---	--- ---	--- ---
300	6VAF343300C	C	64.60 (1641)	63.60 (1615)	63.20 (1605)	--- ---	32.40 (823)	--- ---	--- ---	--- ---

Inverter HP	Catalog Number	View	Dimension in Inches and (mm) W	W1	W2	W3	W4	D	D1	d
40	6VAF343040C	A, B	13.4 (340)	9.45 (240)	13.30 (338)	13.00 (331)	2.55 (65)	9.85 (250)	5.70 (145)	.40 (10)
50	6VAF343050C	A, B	14.8 (375)	10.80 (275)	14.70 (373)	14.40 (366)	2.55 (65)	9.85 (250)	5.70 (145)	.40 (10)
60	6VAF343060C	A, B	14.8 (375)	10.80 (275)	14.70 (373)	14.40 (366)	2.55 (65)	9.85 (250)	5.70 (145)	.40 (10)
75	6VAF343075C	A, B	14.8 (375)	10.80 (275)	14.70 (373)	14.40 (368)	2.55 (65)	9.85 (250)	5.70 (145)	.40 (10)
100	6VAF243100C	A, B	20.90 (530)	16.90 (430)	20.75 (527)	20.50 (520)	3.14 (80)	12.60 (320)	7.10 (180)	.60 (15)
125	6VAF343125C	A, B	20.90 (530)	16.90 (430)	20.75 (527)	20.50 (520)	3.11 (79)	12.60 (320)	7.10 (180)	.60 (15)
150	6VAF343150C	A, B	20.90 (530)	16.90 (430)	20.75 (527)	20.50 (520)	3.11 (79)	12.60 (320)	7.10 (180)	.60 (15)
200	6VAF343200C	C	26.80 (680)	22.80 (580)	26.00 (660)	24.00 (610)	3.10 (79)	14.40 (366)	9.25 (235)	--- ---
250	6VAF343250C	C	33.50 (850)	29.50 (750)	32.70 (830)	30.70 (780)	3.10 (79)	14.40 (366)	9.25 (235)	--- ---
300	6VAF343300C	C	33.50 (850)	29.50 (750)	32.70 (830)	30.70 (780)	3.10 (79)	14.40 (366)	9.25 (235)	--- ---

4. WIRING PROCEDURES

Wiring should be implemented in accordance with the following procedures. If these procedures disagree with local electrical codes, the local codes should be followed instead. After completing wiring, check that all wires have been properly connected.

CAUTION: Incorrect wiring may cause damaae to the inverter and/or result in improper operation.

The Main Circuit Terminal Block is located at the bottom portion of the inverter. The Control Circuit Terminal Block is located near the middle of the inverter. (See Figure 4-l)

NOTE: *P1-P* and *CM-THR* terminals are *jumpered together at the factory. A DC Reactor is also shipped* for *connection between terminals P1-P on 100 HP inverters and above.*

MAIN CIRCUIT WIRING

CAUTION: Be sure that the power supply is never connected to the U, V, W terminals nor the N, P1, and P terminals. Never connect N to P1 nor N to P.

Power Supply Connections

1. Connect a circuit breaker between the three-phase power supply and main circuit terminals L1, L2, L3 (see Figure 4-2).
   - An AC Line Reactor may be required for some applications. (See "AC Line Reactor Connection" paragraph in this section.)

2. If a starter or contactor is used between the circuit breaker and the main circuit terminals to shut off power to the inverter if an inverter fault occurs, then wire control power as shown in Figure 43.

NOTE: *See TABLE 2 for description of all terminals and TABLE 3 for application wiring list.*

NOTE: *An MC Contactof (see figure 4-3) should be used when either:*

(7) A dynamic braking option is installed.

(2) The power supply lines can experience an overvoltage condition.

if the current rating of the MC Contactor coil is too large for the Fault Relay Contacts, be sure to use a pilot relay as shown in Figure 4-12.

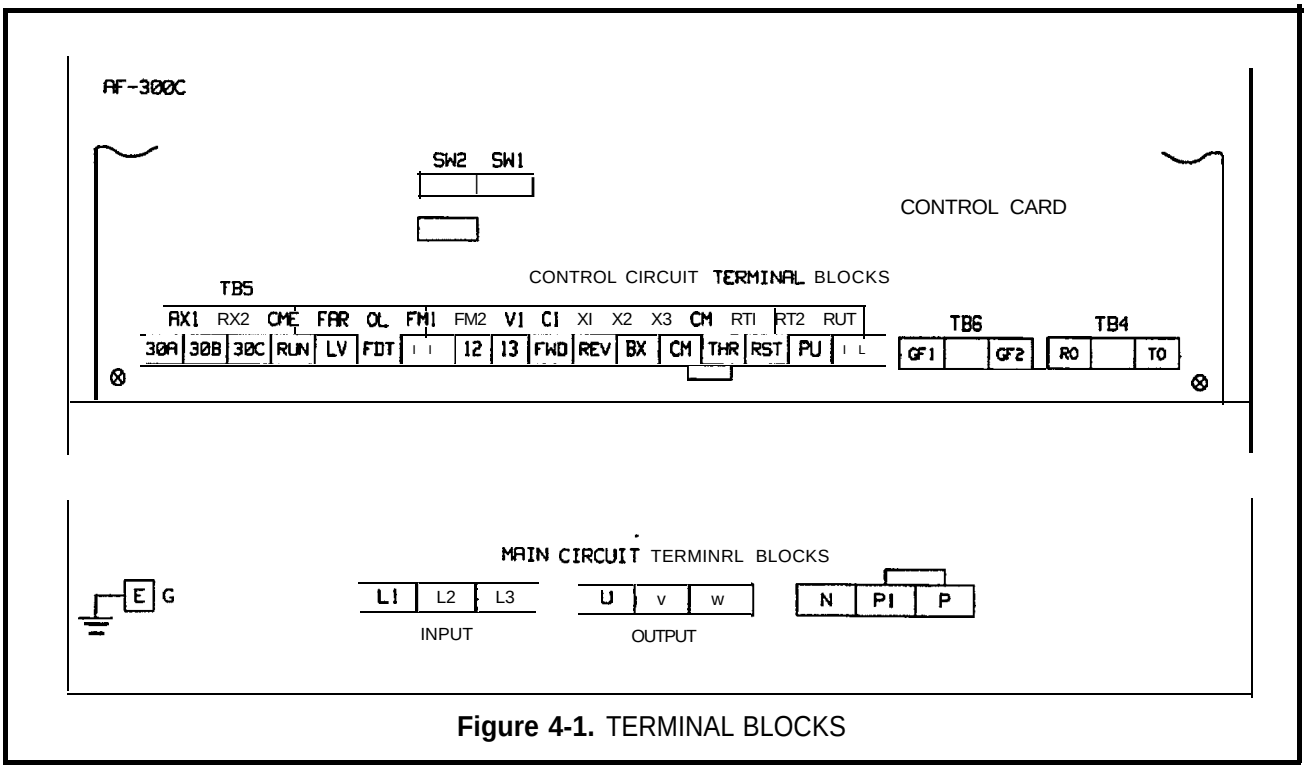


Figure 4-1. TERMINAL BLOCKS

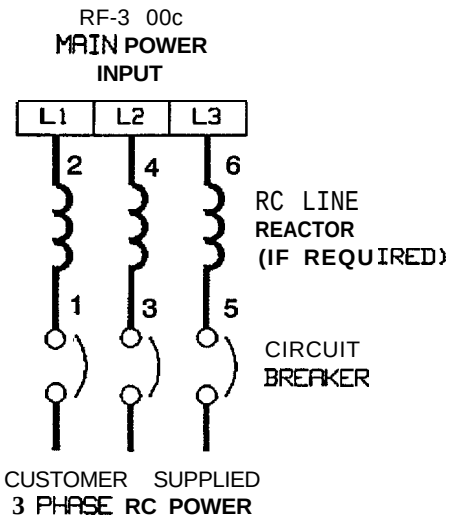


Figure 4-2. AC POWER CONNECTION

AUXILIARY CONTROL POWER SUPPLY CONNECTION

When using the circuit shown in Figure 4-3, it is necessary to connect terminals "RO" and "TO" to the hot side of the MC Contactor.

CAUTION: Do not neglect to make the RO and TO connections. If not made, continuous cycling of the MC Contactor may occur under certain conditions that will stress (or fail) the charge resistor and DC link capacitors.

AC LINE REACTOR CONNECTION

An AC Line Reactor is required on AF-300C inverters if the KVA rating of the supply transformer is greater than 10 times the horsepower rating of the inverter or 500 KVA (whichever is greater). This reactor is considered optional and must be purchased separately when required for specific applications (see "Options' paragraph in Section 2).

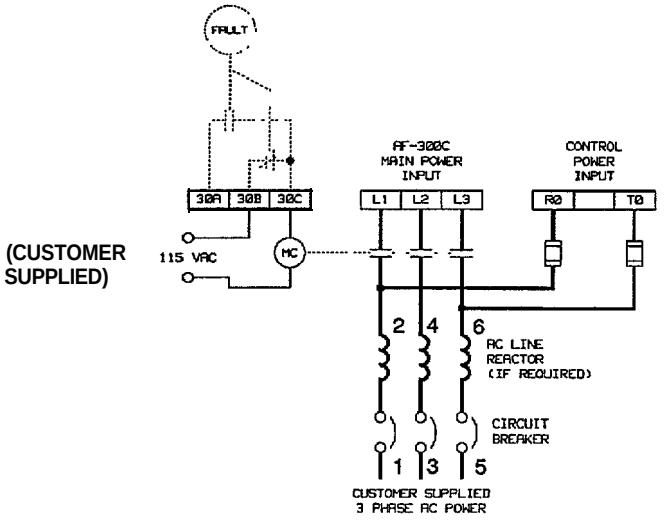


Figure 4-3. AUXILIARY CONTROL POWER SUPPLY CONNECTION

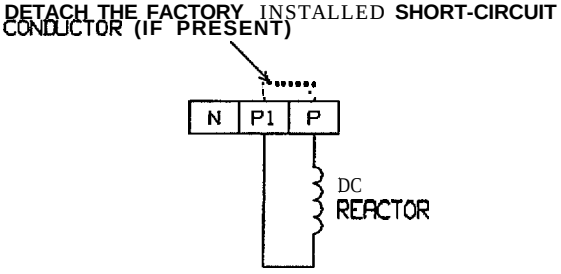


Figure 4-4. DC REACTOR CONNECTION

DC REACTOR CONNECTION

A DC Reactor is required on all AF-300C inverters rated 100 HP and above. This reactor is included in the purchase of the inverter and shipped with the inverter. This **reactor must be installed or the warranty will be voided.**

1. Remove the jumper between the "P" and "P1" terminals (installed at factory).
2. Connect DC Reactor between these terminals as shown in Figure 4-4 (see TABLE 3 for wire sizing).

Motor Connectlons (See Figure 4-3)

WARNING-BURNANDSHOCKHAZARD:
Shut off the power supply before making the output connections. If the power supply is on, a voltage may be present between the output terminals even though the Inverter Is stopped.

1. **Connect the 3-phase motor wires to the U, V, and W terminals of the Main Circuit Terminal Block as shown in Figure 4-5 (View A). (See TABLE 2 for description of all terminals and TABLE 3 for wire sizing.)**

NOTE: *Motor will rotate counterclockwise when viewed from the load side when connected normally. If the motor rotates in reverse direction, interchange any two of the U, V, or W terminal connections.*

2. **If contactors are connected as shown in Figure 4-5 (View B), connect an interlock to prevent closing both contactors (MC-1 and MC-2) at the same time. A contactor on the output for any other purpose must operate such that the contactor closes before the inverter is started and opens after the inverter stops.**

CAUTION: Do not connect anv capacitors in the output wires. The capacitors and inverter will overheat and be damaged due to harmonics.

BRAKING UNIT AND BRAKING RESISTOR CONNECTIONS

NOTE: *Braking Resistor must be used in combination with braking unit. The number of braking units and braking resistors varies per the horsepower of the inverter and the braking requirements of the the application (see Figure 4-6 for wiring).*

CAUTION: Alwavs match terminals "P" and "N" between the inverter and the brakina unit.

1. **Remove the jumper between the THR and CM terminals of the inverter (installed at the factory).**
2. **Connect the thermal contacts in series (to THR and CM) so that both the braking unit and braking resistor will be OFF should overheating occur.**
   -- **Connect braking unit so that thermal contacts can be made for terminals 1 and 2 (of braking unit) to terminals THF? and CM (of the inverter) respectively.**
3. **When two braking units are used, set Switch SW1 on the printed circuit board of the braking unit in the 1-2 position (set in 2-3 position at factory).**

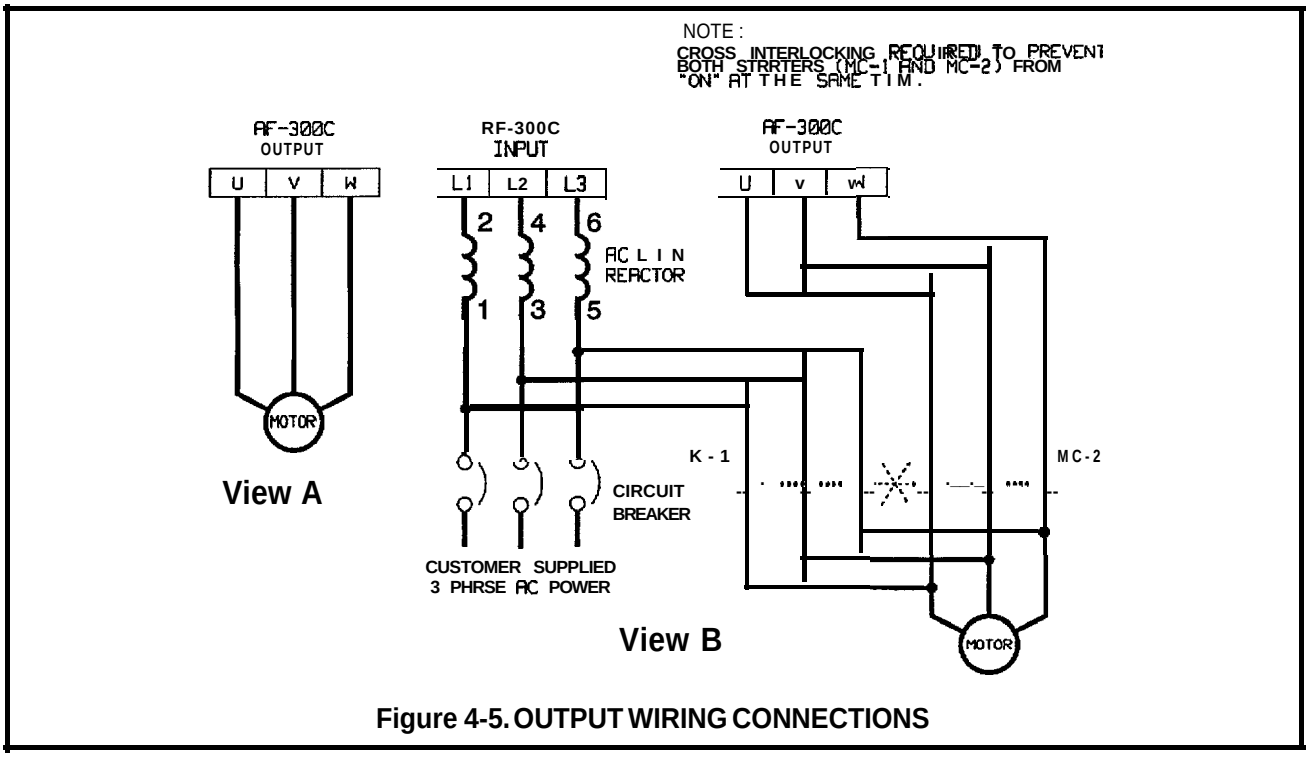
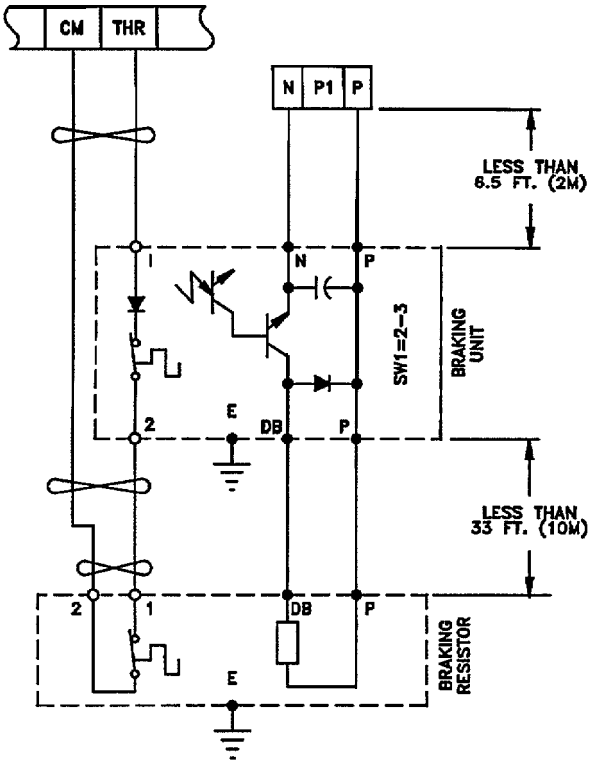


Figure 4-5. OUTPUT WIRING CONNECTIONS

1 Braking Unit
1 Braking Resistor
View "A"

NOTE: Be sure to remove the jumper from between terminals **THR** and **CM** or the Braking Unit will not operate.



1 Braking Unit
2 Braking Resistors
View "B"

NOTE: Be sure to remove the jumper from between terminals THR and **CM** or the Braking Unit will not operate.

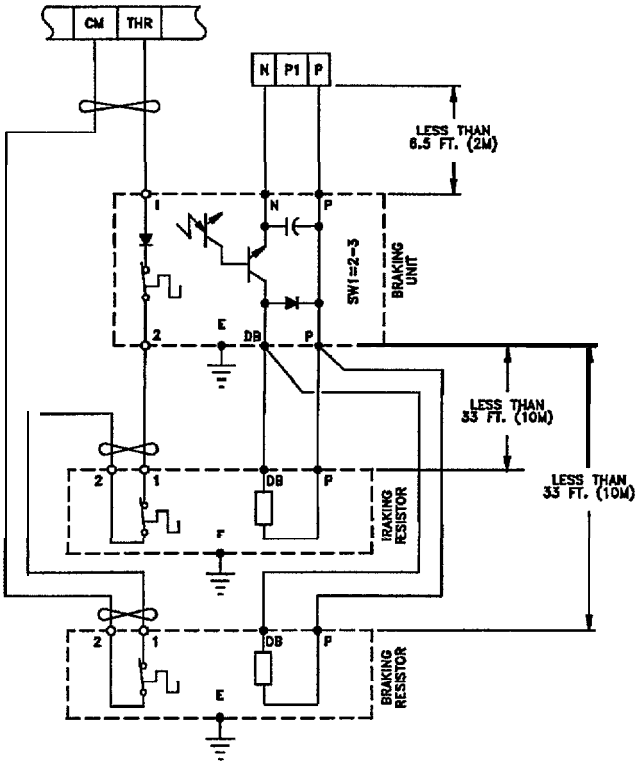
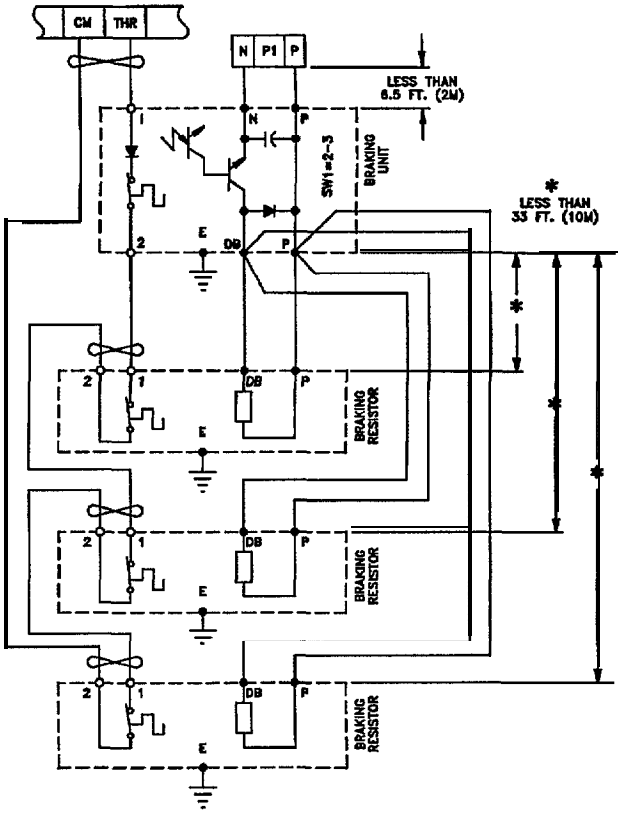


Figure 4-6. BRAKING RESISTOR AND BRAKING UNIT CONNECTIONS (Part **1** of 2)

1 Braking Units
3 Braking Resistors
View "C"

NOTE: Be sure to remove the jumper from between terminals THR and CM or the Braking Unit will not operate.



2 Braking Units
4 Braking Resistors
View "D"

NOTE: Be sure to remove the jumper from between terminals THR and **CM** or the Braking Unit will not operate.

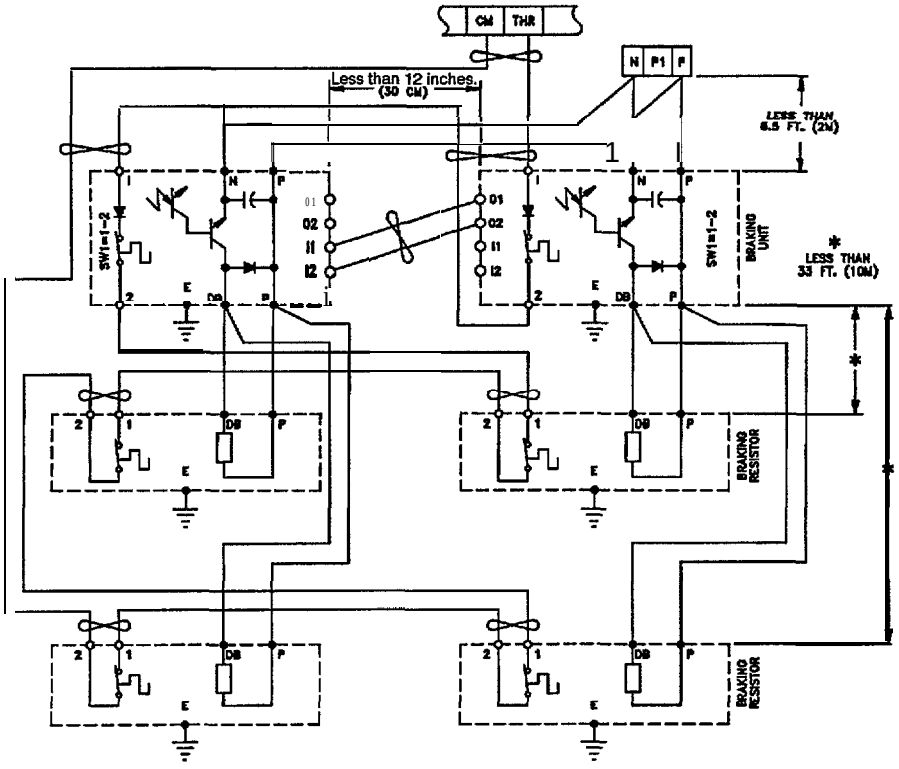


Figure 4-6. BRAKING RESISTOR AND BRAKING UNIT CONNECTIONS (Part 2 of 2)

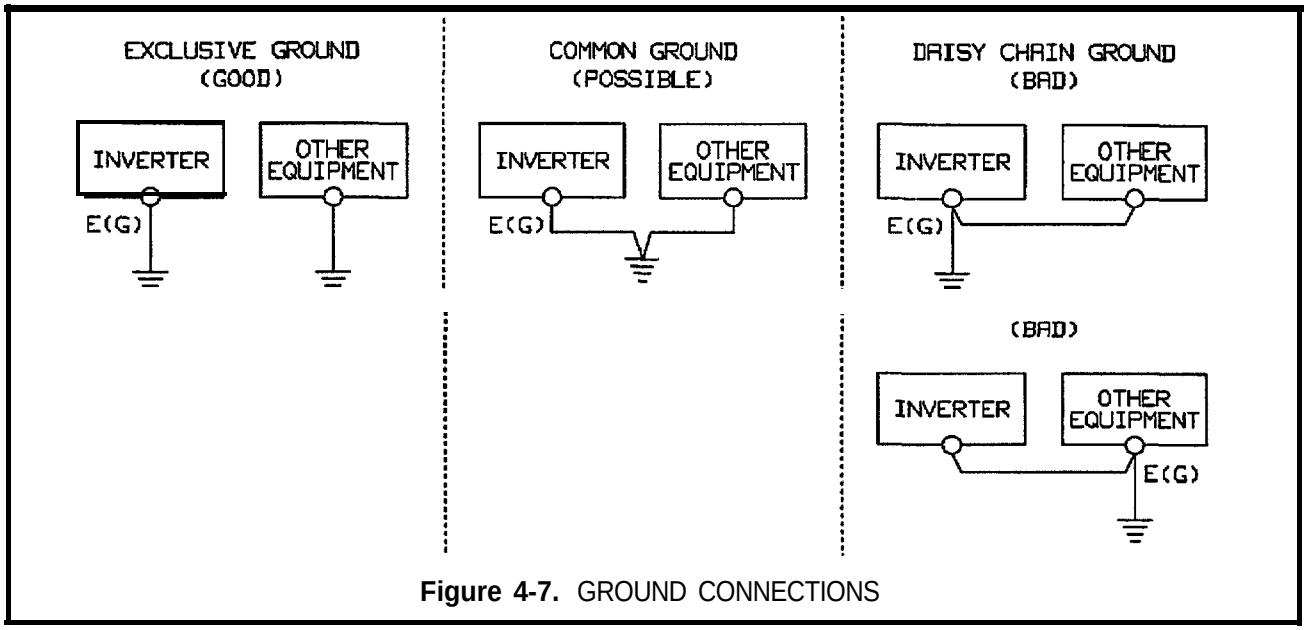


Figure 4-7. GROUND CONNECTIONS

GROUND TERMINAL CONNECTIONS

It is necessary to provide grounding in order to protect against an electral shock due to leakage and to reduce the effects of electrical 'noise". (See Figure **4-7.)**

Preferably, each inverter or other piece of equipment should be grounded individually. If this is not possible, the alternative method is common grounding where the ground wire from each piece of equipment is connected at the ground point.

NOTE: The ground wire *should* be as large a gauge *as possible and the length should be kept as* ***short*** *as possible.*

Avoid grounding method where the ground wire is used in common with other equipment i.e., do not daisy chain ground connections.

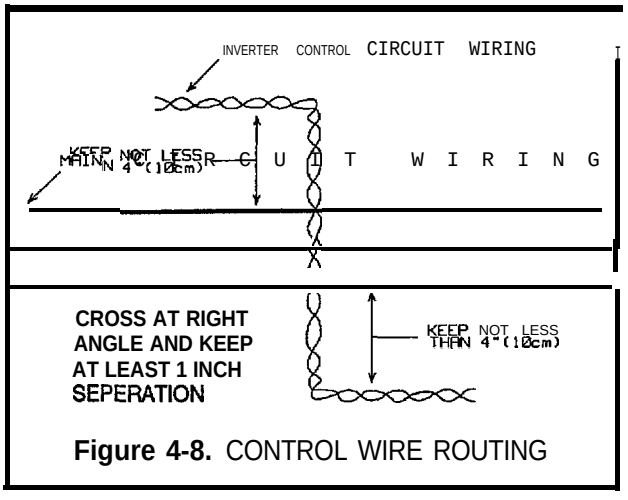


Figure 4-8. CONTROL WIRE ROUTING

CONTROL CIRCUIT WIRING

Perform all control wiring in accordance with the following applicable subparagraphs (and referenced Figures) and the applicable circuit as shown in Figures 4-15 to 4-26 (located at the end of this Section). Refer to TABLE 2for terminal descriptions and TABLE 3 for wire sizing.

CAUTION: High reliabilitv contacts for lowvoltaqe and current sianals must be used on the control circuit terminals (FWD, REV, BX, THR, **RST,** PU, IL, **X1,** X2. X3, **RT1** , RT2, **AUT).** When the contact is open, the voltage is approximatelv **+24 VDC;** when the contact is closed, the current is approximatelv 3 **mA** DC (see **Figure** 4-12 for circuit).

CAUTION: The control circuit terminal wirina should be kept as far **away** as possible from the main power wirinn to prevent operational error due to noise interference. Never install both **types** of wiring in the same duct or conduit. (A separation distance of 10 centimeters **[4** inches1 or more is recommended.) If the control circuit wirina must cross the main power **wir-**inn, it should cross at a riaht **angle** and be seperated **by** at least **1 inch(see Figure** 4-8).

CAUTION: Use shielded twisted wire for the control circuit wirina; wirinn should be as short as **possible,** i.e. 20 meters **(65 feet) or less. Connect** shield to the **inverter ground** terminal and leave the other end open and **electrically** insulated.

CAUTION: Install a suppressor in parallel with any **relays** or **solenoid type** coils, etc. that mav be close to the inverter.

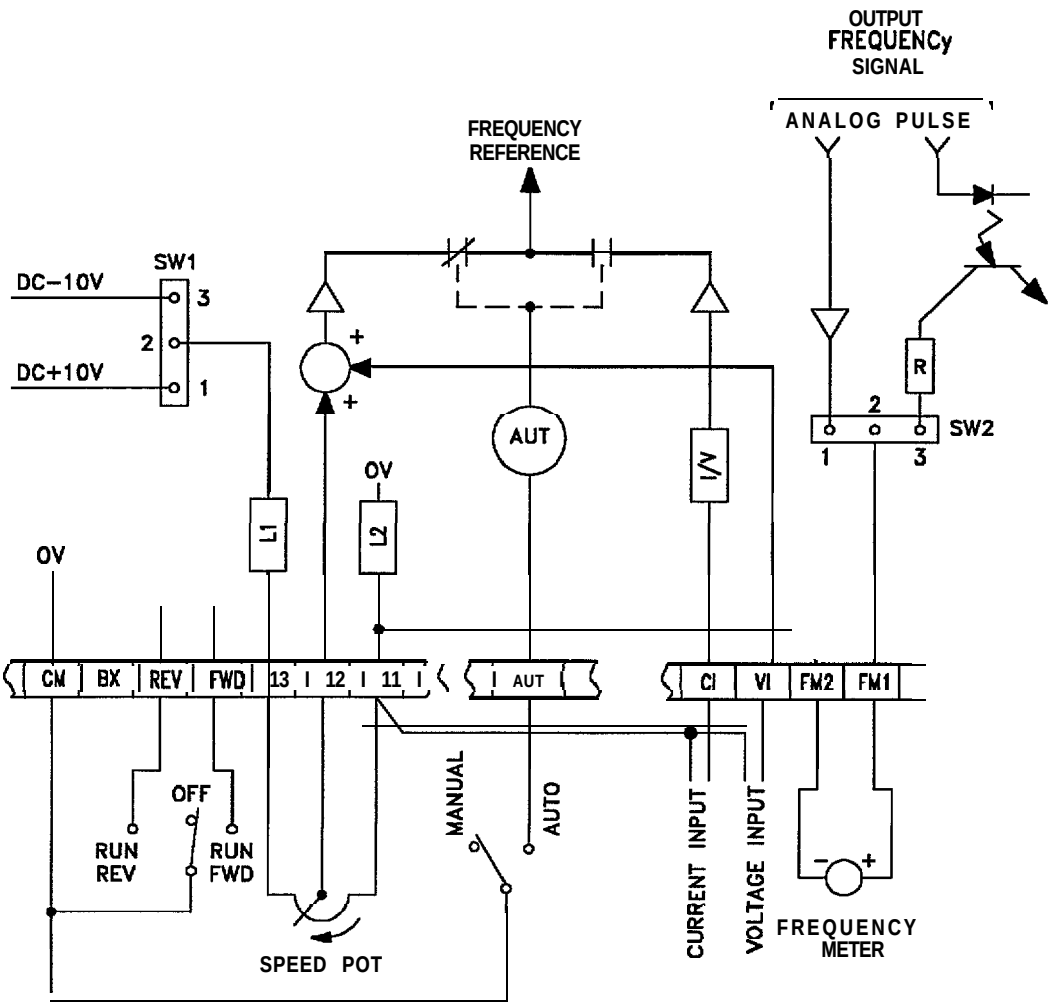


Figure 4-9. FREQ. SETTING/MONITORING EQUIVALENT CIRCUIT

Frequency Setting/Monitoring Equivalent Circuit

Frequency Setting Power Terminal (13) can beset for either positive or negative voltage polarity via jumper SW1 on the Main Control Card. (See Figure 4-9 for equivalent circuit.)

-- Factory setting of polarity is positive (SW1 in l-2 position).

-- Also see Function Code 47.

Although output frequency is digitally displayed on the keypad, when an external display is required, connect a frequency meter to monitoring terminals FM1 and FM2. Either analog or digital meters can be connected, but Switch SW2 must be set in accordance with the type meter used. Set SW2 as follows:

-- **1-2** Position for Analog Output (factory setting)

-- 2-3 Position for Digital Output

-- Also see Function Codes 40 and 50

NOTE: These *output* terminals *(FM1, FM2)* *are designed for meter connection* ***only. If a*** *special application* ***(for*** *control, etc.) is required,* ***contact*** *the distributor that supplied the* ***inverter.***

Alarm Reset Switch Connection

If an external RESET is required, connect a momentary pushbutton to the RST terminal as shown in Figure 4-10.

NOTE: *The application of this terminal allows a parallel operation to be performed with the* ***RESET key*** *on the Keypad Panel. Carelessness when setting Function Codes and/or retrieving faults* **may result** *in inputting a RESET signal.*

Equivalent Control Circuit

The equivalent circuit as shown in Figure 4-11 is present in the control circuits (FWD, REV, BX, THR, RST, PU, IL, X1, X2, X3, RT1, RT2, AUT). The contacts used must be highly reliable contacts for low voltage signals. 24 VDC is present when the switch is open; 3 mA DC is present when the switch is closed. The maximum leakage current that can be tolerated and still maintain an OFF state is 80 uA. The maximum voltage drop that can be tolerated and still maintain an ON state is one volt.

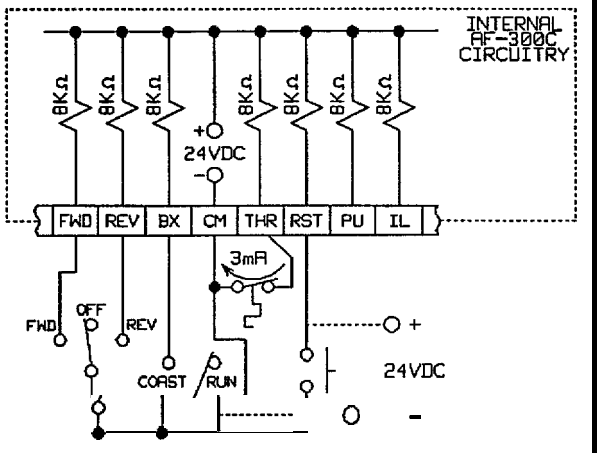


Figure 4-11. EQUIVALENT CONTROL CIRCIUT

Fault Relay Contact Rating

The Fault Relay contacts are rated at 250 VAC, 300 milliamps inductive @ cos $\phi \geq$ 0.3. When a large MC Contactor is used, use a pilot relay with the appropriate contact rating and coil rating. (See Figure 4-12.)

Open Collector Output Terminal Connection

It is recommended to use the optional relay output unit when using these output signals, however it is possible to use a customer furnished power supply and relays. The open collector outputs are "sink circuits. If customer supplied relays are to be controlled, an external DC power supply (27 VDC max.) is required. A diode must also be installed (reverse biased) for relay coil suppression (see Figure 4-13).

CAUTION: Verify that **the polarity** of the power supply connections are correct. Damaue will occur if **polarity is incorrect.**

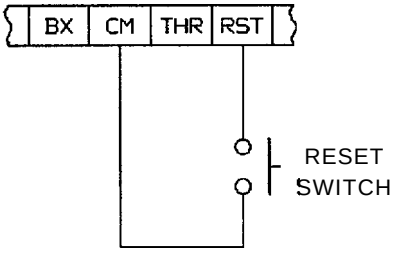


Figure 4-10. FAULT RESET CONNECTION

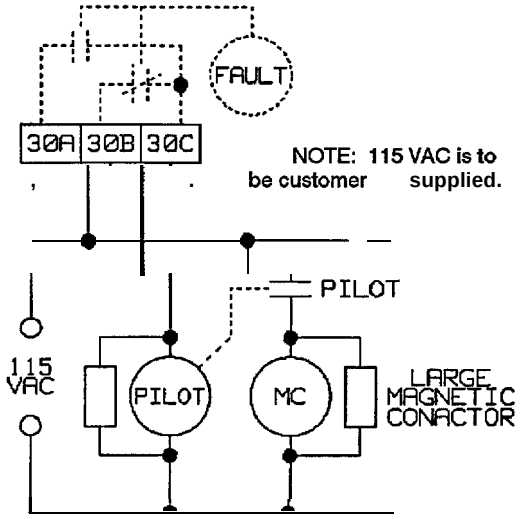


Figure 4-12. FAULT RELAY CONTACTS

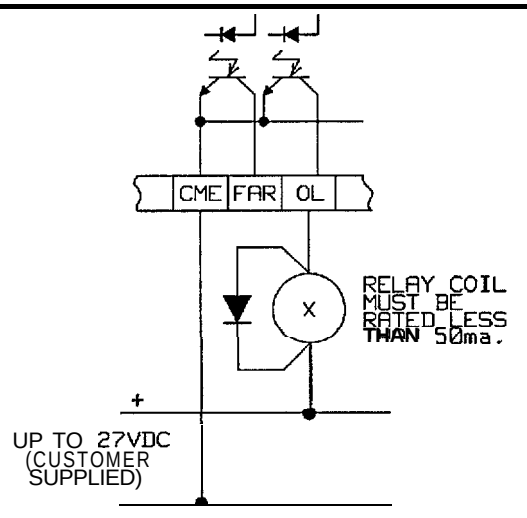


Figure 4-13. OPEN COLLECTOR CONNECTION

Surge Suppressor Connection

When a magnetic coil actuated device such as a starter, control relay, solenoid valve, etc., opens and closes, the current will sharply fluctuate resulting in a voltage spike (or noise). In some instances this voltage spike can cause misoperation of the electrical circuits of the inverter and other peripheral equipment. A surge suppressor should be connected across all magnetic coils connected to the inverter, as well as across all magnetic coils with supply wiring running parallel with inverter wiring. Wiring should be kept as short as possible (8 inches [20 cm] maximum). See Figure 4-14 for **connection** methods and thefollowing chart for recommended RC Filter capacity.

Surge Suppressor (RC Filter) Application

Equipment	RC Filter or Diode
Magnetic Contactor (DC)	0.2 microfarad, 500 VDC Capacitor; 500 Ohm Resistor or diode*.
Main Circuit (AC)	0.2 microfarad, 500 VDC Capacitor; 500 Ohm Resistor.
Auxiliary Relay (DC)	0.1 microfarad, 500 VDC Capacitor; 500 Ohm Resistor or diode*.
Auxiliary Relay (AC)	0.2 microfarad, 500 VDC Capacitor; 500 Ohm Resistor.
Solenoid Brake (DC)	0.2 microfarad, 500 VDC Capacitor; 500 Ohm Resistor or diode*.
Braking Clutch	0.2 microfarad, 500 VDC Capacitor; 500 Ohm Resistor.

**Diode to be used only when the current of the operating coil is no more #an 1 Amp.*

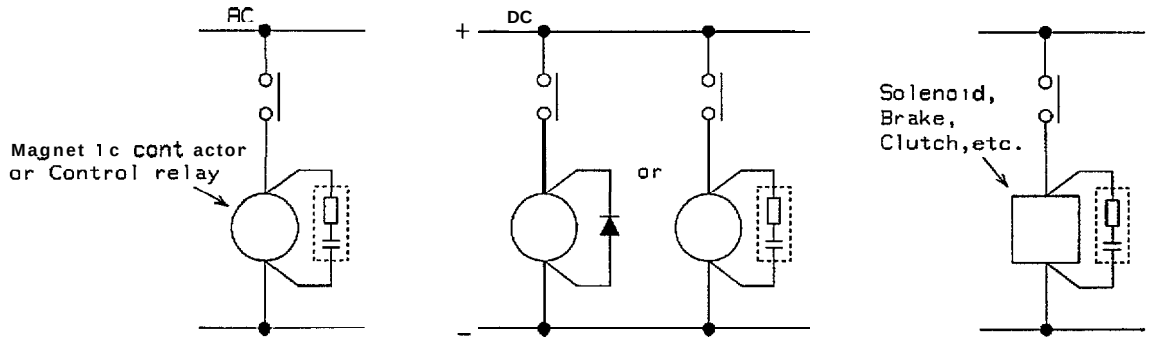


Figure 4-14. SURGE SUPPRESSOR CONNECTION

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION

Terminal Label	Terminal Name	Function
MAIN CIRCUIT:		
L1, L2, L3	AC Supply Line Input Terminals	Connection for commercial power (400-460 VAC @ 60 Hz) or (380-460 VAC @ 50 Hz).
U, V, W	Inverter Output Terminals	Connection for 3-phase induction motor.
P, N	Brake Unit Terminals	Connection for brake unit (optional). (Brake resistor must be connected through brake unit.)
P, P1	DC Reactor Connection Terminals	To be connected when power factor improving DC Reactor (optional on 75 HP and below) is used. (Provided and required on 100 HP and above.)
E (G)	Ground Terminal	Connection for ground.
CONTROL POWER:		
RO, TO	Control Power Auxiliary Input	Connection for single-phase AC power for backing up the control circuit power when input starter is used (460 VAC).
FREQ. SET MONITOR		
11	Frequency Setting Common	Common terminal for frequency setting signals (terminals 12, 13, V1, and Cl). (Speed Pot Common)
12	Frequency Setting Voltage Input	When 0 to + 10 VDC is input, the maximum frequency is reached at +10 VDC. The output frequency is proportional down to 0 VDC. Input impedance is 22K ohm. (Speed Pot Wiper)
13	Frequency Setting Voltage Output Term.	+ 10 VDC power supply for Speed Pot (10mA maximum). (Speed Pot Power)
Cl	Frequency Setting Current Input	When 4 to 20mA DC is input, the maximum frequency is reached at 20mA. The output frequency is proportional down to 4mA.
Vi	Voltage Input Auxiliary Terminal	Reaches maximum output frequency at +/-10 VDC (the output freq. is proportional down to 0 VDC). Input impedance is 22K ohms.
FM1 FM2	Frequency Meter Connection Points	Provides an output of 0 to +10 VDC (+10 VDC at max. frequency). Available for connection of a voltmeter (with internal resistance of 10K ohms). FM1 = Pos., FM2 = Neg. To output a digital signal, SW2 must be changed to 2-3 position. (See Figure 4-9 and Function Codes 40 and 50.)

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION (continued)

Terminal Label	Terminal Name	Function
CONTACT INPUT:		
CM	Contact Input Corn.	Common terminal of contact input signals.
AUT	Current Input Select	When the frequency setting signal is provided with voltage or current, this changes over the input signal to be used. AUT-CM On = Selects current input (4 to 20 mA). AUT-CM Off = Selects voltage input and frequency setting input,
FWD REV	Forward Command Input Terminal Reverse Command Input Terminal	Forward command via FWD-CM (closed). Reverse command via REV-CM (closed). When FWD-CM is closed and REV-CM is closed at the same time, the inverter will decelerate and stop (only when Function Code 41 is set inactive).
BX	Motor Coast-To-Stop Input Terminal	Motor will coast-to-stop with BX-CM (closed). No alarm signal will be output. If BX-CM is opened while FWD or REV closed , the inverter will start to operate because a RUN command is present.
XI x2 *x3*	Multi-Step Frequency Setting Terminals	Eight differenct frequency settings can be made with the combinations of ON and OFF of each terminal.
	Increase/Decrease Frequency Control	By changing Function Code 42 to YES, the X1 and X2 terminals are changed from multi-step speed commands to increase and decrease commands. The increase and decrease commands simulate a motorized speed pot.
	Momentary START/ STOP Contact Selection	By changing Functtion Code 41 to YES, the X3 terminal is changed from a multi-step speed command to a momentary pushbutton common input for momentary START/STOP commands. With X3-CM closed, FWD and REV signals input by a pulse signal (pulse width of 50 msec or greater) can be self-sealed.
RT1 RT2	Multi-Step Accel/ Decel Time Command Input Terminals	RT1-CM (closed), RT2-CM (closed), RT1-RT2-CM (closed) are the second, third, and fourth types of acceleration/deceleration time selections. NOTE: *When there is no input to RT1, RT2, operation is by normal first acceleration/deceleration time.*
THR	External Thermal Relay External Braking Resistor Thermostat Terminal	With THR-CM open, OH2 trip will occur and motor will coast-to-stop. NOTE: *With no external thermal relay or external braking resistor thermostat, the THR-CM terminals must be jumpered or the inverter will not operate.*
RST	Reset Signal Input Terminal	Faults are reset when a momentary contact is made between the RST-CM terminals for more than 0.1 seconds. NOTE: *If there is an input to the FWD or REV terminals, the unit will not RESET.*

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION (continued)

Terminal Label	Terminal Name	Function
CONTACT INPUT: (continued)		
PU	Commercial To Inverter	When PU-CM is closed, reference increases and preparation for inverter operation is made. When set to OFF after a fixed time, inverter will operate. (Used for open transfer Line-Inverter-Line; also used with terminal **"IL"** for transfer.)
IL	Output Starter Interlock	When providing an output starter, its auxiliary contact (normally closed) must be between IL-CM (closed). When the starter picks-up, the auxiliary contact must open (IL-CM open) permitting inverter operation. IL-CM On - interrupts inverter operation; IL-CM Off - continues inverter operation. Used with terminal **"PU"** for transfer.
OPEN COL-LECTOR OUTPUT*:		
CME	Open Collector Comm.	Common terminal for output of open collectors.
RUN	**Inverter** In Operation	Transistor turns ON between RUN-CME at start frequency and above. Transistor turns OFF when inverter is stopped, during coast stop, and DC braking.
FAR	Set Frequency Detection Signal	When the set frequency is reached (inverter is up-to-speed), FAR-CME will be ON. See Function Code 36.
FDT	Optional Detection of Frequency Level	FDT-CME will be on when the output frequency is larger than the set detection frequency; FDT-CME will be OFF when output frequency is less than set frequency. See **Functtion** Code 35.
OL	Inverter Overload Early Warning Signal	OL-CME will be on when the output current is larger than the set value; OL-CME will be OFF when output current is less than set current. See Function Code 32.
LV	Undervoltage Stop	LV-CME will be on when an undervoltage condition exists. LV-CME will be off when **the** DC link voltage is above the trip threshold. No output will occur for a set time after power-up.
CONTACT OUTPUT:		
AX1 **AX2**	Run Relay Dry Contact output	Contact is closed when the inverter is running. (Contact rating: 250 VAC, 0.3 Amps inductive @ cos theta $\geq$ 0.3)
30A **30B** 3oc	Fault Relay Dry Contact Output	During normal operation, the relay is not energized and contact is made between **30B** and 30C. When a fault is detected, the relay is energized and contact is made between 30A and 30C. (Contact rating: 250 VAC, 0.3 Amps inductive @ cos theta $\geq$ 0.3)

Open collector outputs are "Sink" circuits with +27 VDC, 50mA maximum ratings. Power for these circuits* **must *be provided via an external power supply when using them to control external relays (see Figure 4-13.) The Relay Option 6VKC3RY has this power supply built in.*

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION (continued)

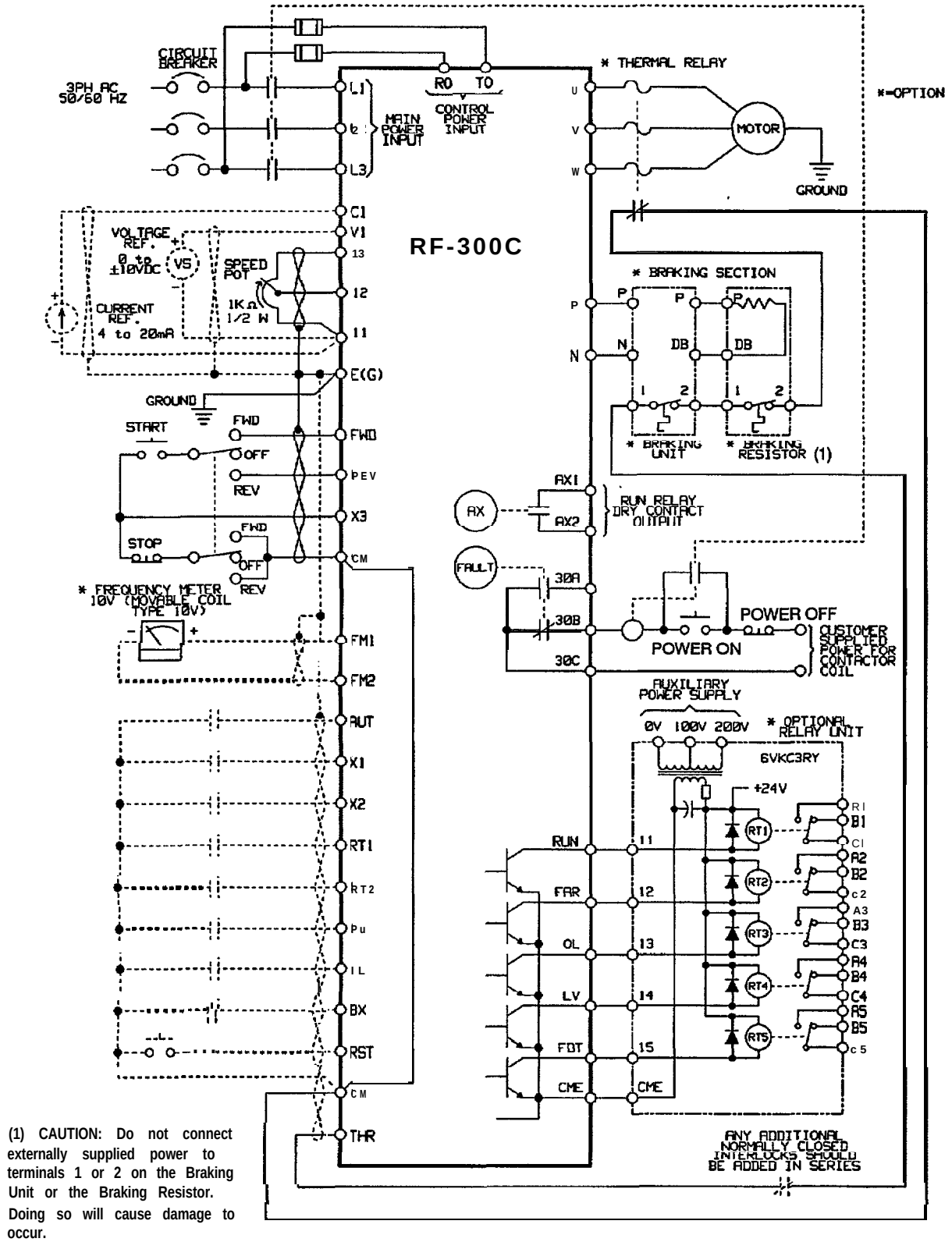
Terminal Label	Terminal Name	Function
PROTEC- TION		
GF1 GF2	**Ground Fault Protection Input**	**Input terminals exclusively used for internal ground fault detection unit.**

TABLE 3: WIRING AND EQUIPMENT SIZING (Power Wire, AWG*)

Inverter HP	Qty.**	Input Power	Output Power	DC Reactor	Stud Diameter	GE Terminal 104X161AA-	Amp Terminal Number	Amp Crimp Tool #
40	1	2	--	--	0.236	057	320383	69062
40	1	--	2	--	0.236	057	320383	69062
An	--	--	--	--				
50	1	2	--	--	0.315	056	321600	69062
50	1	--	2	--	0.315	056	321600	69062
50	--	--	--	--				
60	1	1/0	--	--	0.315	060	321868	69120
60	1	--	2	--	0.315	056	321600	69062
60	--	--	--	--				
75	1	2/0	--	--	0.315	023	36923	69120
75	1	--	1/0	--	0.315	060	321868	69120
75	--	--	--	--				
100	1	2/0	--	--	0.393	023	36923	69120
100	1	--	3/0	--	0.393	025	36927	69120
100	1	--	--	4/0	0.393	068	321878	69120
125	1	4/0	--	--	0.393	068	321878	69120
125	1	--	4/0	--	0.393	068	321878	69120
125	2	--	--	1/0	0.393	060	321868	69120
150	2	1/0	--	--	0.393	060	321868	69120
150	2	--	1/0	--	0.393	060	321868	69120
150	2	--	--	2/0	0.433	024	36925	69120
200	2	3/0	--	--	0.512	026	36929	69120
200	2	--	3/0	--	0.512	026	36929	69120
200	2	--	--	4/0	0.512	028	36934	69120
250	2	4/0	--	--	0.512	028	38934	69120
250	3	--	2/0	--	0.512	024	36925	69120
250	3	--	--	3/0	0.512	026	36929	69120
300	3	2/0	--	--	0.512	024	36925	69120
300	3	--	2/0	--	0.512	024	36925	69120
300	3	--	--	4/0	0.512	028	36934	69120

**Wire size from NEC Table 310-16. Copper wire rated 60°C for 100 amps and less, 75°C for over 100 amps in 30°C ambient and 1.25 times inverter rated amps. These are minimum wire sizes; consult and conform to local and national codes.*

***Quantity of wires in parallel per terminal.*



(1) CAUTION: Do not connect externally supplied power to terminals 1 or 2 on the Braking Unit or the Braking Resistor. Doing so will cause damage to occur.

Figure 4-t 5. BASIC CONNECTION DIAGRAM

IMPORTANT: The following CAUTIONS apply to Figures 4-l 6 through 4-26. Other NOTES that apply to a specific Figure are shown on that Figure.

***1 - CAUTION:** If it is desired to use thermal interlocks, remove the jumper from between terminals THR and CM and connect **Normally** Closed thermal interlock contacts in series between THR and CM.

***2 - CAUTION:** If the "LOCAL/REMOTE' switch on the Kevpad Panel is in the 'LOCAL' position, START/STOP commands from remote are **ignored.**

NOTE: LOCAL/REMOTE switch should be in the 'LOCAL' position **(*2).**

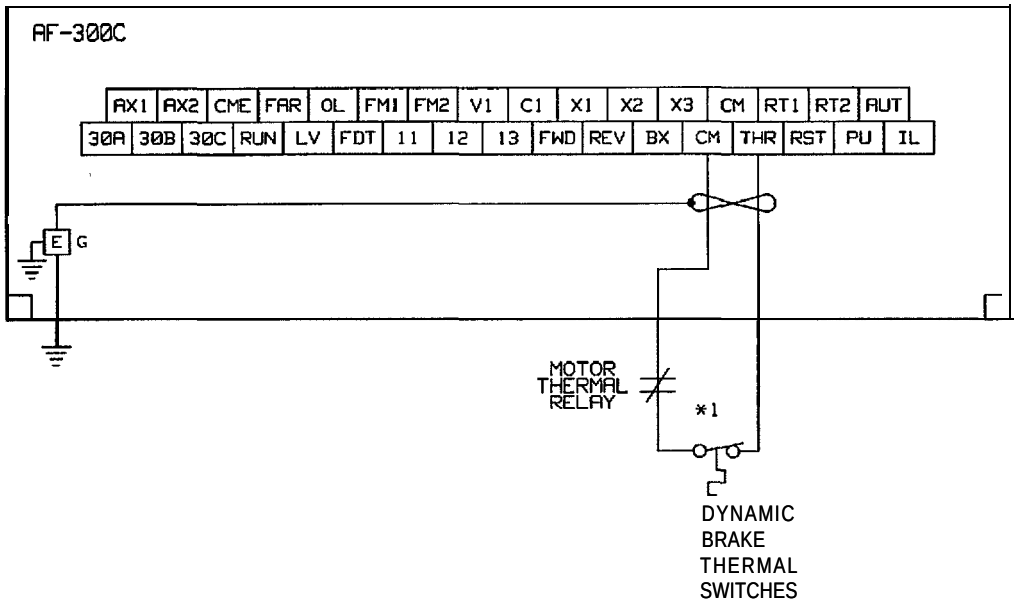


Figure 4-18. OPERATION FROM KEYPAD

NOTE: LOCAL/REMOTE switch should be in the 'REMOTE' position **(*2).** Set Function Code 41 to **"0"** (Yes).

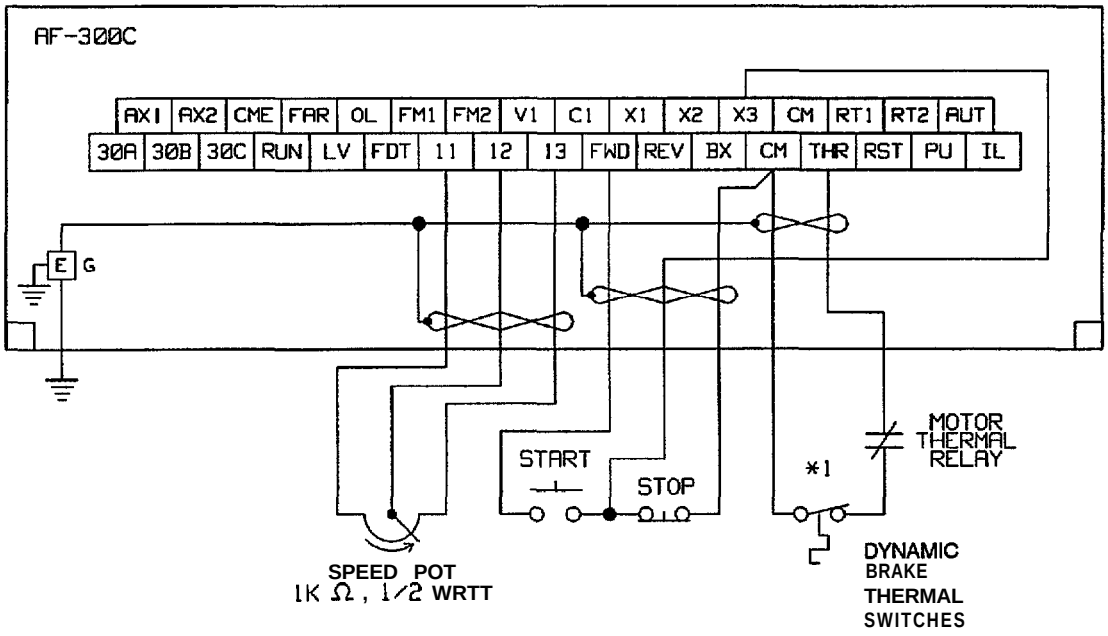


Figure 4-17. REMOTE OPERATION WITH MOMENTARY CONTACTS

NOTE: LOCAL/REMOTE switch should be in the 'REMOTE" position **(*2).** Set Function Code 41 to **"0"** (Yes).

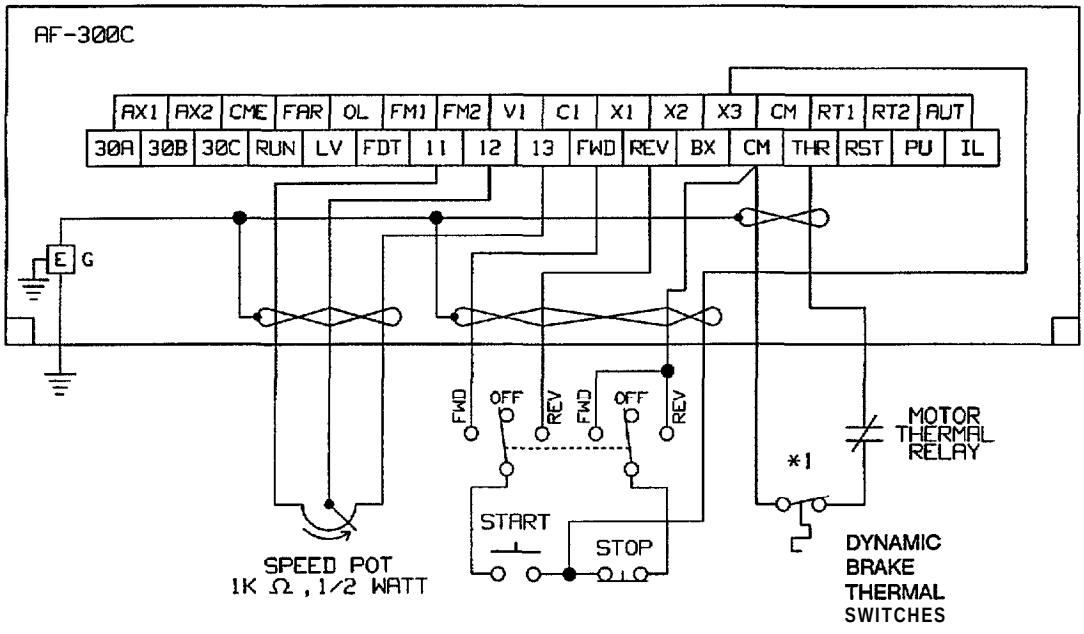


Figure 4-18. REMOTE OPERATION WITH MOMENTARY CONTACTS AND REVERSING CAPABILITY

NOTE: LOCAL/REMOTE switch should be in the "REMOTE' position **(*2).** Set Function Code 41 to "--" (No).

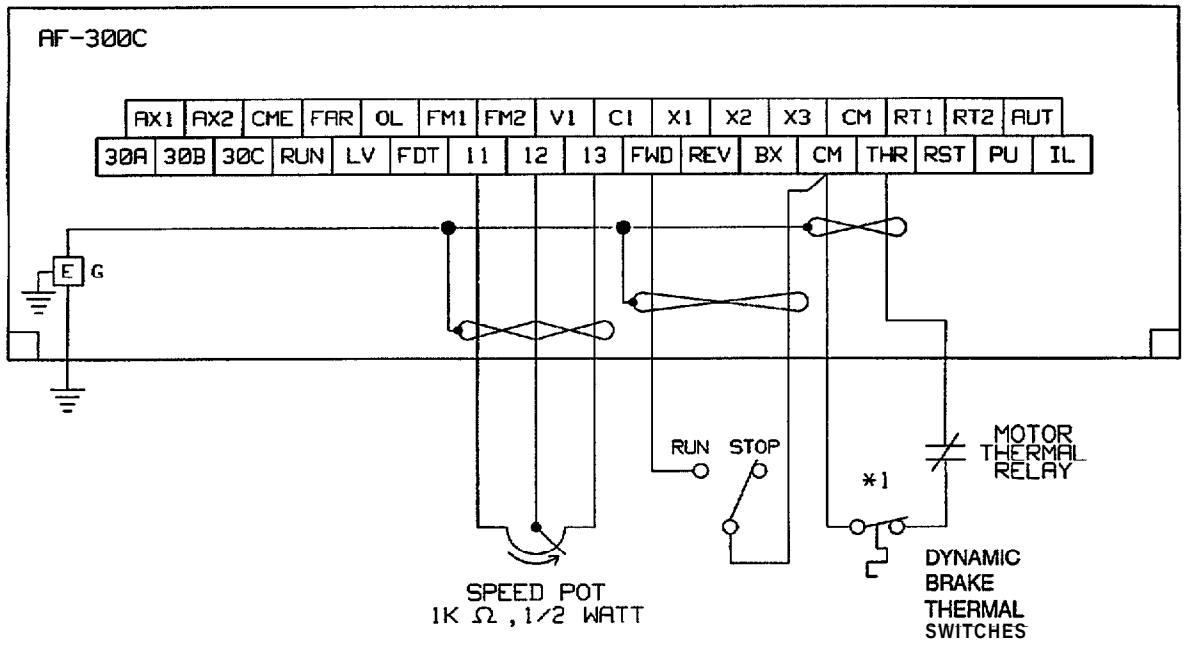


Figure 4-19. REMOTE OPERATION WITH MAINTAINED CONTACTS

NOTE: LOCAL/REMOTE switch should be in the "REMOTE" position **(*2).** Set Function Code 41 to "--" (No).

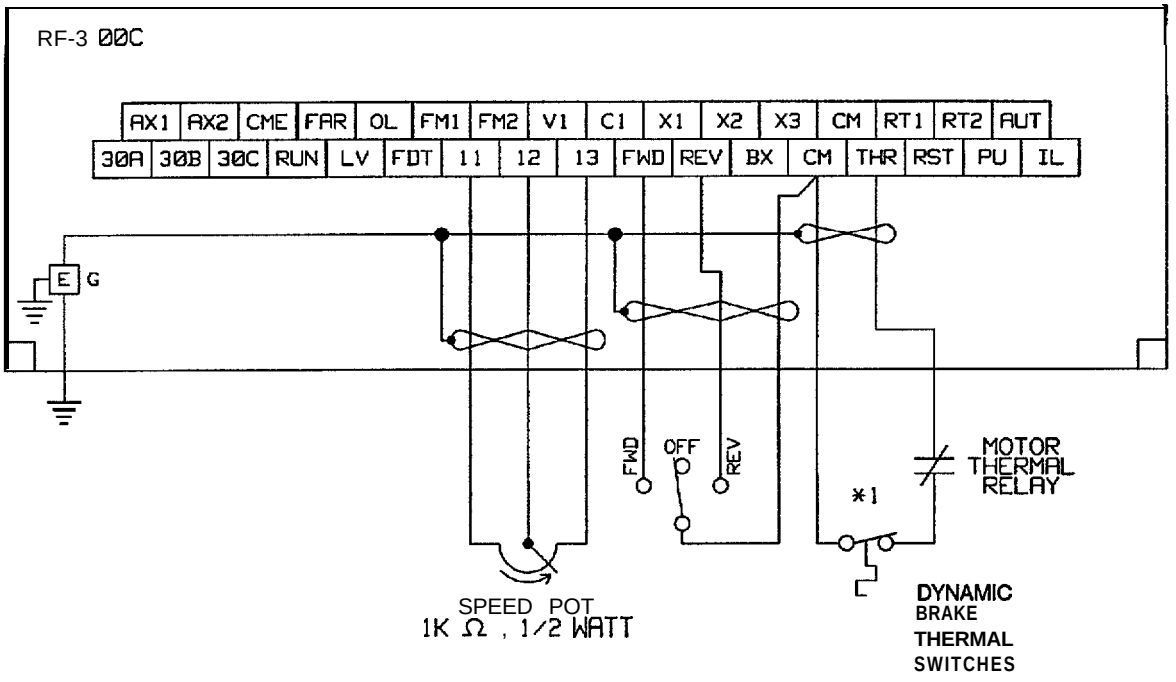
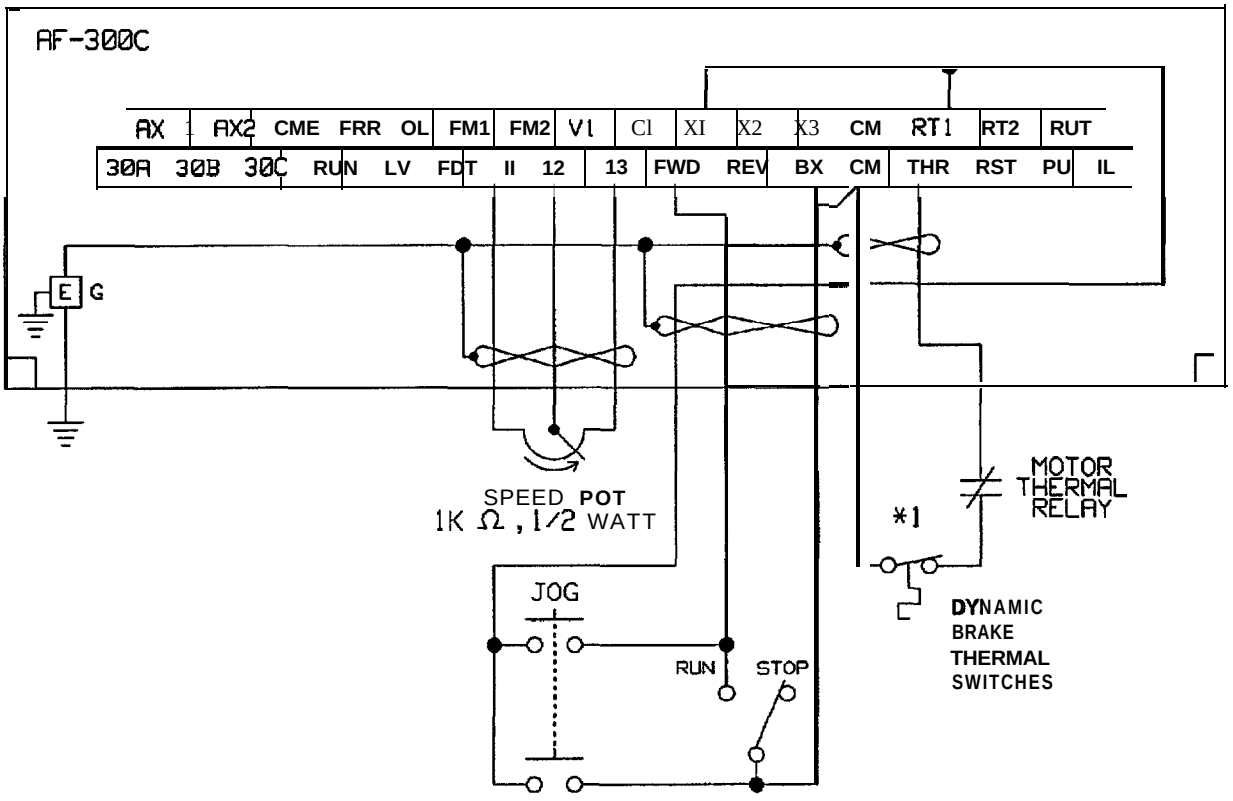


Figure 4-20. REMOTE OPERATION WITH MAINTAINED CONTACTS AND REVERSING CAPABILITY

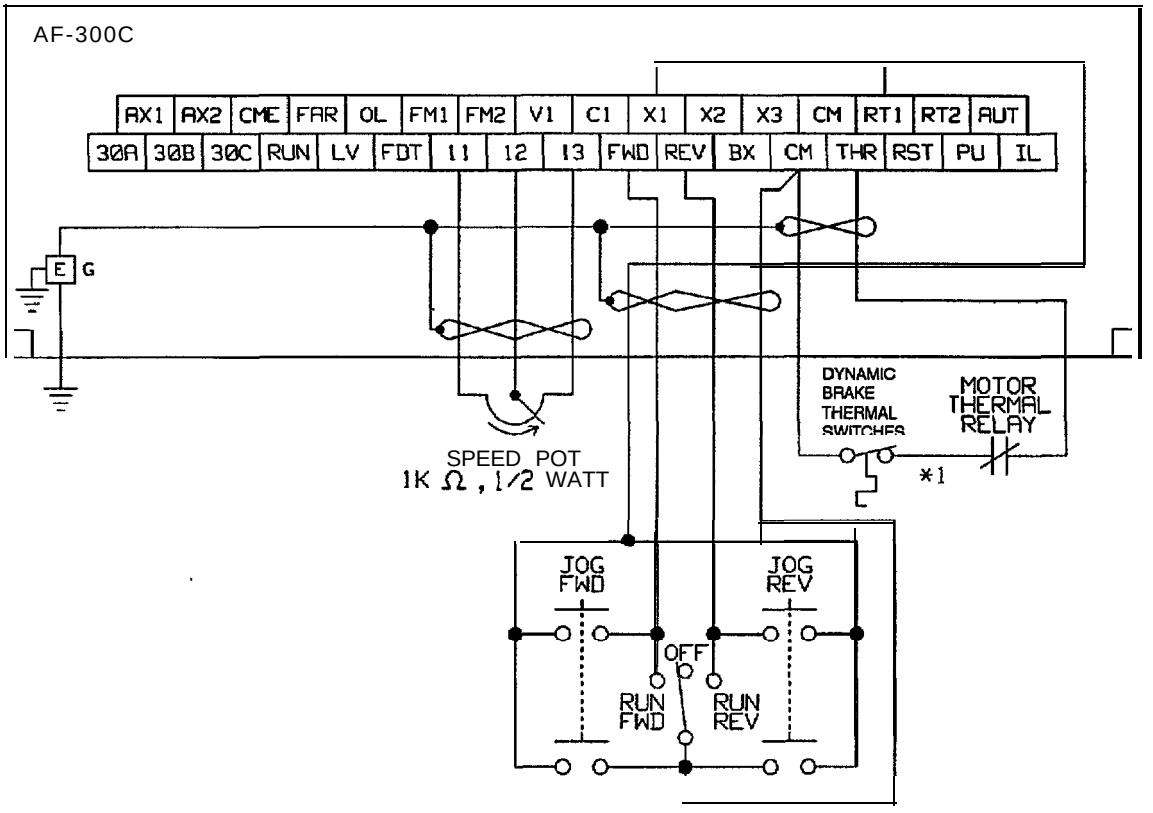


NOTE: **LOCAL/REMOTE** switch should be in the "REMOTE' position **(*2).** Set Function Code 41 to **"--"** (No).

NOTE: JOG Decel follows the same slope as RUN Decel, except when JOG is commanded during operation above the JOG frequency setting. Check the following Function Codes:

Function Code 23 = JOG Speed (Frequency) Setting.
Function Code 2A = JOG Accel Time Setting.
Function Code 26 = JOG Decel Time Setting.

Figure 4-21. REMOTE OPERATION WITH MAINTAINED CONTACTS AND JOG FUNCTION



NOTE: LOCAL/REMOTE switch should be in the "REMOTE' position **(*2).** Set Function Code 41 to "--" (No).

NOTE: JOG Decel follows the same slope as RUN Decel, except when JOG is commanded during operation above the JOG frequency setting. Check the following Function Codes:

Function Code 23 = JOG Speed (Frequency) Setting.
Function Code 2A = JOG Accel Time Setting.
Function Code 26 = JOG Decel Time Setting.

Figure 4-22. REMOTE OPERATION WITH MAINTAINED CONTACTS AND JOG FUNCTION WITH REVERSING CAPABILITY

NOTE: LOCAL/REMOTE switch should be in the 'REMOTE" position **(*2).** Set Function Code 41 to **"0"** (Yes).

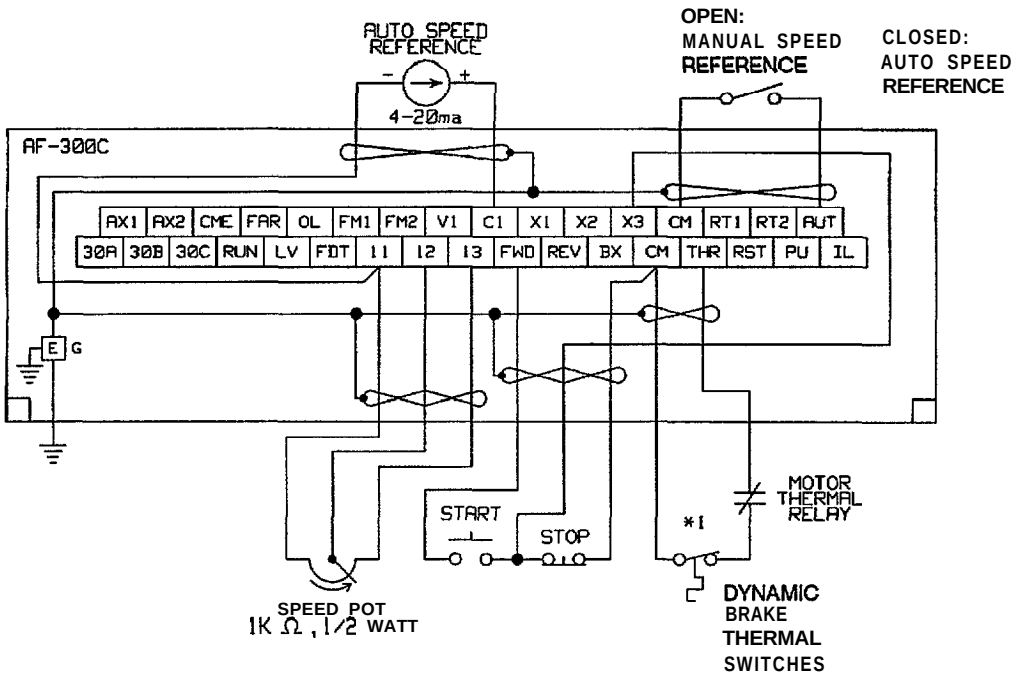


Figure 4-23. REMOTE OPERATION WITH MOMENTARY CONTACTS AND AUTO/MANUAL OPERATION

NOTE: LOCAL/REMOTE switch should be in the "REMOTE" position **(*2).** Set Function Code 41 to **"0"** (Yes).

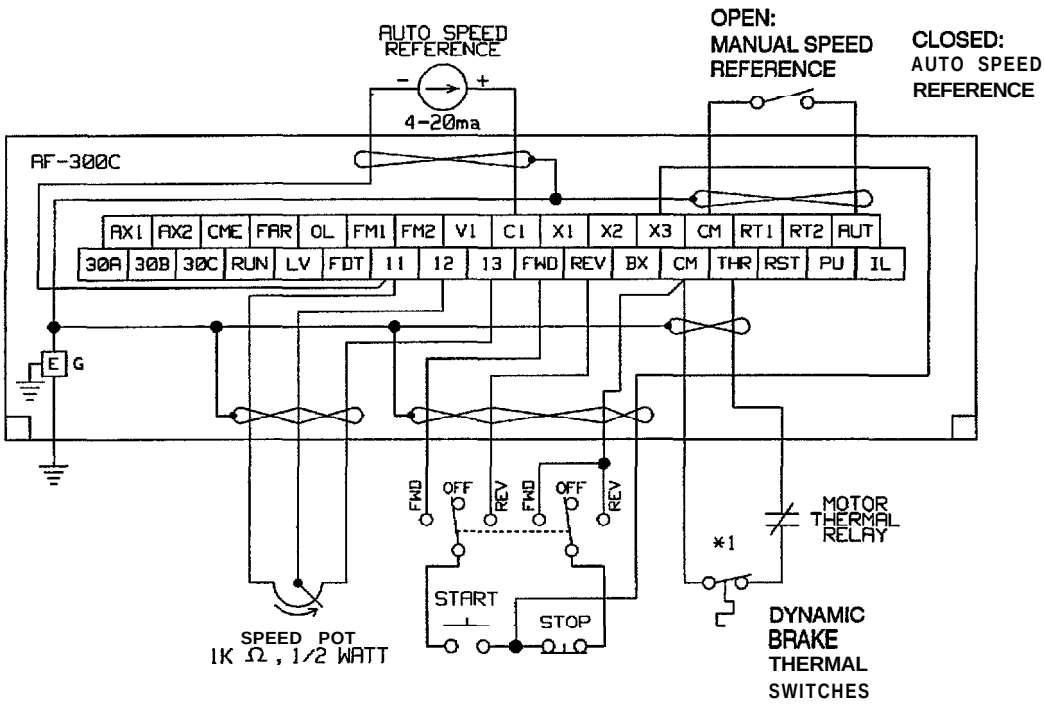


Figure 4-24. REMOTE OPERATION WITH MOMENTARY CONTACTS AND REVERSING CAPABILITY AND AUTO/MANUAL OPERATION

NOTE: LOCAL/REMOTE switch should be in the "REMOTE" position **(*2).** Set Function Code 41 to "--" (No).

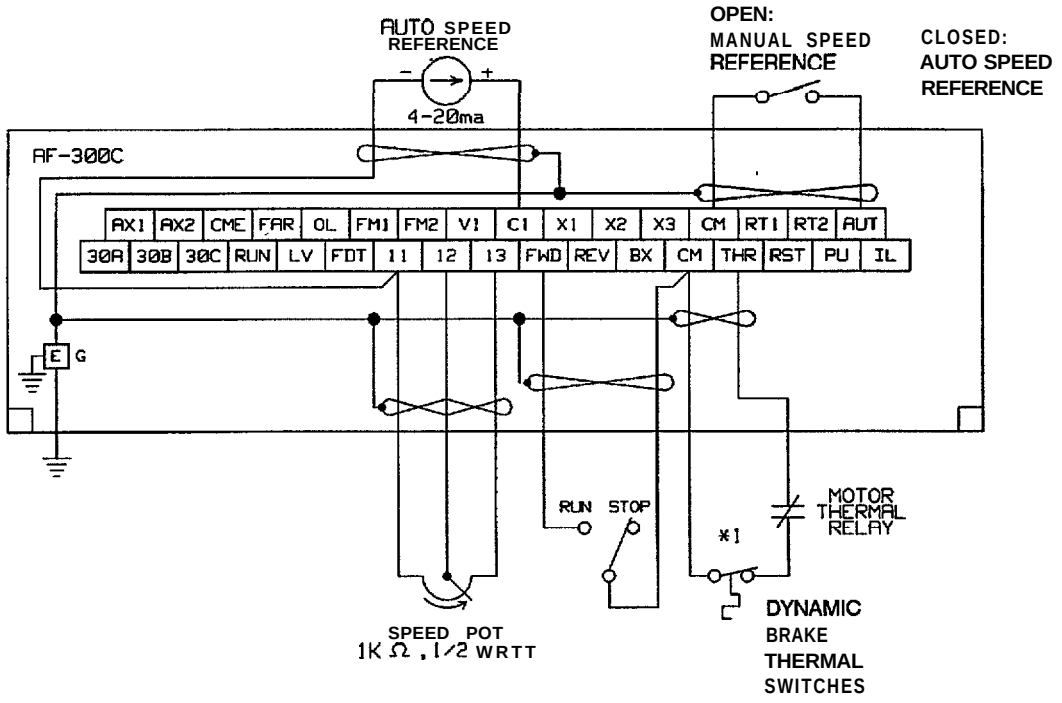


Figure 4-25. REMOTE OPERATION WITH MAINTAINED CONTACTS AND AUTO/MANUAL OPERATION

NOTE: **LOCAL/REMOTE** switch should be in the "REMOTE position **(*2).** Set Function Code 41 to "--" (No).

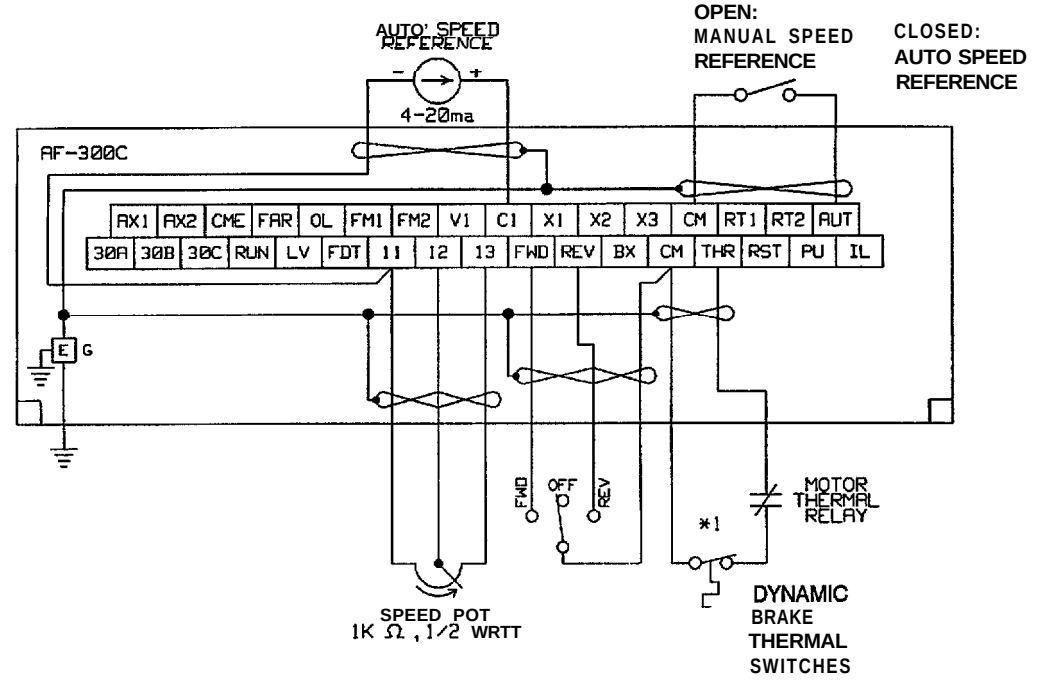


Figure 4-26. REMOTE OPERATION WITH MAINTAINED CONTACTS AND REVERSING CAPABILITY AND AUTO/MANUAL OPERATION

NOTES:

5. INVERTER OPERATION

BEFORE APPLYING AC POWER INSPECTION

After mounting and wiring has been completed, check the inverterfor the following items before applying AC power:

- -- Check for wiring errors (especially main circuit wiring).
- -- Verify that there are no wiring chips, screws, etc. remaining in the inverter.
- -- Check that all screw and terminal connections are tight.
- -- Verify that no exposed wire ends are touching other terminals.

Before a test run is performed, be sure to review and understand the operation of the Keypad Panel as described in the following charts and paragraphs. Refer to Trial Operation " paragraph near the end of this Section for recommended procedures.

CAUTION: Do not conduct megger testsbetween the inverter terminals or control circuit terminals. For megger testina method, refer to Section 7, Maintenance and Inspection.

KEYPAD PANEL IDENTIFICATION/ FUNCTION

The Keypad Panel is installed on the front panel of the inverter and is used for data display and Function Code setting and modifications. The inverter is operated with the Function Codes set by this Keypad Panel and by external operation/control commands.

Also, FORWARD/STOP and REVERSE/STOP can be input and the reference frequency can be set from the Keypad Panel. A Mode Change Switch is also provided to switch operation between LOCAL (Keypad Panel) and REMOTE (control circuit terminals). See the following chart for a description of all Keypad Panel functions and TABLE 4 for Keypad Panel operation procedures. Refer to Figure 5-1 for identification of the components of the Keypad Panel.

KEYPAD PANEL FUNCTIONS CHART

Function		Description
Operation Monitoring		Monitor the operation states of the inverter.
Function Code Setting:	Basic Function Codes	Set data required for operation.
	Auxiliary Function Codes	Set data required for control.
	Calibration Function Codes	Adjust output signals to match with instruments externally installed.
Fault Monitoring		Display and retrieval of a fault and the operating conditions at the time of the fault.
Set Data Protection		Protect set data against accidental modification.
Reset		Data reset at Function Code setting, reset to return to operation monitoring mode after completing the setting, fault reset and error display reset.
Operation Mode Selection		Switch for changing between “Local and ‘Remote’ modes of operation.
Operation Commands		Forward, Reverse, and Stop commands can be entered in the "Local" mode.
Frequency Setting Commands		Increase and decrease reference frequency commands can be entered in the “Local’ mode.

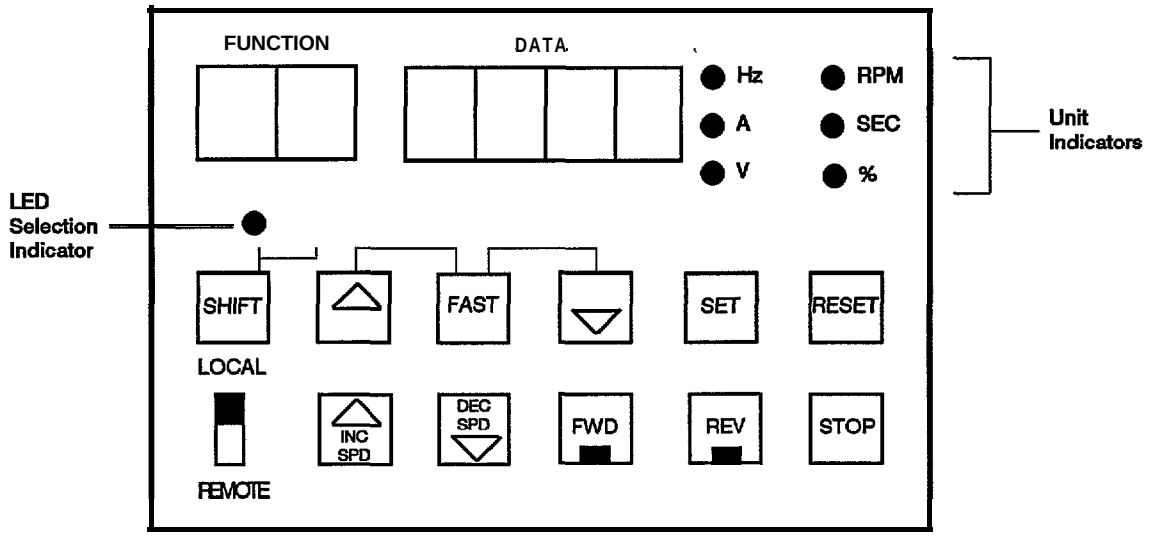


Figure 5-1. KEYPAD PANEL COMPONENT IDENTIFICATION

TABLE 4: KEYPAD AND DISPLAY OPERATION

LED indicator -The LED indicator is made up of three sections as follows:

Function Code Display-Displays in two digits the selected Function Code number. LED Selection Indicator under each digit lights when that digit is selected to be changed.

Data Display - Display in four digits the operation data, Function Code setting data, and the fault codes.

Unit Indicator -The LED on the left of each unit symbol corresponds to the contents of the data display.

NOTE: All letters are displayed *in upper case except* ***"b", "d", and "r".***

Touch Kevs - There are eleven Touch Keys and one switch that perform as follows:

SHIFT **Key** - Selects the Function Selecting Mode and the Parameter Setting Mode.

-- The order of selecting each digit is fixed as 2nd Function digit - 1st Function digit - Data digits in sequence, except the order is 2nd Function Digit - 1st Function digit for Function Codes 00-05, **F1-F3,** and **F5-F7** (see Figure **5-2).**

-- When the data on the Data Display is blinking, you cannot change the selection by pressing the **SHIFT** Key. Press the **RESET** key to recover original data and stop the blinking or press the **SET** key to store the newly changed data and stop the blinking. The **SHIFT** key can then be used to make the selection.

TABLE 4: KEYPAD AND DISPLAY OPERATION (continued)

UP and DOWN ARROW Keys - Used to modify **values** of each mode selected **by SHIFT key.**

- The **UP ARROW** key increases values.
- The **DOWN ARROW** key **decreased values.**
- The increase and decrease of values can be set anywhere between the upper and lower limits (see TABLE 5, Function Code Data).

FAST Key - Used in conjunction with the **UP** and **DOWN ARROW** keys to increase speed of Function Code data change.

- Pressing the **UP** or **DOWN ARROW key** (without pressing FAST key) will obtain low speed.
- While pressing the **UP** or **DOWN** ARROW key, press **FAST** key once to obtain moderate speed and twice to obtain high speed changes. Release the **UP or DOWN ARROW** key to clear the moderate and high speed changes.
- Pressing **SET or RESET** key **while** pressing **FAST** key allows selection/setting of 'Set Data Protection' Function (see "Set Data Protection" paragraph in this Section).

SET Key - Used when setting parameters to enter set data into memory.

- When new data is being set, the Data Display will blink; pressing SET key will enter the new data into the memory and the blinking will stop.
- Data that has been set in the memory is not erased when power supply is turned off.

RESET Key - Used in the Parameter Setting Mode and Fault Monitoring Mode as follows:

Parameter Settlng Mode

- Reset of Set Data - Press **RESET** key prior to **SET** key and the displayed data will be cleared so that you can return to the original data.
- Reset of Set Error Display - Press **RESET key** when 'Err" is displayed due to incorrect setting and the set data will be cleared so that you can return to the previous data setting.
- Return to Operation Monitoring Mode - After setting parameters, press **RESET** key to return to Operation Monitoring Function Code that had been selected and displayed before setting parameters.

Fault Monitoring Mode

- Fault Reset - Press RESET key after solving problem and the fault will be cleared so that you can return to the mode selected before the fault occurred. (Faults **cannot be reset while** a RUN command is present.)

LOCAL/REMOTE Switch - Used for operation mode selection.

- **Do not change while running.** The mode will not change if operated while running; the mode changes only after stopping.
- Local Mode accepts input from the upper and lower rows of keys of the keypad panel.
- Remote Mode accepts input from the FWD, REV, BX, and frequency setting terminals (lower row keys on the keypad are not accepted except STOP key; see "STOP **Key").**

INC SPEED Key - Increases reference frequency when in the "Local mode.

- When power is off or **a fault** occurs, the reference frequency is stored.

DEC SPEED Key - Decreases reference frequency when in the **"Local"** mode.

- When the **INC SPEED** and DEC SPEED **keys are** pressed at the same time, the DEC SPEED key is given priority.
- When power is off or a fault occurs, the reference frequency is stored.

FWD Key - Forward operation command when in the **"Local"** mode.

- FWD Command is not accepted when operating by a **REV** operation command. (Press **STOP** first.}

REV Key - Reverse operation command when in the "Local" mode.

- **REV Command** is not accepted when operating **by** a **FWD** operation command. (Press **STOP** first.)

STOP Key - Stop command when in either **"Local"** or "Remote" mode and "Serial Link" mode.

- When pressed while running in **"Local"** mode, **FWD/REV** LED goes off, inverter output frequency decreases, and motor stops.
- When pressed while running in "Remote' or 'Serial Link" mode, inverter output is turned off and motor coasts to a stop (OH2 Fault Trip).

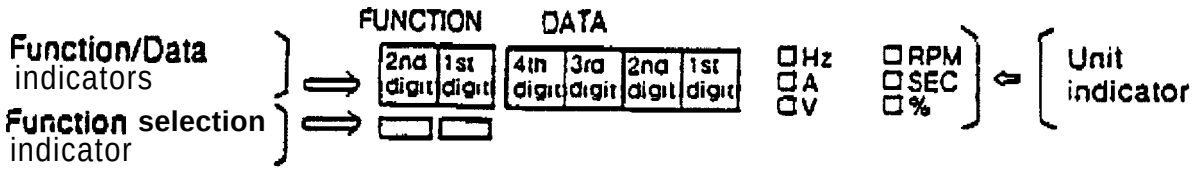


Figure 5-2. DISPLAY LED INDICATOR IDENTIFICATION

DISPLAY EXAMPLES CHART

Display Item	Function Code Digit		Data Digit				Unit Disp. LED Lit
	2nd	1st	4th	3rd	2nd	1st	
Frequency (When output frequency is 60 Hz)	0	0		6	0.	0	Hz
Current (When output current is 100 Amps)	0	3		1	0	0.	A
Voltage (When output voltage is 400 VAC)	0	4		4	0	0.	V
Speed (When machine speed is 150 RPM)	0	5		1	5	0.	RPM
Time (When acceleration time is set at 10 seconds)	1	6		1	0.	0	SEC
Percentage (When torque limit is set at 100%)	3	3		1	0	0.	%
Code (When torque boost is set at "3")	1	9	C	-	-	3	
Selection (When automatic and energy conservation operations are set "Active")	1	A				o	
Coefficient (When pulse freq. monitor coefficient set at 50)	4	0			5	0	
Number of Motor Poles (When set at 4)	3	d				4	
State (When state of input terminal is "Forward Operation")	0	6	A	o	-	-	
Fault (When the 4th digit indicates the first fault in overcurrent during deceleration)	F	0	1.	O	c	2	
Setting Error (When the setting of the lower limit of frequency exceeds that of the upper limit frequency)			E	r	r	1	
Setting Error (When a Function Code that cannot be set during operation has been set)			E	r	r	2	

The **display for Synchronous **Speed** (Function Code 02) requires that the Number of Electric Motor Poles (Function Code **3d**) be set in advance. (Also, for Machine Speed [Function Code **05**], the machine speed conversion factor must be entered in advance.)*

MONITORING CONTROL SIGNALS AND INVERTER OUTPUT

When monitoring functions other than faults, perform the following procedures. (See Figures 5-2 and 5-3.)

OPERATION PROCEDURE

Function Codes 00 through OA provide "Read Only" access to inverter control signals and output information. These can be very useful in troubleshooting problems quickly.

Function Codes 06 and 07 will be covered first because they take an additional step, due to the Function Code extension, to get to the signal status information,

1. Press RESET key and confirm that the Function Selection LED under the second digit of the Function Code Display is lit and that the value of it is "0".
2. Press the SHIFT key once to move the Function Selection LED to the first digit of the Function Code Display.
   -- The Function Selection LED under the first digit will now be lit and the value of this first digit will be changed in the next step.
3. Press the UP ARROW or DOWN ARROW key to set the first digit of the Function Code to "6".
4. Press the SHIFT key once to turn off the Function Selection LED and permit data retrieval.
   -- The 4th digit of the Data Display will indicate the Function Code Extension (A, b, C, d, or E) of Function Code "06".
   -- Function Code data will be displayed in the 3rd, 2nd, and 1 st digits of the Data Display and data units are shown by the Unit Indicator.
   -- Display symbols are "Signal Received = [0], No Signal Received = [-]".
5. Press the UP ARROW key to display (in order) the contents of b, C, d, and E (of Function Code 06).
6. Press the SHIFT key once to light the Function Selection LED under the second digit of the Function Code Display.
7. Press the SHIFT key a second time to move the Function Selection LED to the first digit of the Function Code Display.
   -- The Function Selection LED under the first digit will now be lit and the value of this first digit will be changed in the next step.
6. Press the UP ARROW key once to set the first digit of the Function Code to "7".

Function	Description	Ind. Unit
00	Output Frequency	HZ
01	Reference Frequency	Hz
02	Synchronous Speed	RPM
03	Output Current	A
04	Output Voltage	V
05	Machine Speed	RPM
06	Input Signal Status	
07	Output Signal Status	
08	Motoring Torque Lim.	
09	Braking Torque Limit	
OA	Calculated Torque	

Function	Explanation of function	Indication data			
		4th digit	3rd digit	2nd digit	1st digit
06	Input signal status	A	F WD	REV	BX
		b		THR	RST
		C	X1	X2	X3
		d	RT1	RT2	AUT
		E		IL	PU
07	Output signal status	A	AX	OL	LV
		b	RUN	FAR	FDT

Figure 6-3. FUNCTION CODE 06 AND 07 DATA

9. Press the SHIFT key once to turn off the Function Selection LED and permit data retrieval.
   -- The 4th digit of the Data Display will indicate the Function Code Extension (A or b) of Function Code "07".
   -- Function Code data will be displayed in the 3rd, 2nd, and 1 st digits of the Data Display and data units are shown by the Unit Indicator.
   -- Display symbols are 'Signal Received = [0], No Signal Received = [-]".

All other signals from Function Codes 00 through OA are obtained using the same procedure as Function Codes 06 and 07, except that the Function Code extension does not exist.

NOTE: After *any operations with the Keypad Panel have been completed, it is recommended to return the display to Function Code* **"00"** *(Output Frequency) unless your particular application deems otherwise.*

SETTING FUNCTION CODES

To change Function Codes settings, perform the following procedures. (See Figure **5-2)**

NOTE: *For convenience, always change the second digit of* a *Function Code first since changing the second digit* always *returns the* ***first*** *digit to a value* of *zero.*

NOTE: ***If*** *the inverter is running and a Function Code is changed that is not permitted to be changed while running, an* **"Err2"** *fault* ***will*** *be displayed. Press* **RE-**SET *key to clear this fault and continue.*

NOTE: ***If*** *Function Code data is "protected' (Function Code* ***99),*** *it cannot be changed until Function Code 99 is changed to* ***"unprotected".*** *See 'Data Protection' paragraph in this section.*

OPERATION PROCEDURE

1. Press RESET key and confirm that the Function Selection LED under the second digit of the Function Display is lit and that the value of it is **"0".**
2. Press the **UP** ARROW or DOWN ARROW key to set the second digit of the Function Display to the second digit of the Function Code to be changed.
3. Press the SHIFT key once to move the Function Selection LED to the first digit of the Function Display.
   -- The Function Selection LED under the first digit will now be lit and the value of this first digit will be changed in the next step.
4. Press the **UP** ARROW or DOWN ARROW key to set the first digit of the Function Display to the first digit of the Function Code to be changed.
5. Press the **SHIFT** key once to turn off the Function Selection LED and permit Function Code data changes.
6. Press the UP ARROW or DOWN ARROW key to change data of the Function Code selected to the desired new value.
   -- To make large changes, use FAST key together with **UP** or DOWN ARROW keys. Press UP or DOWN ARROW key first (data will change in tenths of units), then press FAST key once to change data in single units, or press FAST key twice to change data in tens of units.
   -- Data Display data will blink durning change procedure and when desired value is reached.
7. Press the SET key to set the new Function Code value into memory.
   -- Data Display will cease to blink after the data has been set into memory.
8. If additional Function Codes changes are desired, press SHIFT key and repeat steps 2 thru 7 of this procedure.

NOTE: *After any operations with the Keypad Panel have been completed, it is recommended to return the display to Function Code* **"00"** *(Output Frequency) unless your particular application deems otherwise.*

FUNCTION OF RESET KEY AFTER CHANGING FUNCTION CODE DATA

Pressing RESET after finishing Function Code changes will always return the display to one of the Operation Monitoring Function Codes (Function Codes that begin with zero). The particular one that it is returned to can be changed by selecting the desired Function Code and leaving it there for approximately two or three minutes without pressing any of the keys on the keypad.

NOTE: *The "Power-Up* ***Default"*** *will be the same Function Code as the 'Return Via* ***Reset"*** *Function Code.*

FAULT INDICATION AND RETRIEVAL

When a fault occurs, the mode is automatically switched to the Fault Monitoring mode. The fault Function Code will be displayed in the Function Display, the number (in order) and type of fault will be shown in the Data Display (blinking), and the Function Selection LED for the first digit of the Function Code will be lit.

The fault indication function is composed of the following three parts:

1. Indication of present fault.
2. Indication of operating conditions at the time that the fault took place.
3. Indication of previous faults.

If the fault is a multiple fault, indication can be retrieved in the order of **occurence** (up to seven faults) by Function Code **FO, 1-7.**

In addition, the operating conditions at the time of the first fault can be indicated by Function Codes **F1** through **F4** (only for present fault; after RESET, this information is lost).

Also, the first fault for the past three fault trips can be indicated by F5 through **F7.**

NOTE: *First fault means the first detected fault when mu/tip/e faults occur at the same time.*

NOTE: ***If*** **control power** *is* ***turned*** *off during fault display, fault output signal* **will** *not be he/d.*

Perform either of the following procedures to obtain fault information. (See Figures 5-2 and 5-4.)

WHEN FAULT OCCURS AND IS STILL PRESENT

1. Press **SHIFT** key once to turn off the Function Selection LED under the first digit of the Function display and advance control to the Fault Retrieval mode.
   -- The fault number **(1)** will be shown in the fourth digit of the Data Display and the fault type will be shown in the third, second, and first digits of the Data Display.
2. Press the **UP ARROW** key once to display the second fault in the Data Display.
   -- The fault number (2) will be shown in the fourth digit of the Data Display and the fault type will be shown in the third, second, and first digits of **the** Data Display.
3. Press the **UP ARROW** key again to display the third fault in the Data Display,
   -- The fault number (3) will be shown in the fourth digit of the Data Display and the fault type will be shown in the third, second, and first digits of the Data Display.
4. Repeat step 3 until no fault is shown in the Data Display (Example: **"3. - - -"** indicates that no third fault exists.)

NOTE: ***If*** *you plan on investigating the fault, history and operation data is required. Also record the date, time, and any information about the machine operation prior to the* ***fault.***

Function Code	Indication Data (by digit) 4th	3rd	2nd	1st	Name of Fault
	1 - 7 **in** order	0	c	**1**	Overcurrent At Acceleration
	of **occurence.**	0	**c**	**2**	Overcurrent At Deceleration
		0	c	3	Overcurrent At Fixed Speed
FO			0	U	Overvoltage
			L	u	Undervoltage
		0	H	1	Inverter Oveheat
		0	H	2	External Alarm
		0	L	1	Inverter Overload
		0	L	2	Motor Overload
	E	r	r	0	CPU Error
----	E	r	r	d	Memorv Error
F1		6	0.	0	**Output** Frequency
F2		6	0.	0	Setting Frequency
F3	**1**	0	0.	0	Output Current
F4	1 - 4 in order		F	0	Normal Operation
	of indication		**r**	E	Reverse Operation
			C	L	Current Restriction
			U	L	Voltage Restriction
			u	u	Undervoltage Restriction
F5					1st Prev. Fault (Pres. Fault - 1)
F8					2nd Prev. Fault (Pres. Fault - 2)
F7					3rd Prev. Fault (Pres. Fault - 3)

Figure 54. FAULT INDICATIONS

5. History and Operation Data Not Rewired:

The confirmation of the fault has been completed at this point. If the operation data and fault history are not required, press RESET key after fault cause has been corrected and do not perform the remaining steps of this procedure.

-- This will clear the fault and the control will automatically return to the mode it was in prior to the fault and be ready for operation. Please note that thesecond, third, etc. faults have **not** been stored in the memory. It is recommended that they be recorded for possible future reference.

HIstory and Operation Data Rewired:

Press **SHIFT** key until the Function Selection LED for the first digit of the Function Code Display is lit to retrieve operation data at the time of the fault.

6. Press the **UP ARROW** key to display the following information in the Data Display: (See Figure 5-4.)

a. Press once to select **F1** and the Output Frequency will be displayed.

b. Press a second time to select F2 and the Sat Frequency will be displayed.

c. Press a third time to select F3 and the Output Current will be displayed.

d. Press a fourth time to select F4 and the Operation State will be displayed.

e. Press **SHIFT** key once to turn off Function Selection LED under the firstdigit of the Function Display and advance control the the Operation State Retrieval Mode. (The Operation State Number (1) will be shown in the fourth digit of the Data Display and the Operation State will be shown in the third, second, and first digits of the Data Display.)

f. Press the **UP ARROW** key once to display the Operation State Number (2) in the fourth digit of the Data Display and the Operation State will be shown in the third, second, and first digits of the Data Display.

g. Press the **UP ARROW** key again to display the Operation State Number (3) and the Operation State will be displayed as in (f) above.

h. Press the UP ARROW key again to display the Operaton State Number (4) and the Operation State will be displayed as in (f) above.

NOTE: *If you plan on investigating the fault, history and operation data is required. Also record the date, time, and any information about the machine operation prior to the fault.*

7. **Fault History Data Not Required:**

If the fault history is not required, press **RESET** key after fault cause has been corrected and do not perform the remaining steps of this procedure.

-- This will clear the fault and the control will automatically return to the mode it was in prior to the fault and be ready for operation.

Fault History Data Required:

Press **SHIFT** key until the Function Selection LED for the first digit of the Function Code Dispfay is lit to retrieve fault history data.

8. Press the UP **ARROW** key to display the following information in the Data Display:

a. Press once to select F5 and the last fault (first of a multiple fault occurence) to occur will be displayed.

b. Press a second time to select F6 and the fault one time previous to the last fault will be displayed.

c. Press a third time to select F7 and the fault two times previous to the last fault will be displayed.

NOTE: *When* **RESET** *key is pressed, a 'moving up" of the fault history will occur. The inverter can not be reset when a* **RUN** *command is present.*

9. Press **RESET** key after fault cause has been corrected. Data retrieval for fault monitoring is complete at this point.

-- This will clear the fault and the control will automatically return to the mode it was in prior to the fault and be ready for operation.

OBTAINING FAULT HISTORY WHEN FAULT IS NOT PRESENT

1. Press **SHIFT** key until the Function Selection LED under the second digit of the Function Code display lights.

-- This selects the second digit.

2. Press the UP ARROW key or DOWN ARROW key to set **"F".**

3. Press the **SHIFT** key to select the first digit of the Function Code Display.

-- The Function Selection LED under the first digit will light.

NOTE: *There are no fault inputs for Function Codes F0 to F4 and the Data Display will show "- - -" when the inverter is not presently faulted.*

4. Press the UP or DOWN ARROW keys to set the desired number (F5, F6, F7). (See Figure 5-4.)
   -- Each fault history will be shown on the Data Display when selected.
5. Press RESET key to return to Operation Monitoring mode (no 'move up' of fault history will occur).

DATA PROTECTION

Function Code data can be protected against accidental changes by the setting of Function Code 99. When data is 'Protected', it cannot be changed until Function Code 99 is changed to "Unprotected". There are two different methods of setting the data protection. Perform either of the following procedures that best suits your particular application.

OPERATION PROCEDURE #1

To Set Protection:

Press FAST key first and then SET key (while FAST key is still depressed) to set Function Code 99 to 'Protected'.

-- Function Code Display will display "99" with neither of the Function Selection LEDs lit.
-- Data Display will show (0) in the first digit position to indicate 'Protected*. (This will be displayed for approximately 5 seconds after keys are released.)
-- Data Display will then return to its previous mode.
-- All Function Code data currently set will be protected.

To Release Protection:

Press FAST key first and then RESET key (while FAST key is still depressed) to set Function Code 99 to "Unprotected'.

-- Function Code Display will display '99" with neither of the Function Selection LEDs lit.
-- Data Display will show (-) in the first digit position to indicate 'Unprotected". (This will be displayed for approximately 5 seconds after keys are released.)
-- Data Display will then return to its previous mode.
-- All Function Code data currently set will no longer be protected.

OPERATION PROCEDURE #2

1. Press SHIFT key until Function Selection Led under the second digit of Function Code Display is lit.
   -- This selects the second digit.
2. Press UP or DOWN ARROW keys until "9" is displayed in the second digit position.
   -- First digit of Function Code Display will automatically be set to "9" also.
   -- Current status of "Protected' (0) or 'Unprotected" (-) will be be displayed in the first digit position of the Data Display.
3. Press SHIFT key to advance to Data Setting Mode.
   -- Both Function Selection LEDs will go out.
4. Press UP ARROW key to set to "Protected" status.
   -- (0) will be displayed in the first digit position of Data Display.

   or

   Press DOWN ARROW key to set to "Unprotected status.
   -- (-) will be displayed in the first digit position of Data Display.
5. Press SET key to store selection into memory.
6. Press RESET key to return to Operation Monitoring Mode.

TRIAL KEYPAD OPERATION (Optional)

Before trial operaton of the inverter, ensure that all inspections outlined in the "Before Applying AC Power Inspections' paragraph have been performed.

WARNING - MECHANICAL MOTION HAZARD: When running from the Keypad Panel wlth the "Local/Remote" switch Inthe "Local" position, pressing the FWD or REV key will cause the motor to rotate. Any RUN permissives that may be wired to the FWD or REV terminals are Ignored.

If a test run Is desired, run the inverter at a low frequency of around 5 Hertz. Conduct the trial operation per the following procedure:

1. Disconnect all couplings and belts that connect the electric motors and machinery so as to allow independent operation of the electric motors.
   -- If the electric motors are connected directly to the machinery, verify that any mechanical motion will not pose a safety hazard.

2. Set all operating switches to OFF and set "Local/ Remote' switch to "Local" position.
3. Apply AC power.
   - Control circuits and sequence circuits will be activated and charge LED will light.
   - Data Display portion of Touchpad Panel will blink "LoAd" while the CPU is checking the internal data.
   - "LoAd" ceases to blink while CPU is checking that Function Code set data matches the specification.
   - Touchpad Panel will display last operation mode of invetter prior to removal of AC power.
4. Use "Increase Speed" or 'Decrease Speed' keys to set desired reequency. (A low frequency of about 5 Hz is recommended for the first run.)
5. Give a Forward or Reverse command and check for the following: (See Figure 5-5.)
   - Correct direction of rotation.
   - Smooth rotation of motor.
   - Any abnormal vibration or noise in the motor.
   - Smooth speed increase and speed reduction.
6. Check for any abnormal heating, fumes, etc. while motor is rotating.
7. Perform the following using the Keypad Panel:
   - Raise the reference frequency gradually and verify that it reaches top frequency for the particular application (typically 60 Hz, default setting from factory).
   - Check AC motor current (should be approximately 3040% of rated current with no load).
   - Lower the reference frequency and stop the motor.
8. Reconnect any equipment disconnected in step 1 of this procedure.
9. Perform steps 5 thru 7 with the load connected.

NOTE: *If any Function Code settings need to be changed, refer to 'Setting Function Codes' paragraph in this Section.*

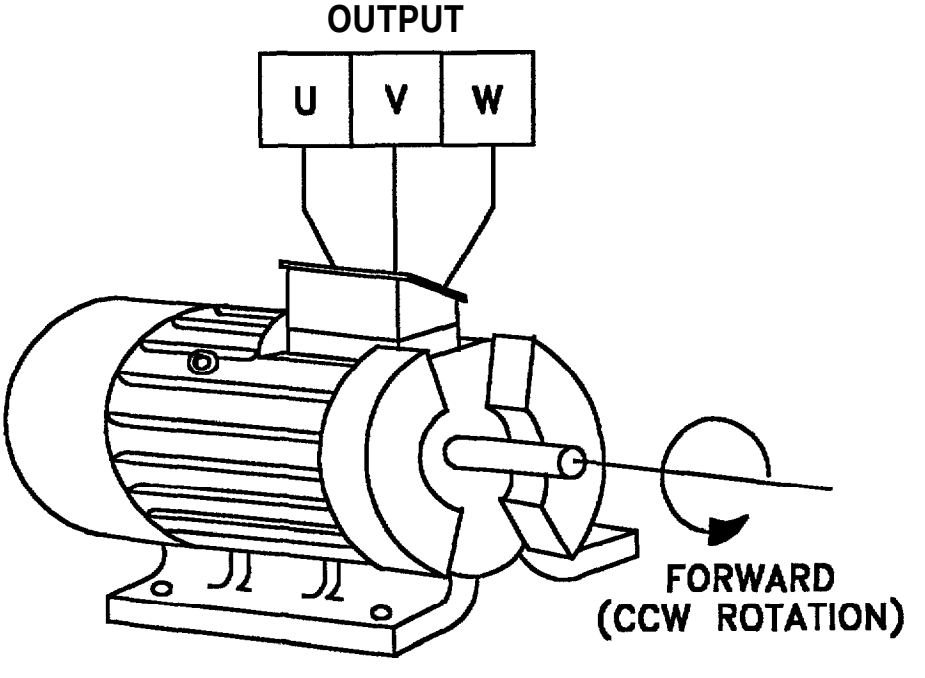


Figure 5-5. ELECTRIC MOTOR ROTATION

CONTROL CIRCUIT CONNECTION AND OPERATION

Basic connectlons that are required for operation from the Keypad Panel are shown in Figure 5-6 (this is a factory installed jumper).

A wiring example of commonly used signals that are optional for either Keypad Panel Operation or External Operation (through control circuit terminals) is shown in Figure 5-7.

Connection description is as follows:

-- **30A, 308, 30C** - Fault Relay Output used for alarm circuits, etc.

-- **FM1, FM2** - Used for external analog meter when frequency indication is needed.

-- **THR** - Connected to **CM** terminal at factory. Connects to a normally closed contact. When inverter STOP is desired through external alarm, open this contact. The inverter will trip on **"OH2"** and the motor will coast stop.

-- **RST** -Connects to a normally open momentary contact for resetting a fault externally.

-- **BX** - Connects to a normally open contact. When motor coast stop is desired during normal operation, close this contact **and remove** the RUN command (FWD or REV).

Termlnal Connectlons required for operatlon from Keypad Panel are shown in Figure 5-8.

Operation is as follows:

-- Rotation Direction - Press **FWD** on the Keypad for forward rotation or **REV** on the Keypad for reverse rotation.

-- Frequency Setting - The frequency reference can come into the inverter in one of the following ways when the **LOCAL/REMOTE** switch is in the **'LOCAL'** position.:

1. Set the motor speed by pressing the INCREASE SPEED or DECREASE SPEED kevs on the Keypad Panel to increase or decrease the inverter output frequency.
2. Multi-Step Speed Settings can be selected by terminals **X1,** X2, and X3. (Programmable preset frequencies in the Function Codes can be selected.)

-- See Figure 5-9 for RUN Operation example.

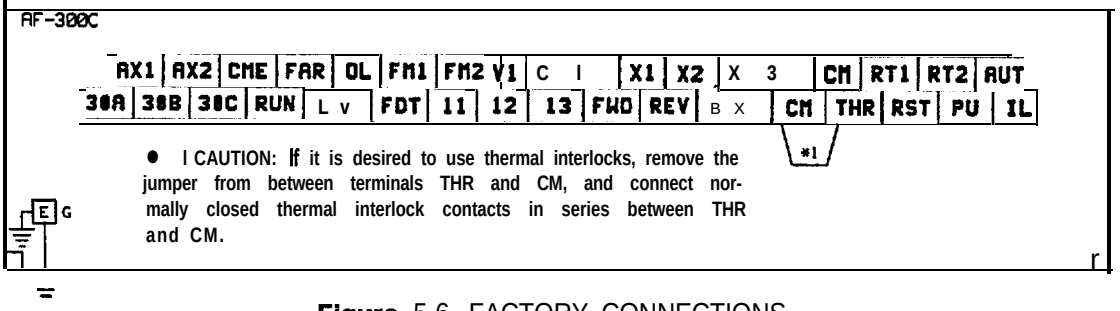


Figure 5-6. FACTORY CONNECTIONS

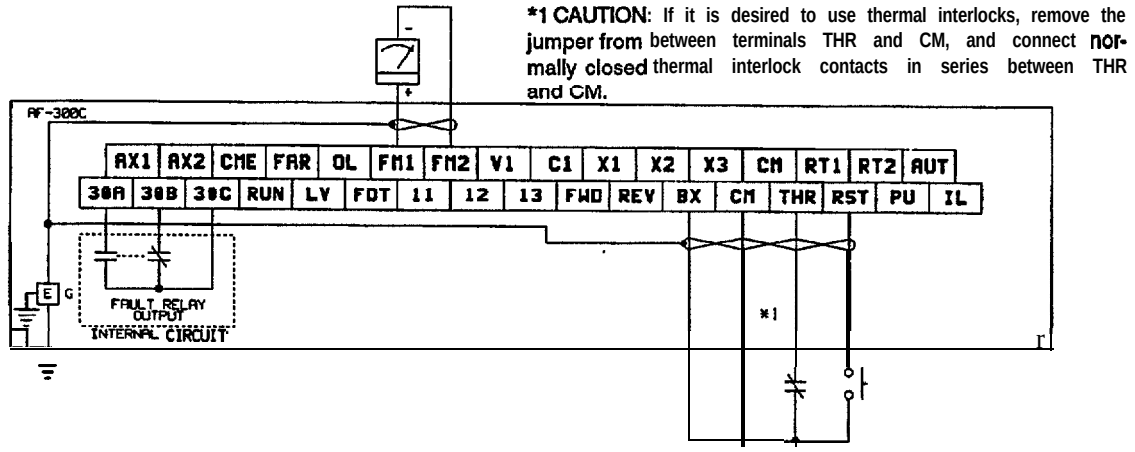
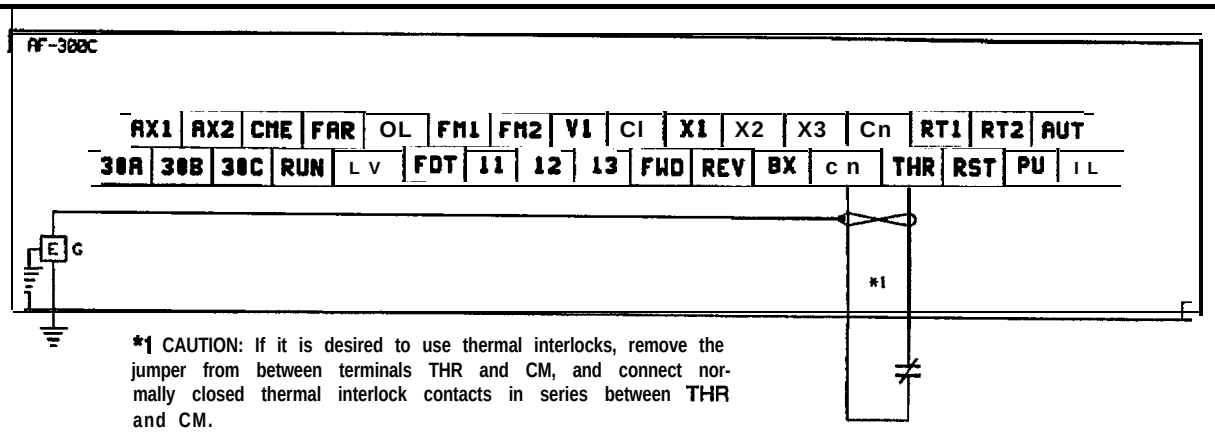
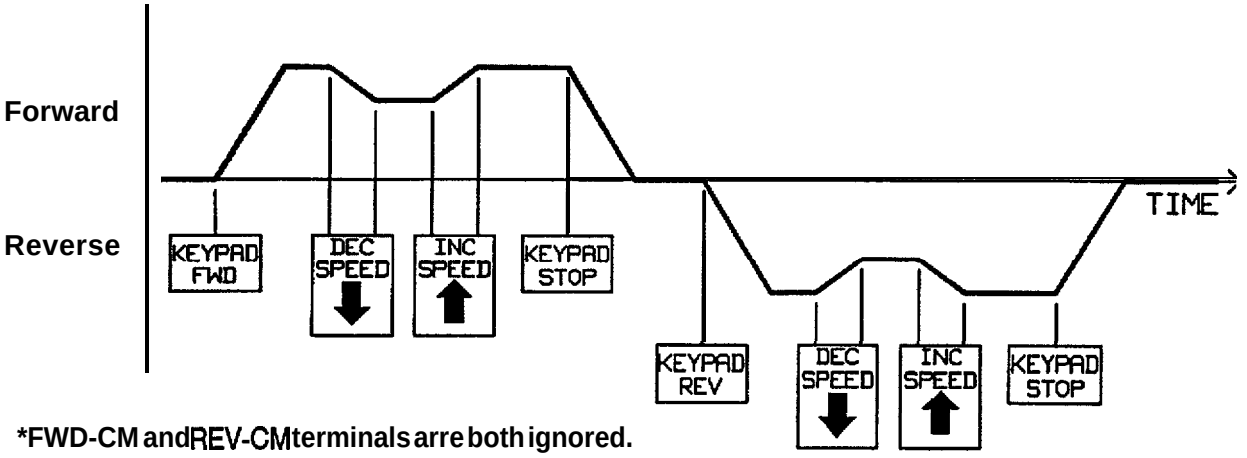


Figure 5-7. EXAMPLE OF COMMONLY USED SIGNALS



*1 CAUTION: If it is desired to use thermal interlocks, remove the jumper from between terminals THR and CM, and connect normally closed thermal interlock contacts in series between THR and CM.

Figure 5-8. KEYPAD PANEL OPERATION CONNECTIONS



*FWD-CM and REV-CM terminals are both ignored.

WARNING - MECHANICAL MOTION HAZARD:
When running from the Keypad Panel with the "Local/Remote' switch in the 'Local" position, pressing the FWD or REV key will cause the motor to rotate. Any RUN permissives that may be wired to the FWD or REV terminals are ignored.

Figure 5-9. KEYPAD PANEL 'RUN' OPERATION EXAMPLE

Terminal Connections required for Remote Operation are shown in Figure 5-l 0.

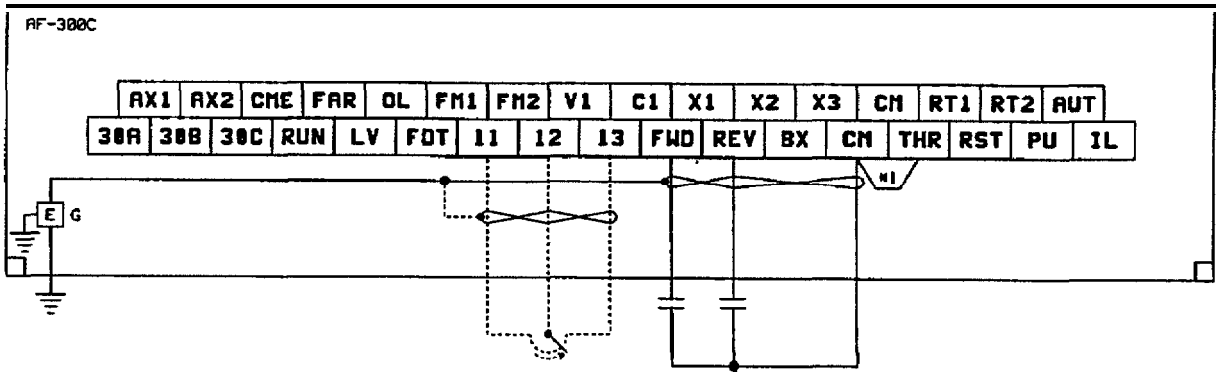
Connection description/operation is as follows:

-- **Rotation Direction** - Make contact between FWD-CM for forward rotation or between REV-CM for reverse rotation.

-- Frequency Setting - The frequency reference can come into the inverter in one of the following ways when the LOCAL/REMOTE switch is in the 'REMOTE' position:

1. Speed Pot - 1 K ohm, 1/2 watt potentiometer connected to terminals 11, 12, and 13.
2. Voltage - 0 to +10 VDC signal to terminals V1 and 11 (11 is common).
3. Current - 4 to 20mA DC current signal to terminals C1 and 11 (11 is common). (Terminal AUT must be connected to CM.)
4. Speed Pot Plus Voltage - 1 K ohm, 1/2 watt potentiometer connected to terminals 11, 12, and 13 is added to 0 to +10 VDC signal connected to terminals V1 and 11.
5. Voltage Plus Voltage - 0 to +10 VDC signal connected to terminals V1 and 11 is added to 0 to + 10 VDC signal connected to terminals 12 and 11 (11 is common).
6. Multi-Step Frequency Settings - Selected by terminals Xi, X2, and X3. (Programmable preset frequencies in the Function Codes can be selected.)
7. UP/DOWN Control (Simulated Motorized Potentiometer) - Selected by terminals Xl and X2.

-- See Figure 5-l 1 for Remote RUN Operation example.



*1 CAUTION: If it is desired to use thermal interlocks, removethe jumper from between terminals THR and CM, and connect normally closed thermal interlock contacts in series between THR and CM.

Figure 5-10. REMOTE OPERATION CONNECTIONS

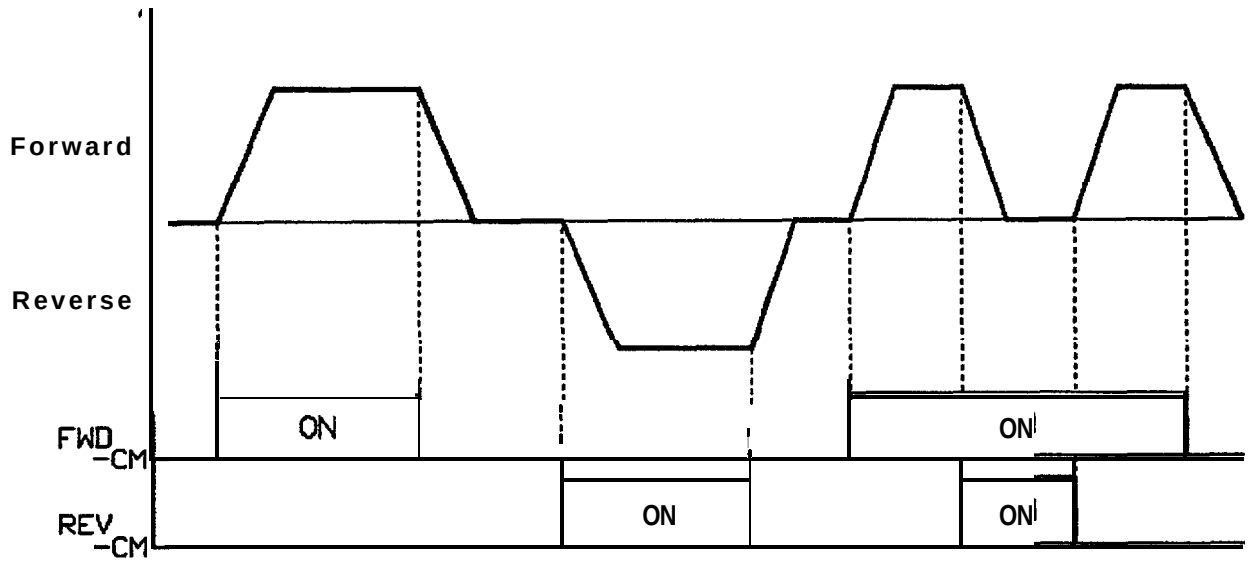
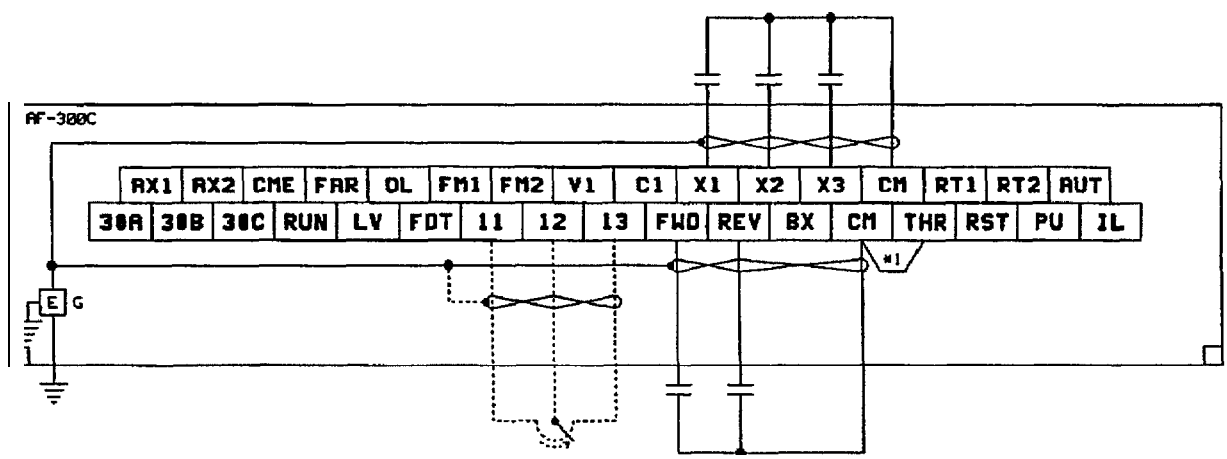


Figure 5-I 1. REMOTE OPERATION EXAMPLE



***1** CAUTION: If it is desired to use thermal interlocks, remove the jumper from between terminals THA and CM, and connect normally closed thermal interlock contacts in series between THR and CM.

Figure 5-12. MULTI-STEP FREQUENCY OPERATION CONNECTIONS

ACTUAL OPERATION

Terminal connections and Function Code selections for Multi-Step Frequency Operation and Multi-Step Acceleration and Deceleration Operation are as described in the following subparagraphs.

Multi-Step Frequency Operation terminal connections (XI, X2, X3 terminals) that are required are shown in Figure 5-12. Multi-step Frequency Operation can be used to select one of eight different speed settings (one manual setting and seven preset settings).

1. The multi-step frequency Function Code settings are:
   - [23] for multi-step frequency 1.
   - [24] for multi-step frequency 2.
   - [25] for multi-step frequency 3.
   - [26] for multi-step frequency 4.
   - [27] for multi-step frequency 5.
   - [28] for multi-step frequency 6.
   - [29] for multi-step frequency 7.
2. See Figure 5-13 for RUN Operation example.

Multi-Step Acceleration and Deceleration Operation terminal connections (RT1, RT2, terminals) are shown in Figure 5-14. Four different acceleration and deceleration times can be externally selected.

1. The multi-step acceleration and deceleration Function Code settings are:
   - [16] for acceleration time 1.
   - [2A] for acceleration time 2.
   - [2C] for acceleration time 3.
   - [2E] for acceleration time 4.
   - [17] for deceleration time 1.
   - [2b] for deceleration time 2.
   - [2d] for deceleration time 3.
   - [2F] for deceleration time 4.
3. See Figure 5-15 for RUN Operation example.

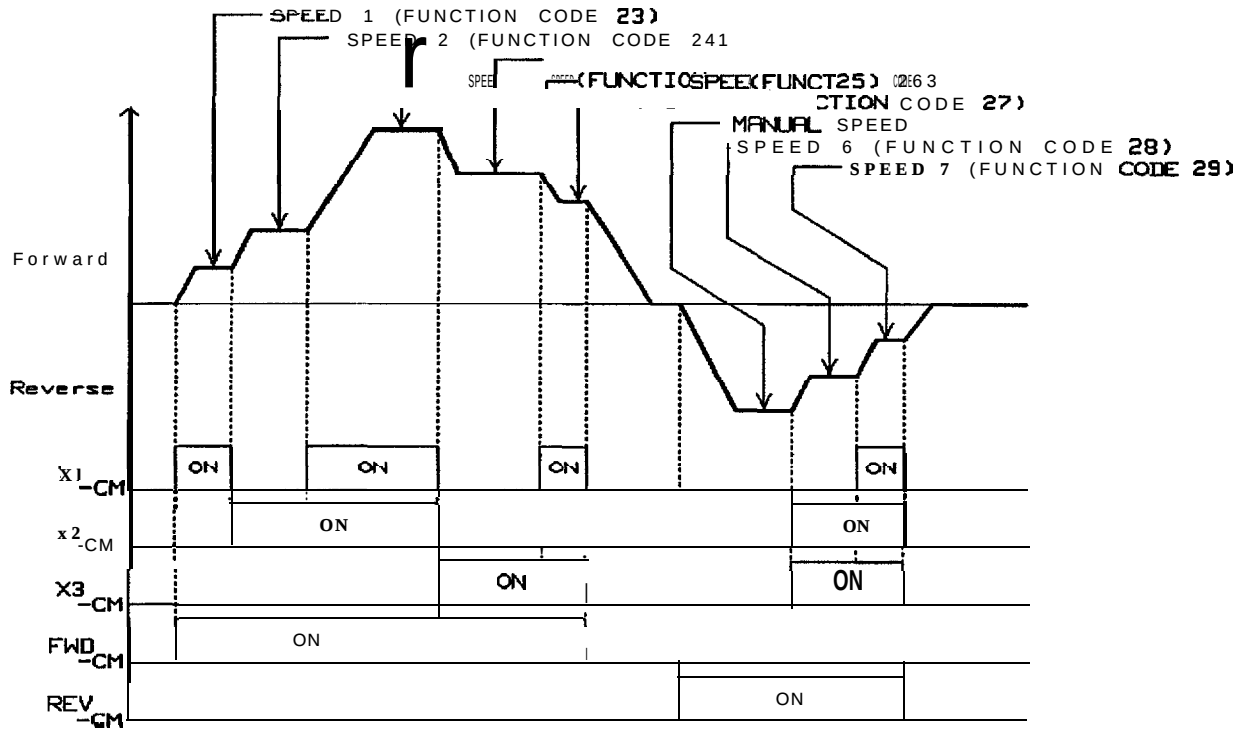
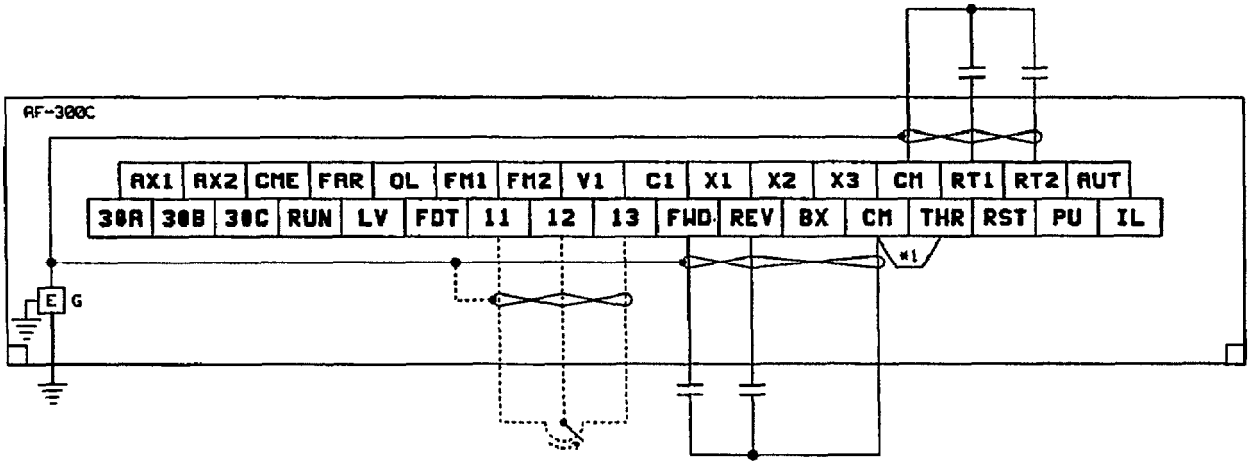


Figure 5-13. MULTI-STEP FREQUENCY 'RUN' OPERATION EXAMPLE



***1 CAUTION:** If it is desired to use thermal interlocks, remove the jumper from between terminals THR and CM, and connect normally closed thermal interlock contacts in series between THR and CM.

Figure 5-14. MULTI-STEP ACCELERATION AND DECELERATION OPERATION CONNECTIONS

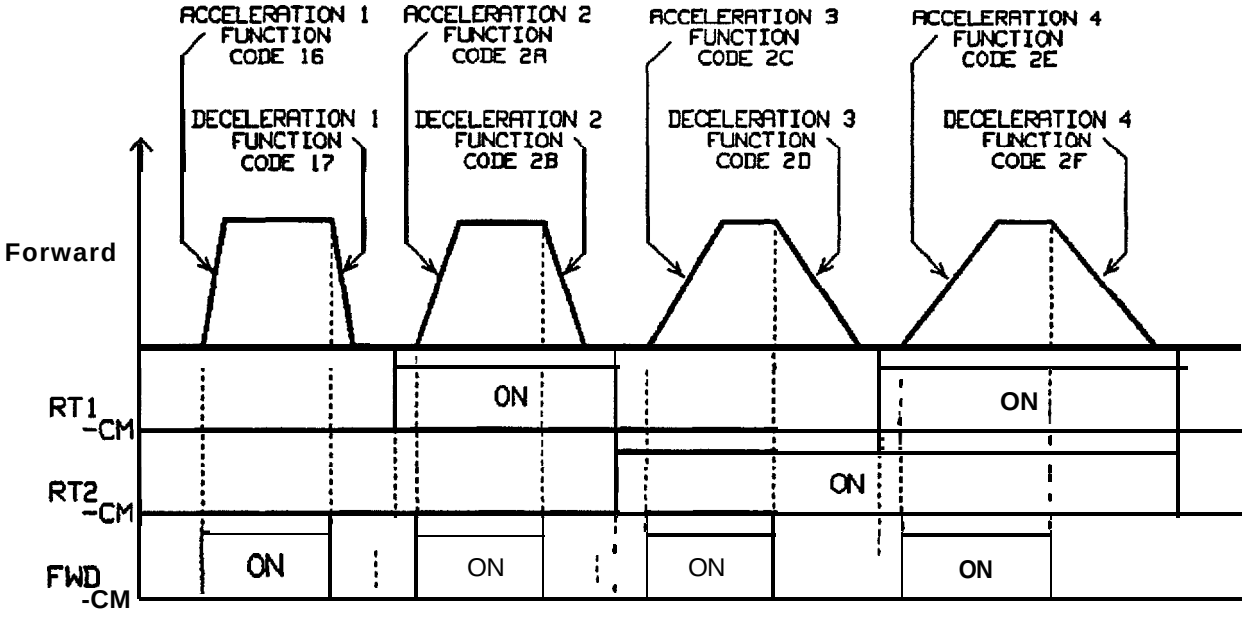


Figure 5-15. MULTI-STEP ACCELERATION AND DECELERATION "RUN' OPERATION EXAMPLE

TABLE 5: FUNCTION CODE DATA

Mode	Funct. Code	Name	Setting and Indication Data: Range	Unit	Increment	Factory Setting
OPERATION MONITOR	00	Output Frequency	Indicates During Operation	Hz	0.1 Hz	----
	01	Reference Frequency	" " "	HZ	0.1 Hz	----
	02(*1)	Synchronous Speed	" " "	RPM	1 RPM	----
	03	Output Current	" " "	Amps	1 Amp	----
	04	Output Voltage	" " "	Volts	2v	****
	05(*1)	Machine Speed	" " "	RPM	1 RPM	----
	06	Input Signal Status	" " "	****	****	****
	07	Output Signal Status	" " "	****	****	****
	08	Motoring Torque Limit	" " "	----	****	****
	09	Braking Torque Limit	" " "	----	****	----
	0A	Calculated Torque	" " "	****	----	****
BASIC PARAMETER	10	Maximum Frequency	50.0 - 400.0 (*5)	HZ	0.1 Hz	60.0 Hz
	11	Base Frequency	50 - 400 (*5)	HZ	1 Hz	60 Hz
	12	Rated Output Voltage	320 - 460	V	1 V	460 V
	13	Frequency Bias	O-400	HZ	1 Hz	0 Hz
	14	High Limiter	O-400	HZ	1 Hz	60 Hz
	15	Low Limiter	O-400	HZ	1 Hz	1 Hz
	16(*3)	Acceleration Time 1	0.2 3600	sec	0.1 sec	20.0 sec
	17(*3)	Deceleration Time 1	0.2 3600	sec	0.1 sec	20.0 sec
	18(*2)	Frequency Gain	O-200	%	0.1%	105.0 %
	19	Torque Boost	O-20	---	1	3
	1A	Auto Energy Saving Op.	Yes - (0), No - (-)	---	---	No (-)
	1b	Elec. Thermal Overload	0 (not used) 50 -105	%	1%	105%
AUXILIARY PARAMETER	20	DC Brake Starting Freq.	0.0, 0.5 - 60.0	Hz	0.1 Hz	0.0 Hz
	21	DC Brake Voltage	0.0 - 10.0	%	0.1 %	10.0%
	22	DC Braking Time	0.0 - 10.0	sec	0.1 sec	0.5 sec
	23(*2)	Multi-Step Freq. Setting 1	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	24(*2)	Multi-Step Freq. Setting 2	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	25(*2)	Multi-Step Freq. Setting 3	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	26(*2)	Multi-Step Freq. Setting 4	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	27(*2)	Multi-Step Freq. Setting 5	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	28(*2)	Multi-Step Freq. Setting 6	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	29(*2)	Multi-Step Freq. Setting 7	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz
	2A	Acceleration Time 2	0.2 - 3600	sec	0.1 sec	100 sec
	2b	Deceleration Time 2	0.2 - 3600	sec	0.1 sec	100 sec
	2c	Acceleration Time 3	0.2 - 3600	sec	0.1 sec	100 sec
	2d	Deceleration Time 3	0.2 - 3600	sec	0.1 sec	100 sec
	2E	Acceleration Time 4	0.2 - 3600	sec	0.1 sec	100 sec
	2F	Deceleration Time 4	0.2 - 3600	sec	0.1 sec	100 sec
	30	Accel / Decel Pattern	0, 1, 2.		---	0

TABLE 5: FUNCTION CODE DATA (continued)

Mode	Funct. Code	Name	Setting and Indication Data: Range	Unit	Increment	Factory Setting
	31	Tone Control Selection	1, 2, 3, 4.	---	---	1
	32	Overload Early Warning	50 - 105	%	1%	105%
	33(*2)	Motoring Torque Limit	0, 20 - 180	%	1%	150%
AUXILIARY	34(*2)	Braking Torque Limit	0, 20 - 180	%	1%	100%
PARAMETER	35(*2)	Freq. Detection Threshold	I-400	HZ	1 Hz	30 Hz
(continued)	36(*2)	Freq. At Reference	0.5 - 5.0	HZ	0.1 Hz	2.5 Hz
	37	Start Frequency	0.5 - 5.0	HZ	0.1 Hz	0.5 Hz
	38	Start Freq. Holding Time	0.0 - 10.0	sec	0.1 sec	0.0 sec
	39	Jump Frequency 1	0.0, 0.5 - 400.0	HZ	0.1 Hz	0.0 Hz
	3A	Jump Frequency 2	0.0, 0.5 - 400.0	HZ	0.1 Hz	0.0 Hz
	3B	Jump Frequency 3	0.0, 0.5 - 400.0	HZ	0.1 Hz	0.0 Hz
	3c	Jump Frequency Range	0.0 - 5.0	HZ	0.1 Hz	2.0 Hz
	3d(*2)	Number of Motor Poles	2, 4, 6, 8, 10, 12	---	2	4
	3E(*2)	Coefficient Mach Spd.	0.1 - 10.0	---	0.1	1.0
	40	Pulse Freq. Meter Coeff ic.	6-100	---	1	30
	41	FWD/REV Command Hold (3 wire control)	Yes (0), No (-)	---	---	Yes (0)
	42	Up / Down Control	Yes (0), No (-)	---	---	No (-)
	43(*4)	Restart After Power Failure	Yes (0), No (-)	---	---	No (-)
	44	Under Voltage Alarm	Yes (0), No (-)	---	---	Yes (0)
	45	Open Coll. Output Select	0, 1, 2.	---	---	0
	46	Slip Compensation Control	0.0 - 2.5	Hz	0.1 Hz	0.0 Hz
	47	Voltage Signal Polarity Op.	Yes (0), No (-)	---	---	No (-)
ADJUSTMENT PARAMETER	50	Freq. Meter Calibration	70.0 - 105.0	%	0.1%	100.0%
PROTECTION	99	Set Data Protection	Yes (0), No (-)	---	---	No (-)

*1 *Because the data display is four digits, if the indication exceeds four digits it will be shown as 1/10 of the actual automatically. (Example: 12000 RPM would be displayed as 1200 RPM.)*

*2 *These parameter values can be changed during inverter operation; other parameters can only be changed when the inverter is stopped.*

*3 *If the setting of Acceleration/Deceleration time exceeds 100 seconds, the minimum setting unit resolution will become 1 second.*

*4 *Momentary power failures only (duration may vary with horsepower and load).*

*5 *Consult distributor for operation above 120 Hz.*

TABLE 5: FUNCTION CODE DATA (continued)

Mode	Funct. Code	Name	Setting and Indication Data Range	Unit	Increment	Factory Setting
FAILURE INDICATION	FO	Fault Message		---	---	---
	F1	Output Frequency at Fault	Indication Failure Condit.	Hz	0.1 Hz	
	F2	Setting Frequency at Fault		HZ	0.1 Hz	
	F3	Output Current at Fault		A	1A	
	F4	Operating Mode at Fault		---	---	
	F5	Fault Memory 1		---	---	----
	F6	Fault Memory 2		---	---	----
	F7	Fault Memory 3		---	---	----

NOTES:

NOTES:

6. FUNCTION CODE DESCRIPTIONS

TABLE 6: FUNCTION CODE DESCRIPTIONS

OPERATION MONITOR FUNCTIONS

Functions 00 . OA

The operation monitor will indicate any of the data selections shown in the chart on the right.

- For monitoring functions 02 (synchronous speed) and 05 (load turning speed), it is necessary to set auxiliary parameters 3d and 3E.

  NOTE: *Because the data display is four digits, if the indication exceeds four digits it will be shown as 1/10 of the actual automatically. (Example: 12000 RPM would be displayed as 1200 RPM.)*

- The input signal status and output signal status can be monitored via the keypad by going to Function Code 06 or 07, then using the UP/ DOWN ARROW keys to change the fourth digit. The chart on the right shows what signals are shown in the first, second, and third digits.A "0" indicates that the signal is present; a "-" means the signal is not present.

Function	Description	Ind. Unit
00	Output Frequency	HZ
01	Reference Frequency	HZ
02	Synchronous Speed	RPM
03	Output Current	A
04	Output Voltage	V
05	Machine Speed	RPM
05	Input Signal Status	
07	Output Signal Status	
08	Motoring Torque Lim.	
09	Braking Torque Limit	
OA	Calculated Torque	

Function	Explanation of function	Indication data			
		4th digit	3rd digit	2nd digit	1st digit
06	Input signal status	A	F WD	REV	BX
		b		THR	RST
		C	X 1	X2	X3
		d	RT 1	RT 2	AUT
		E		IL	PU
07	Output signal status	A	AX	OL	LV
		b	RUN	FAR	FDT

BASIC PARAMETER SETTING

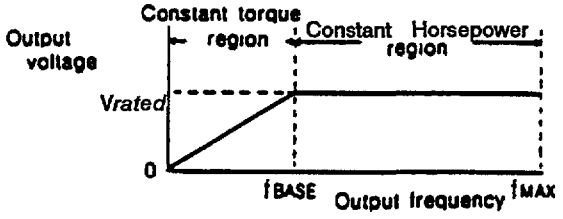
Functions 10 - 19

Voltage/Frequency Setting Characteristics

Function 10 Maximum Frequency (fmax)
Function 11 Base Frequency (fbase)
Function 12 Rated Output Voltage (Vrated)

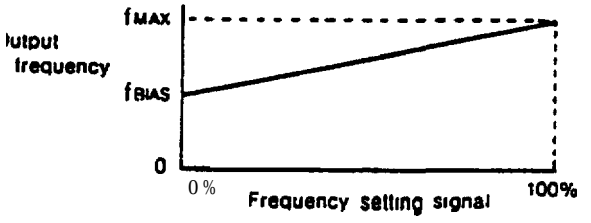
These parameters will be set with fbase, Vrated, and fmax adjusted to the rating of the electric motor. Consult distributor for operation above 120 HZ.

fbase: 50 to 400 Hz
Vrated: *320* to 460 VAC
fmax: 50-400Hz/freq. set signal 100% time



Function 13 Frequency Bias *(fbias)*

When the frequency setting signal is at 0%, bias can be added so that the inverter will run at a specified frequency even with zero reference applied. (Used to add an offset to the low end of the reference signal.}
fbias: 0 to *fmax*



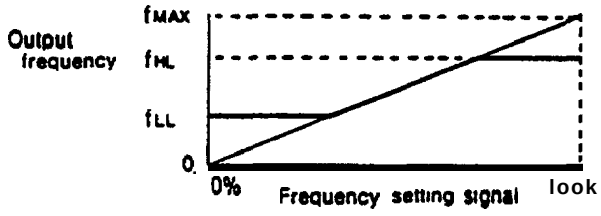
Adjustment of Upper/Lower Limits

Function 14 High Output Freq. Limiter (*fHL*)
Function 15 Low Output Freq. Limiter (*fLL*)

The upper and lower limiters can be set so that the inverter output frequency will stay within these limits. This will prevent motor overspeed and underspeed.

NOTE: *If the High Limit is set lower than the Low Limit, the Low Limit will be ignored and an error indication (Err1) will be displayed.*

fHL: 0 to *fmax*
fLL: 0 to *fmax*

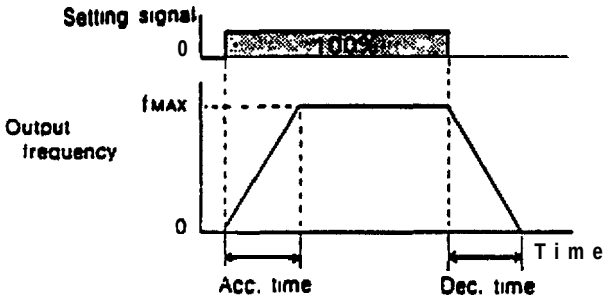


Acc/Dec time setting (*No.* l)

Function 16 Acceleration Time (t_{ACC1})
Function 17 Deceleration Time (t_{DEC1})

The acceleration and deceleration times define the amount of time it takes between 0 Hz and *fmax.*

t_{ACC1}: 0.2 to 3600 seconds
t_{DEC1}: 0.2 to 3600 seconds

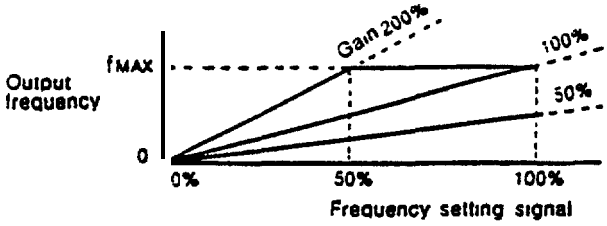


Function 18 Frequency Gain (Gain)

The frequency gain adjustment can be used to scale the analog voltage reference at the top end.

Gain: 0 to 200%

When the freauencv setting signal is applied with a current input, this function cannot be used.



Function 19 Torque Boost

21 patterns can be selected (torq. boost = 0 to 20).

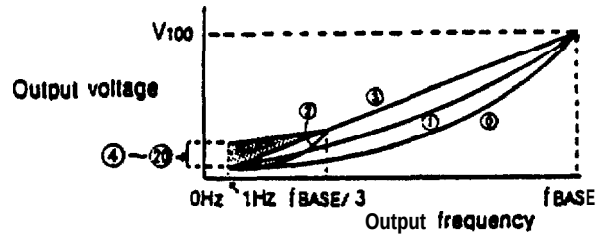
-- Patterns (0) and (1) are used for low torque loads (available for fans and blowers, thus improving characteristics such as vibration, noise, and efficiency.

-- Pattern (2) is characteristic of V proportional to the square of f up to *fBase/3.*

-- Pattern (3) is characteristic of constant V/f up to *fBase.*

-- Patterns (4) through (20) are to increase V/f in a low frequency range between 0 to *fBase/3* to make up the torque at low speed operation and to give torque boost characteristics.

Torque Boost: 0 to 20

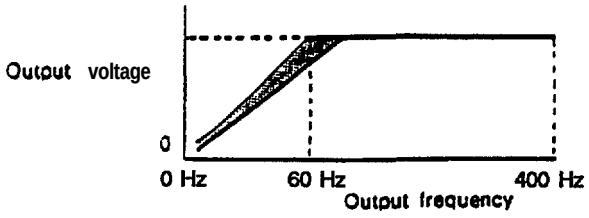


Function 1 A Automatic Energy Saving Oper.

When automatic energy saving operation is selected, voltage is reduced as shown. The voltage is decreased as the load current decreases.

0 = Active

— = Inactive



Function 1 b Electronic Thermal Overload

Electronic thermal setting should be made with the following equation:

Electronic Thermal Overload =

$$\frac{\text{Rated electric motor current}}{\text{Rated inverter current}} \times 100\%$$

--When using a 4 pole, 3 phase motor, the motor can be protected from overload without an external thermal relay.

-- The electronic thermal calculation compensates for a decrease in motor cooling at low speed .

--When using one inverter with more than one motor, an external thermal overload relay should be used for each motor.

-- Ifthe **Electronic Thermal** Function is not used, set at 0%.

CAUTION: The electronic thermal overload relay meets the reauirements of 4-pole standard motors onlv. When anv of the followina conditions exist, use conventional overload **relays** instead of the electronic overload built into the inverter.

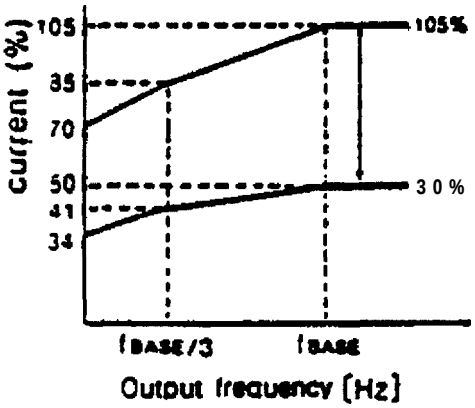
-- Motors other than **4-pole type.**

-- Special motors (anv non-standard motor).

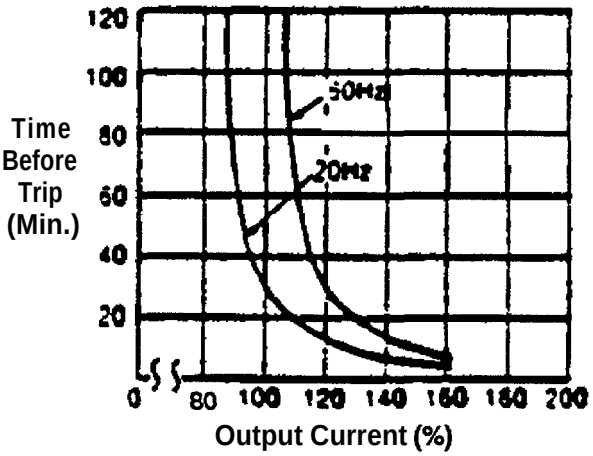
-- For a **group operation** where two or more motors are run **using** a **single** inverter.

-- **Application** has frequent **starting/stopping.**

-- Operation is at **10** Hz or less.



Threshold Where Timing Begins Based On Output Frequency And Current, And Electronic Thermal Overload Setting



Inverse Time Characteristic with Electronic Thermal Overload = 105%

AUXILIARY FUNCTION CODES

Functions 20 - 47

DC Brake

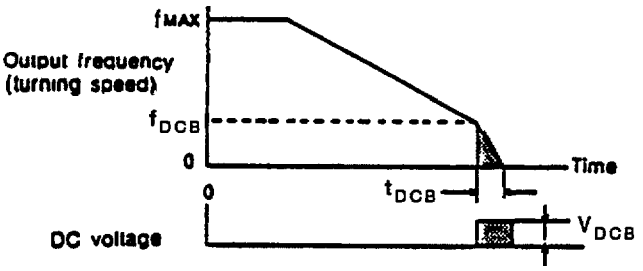
Function 20 DC Brake Start Frequency (f_{DCB})
Function 21 DC Brake Voltage (V_{DCB})
Function 22 DC Brake Time (t_{DCB})

Brake Start Freq.: 0.0, 0.5 to 60 hz.
Brake Voltage: **0 to 10%**
Brake Time: 0 to **10** seconds
Brake Duty Cycle: 5% maximum

The DC brake is applied when a **STOP** command is given and the inverter output frequency is at or below the DC brake start frequency.

-- If the motor/load inertia is so large that the rotation is not stopped aftertheset DC brake time (10 seconds maximum), the motor/load will coast until stopped.

-- When DC brake is not used, set the Brake Start Frequency to **"0".**



Multi-Step *Frequency Setting Selections* ***1*** *- 7*

Function 23 Multi-Step frequency 1 **(MSS1)**
Function 24 Multi-Step frequency 2 (MSS2)
Function 25 Multi-Step frequency 3 (MSS3)
Function 26 Multi-Step frequency 4 (MSS4)
Function 27 Multi-Step frequency 5 (MSS5)
Function 26 Multi-Step frequency 6 **(MSS6)**
Function 29 Multi-Step frequency 7 **(MSS7)**

External contact signals allow a selection of seven speeds.

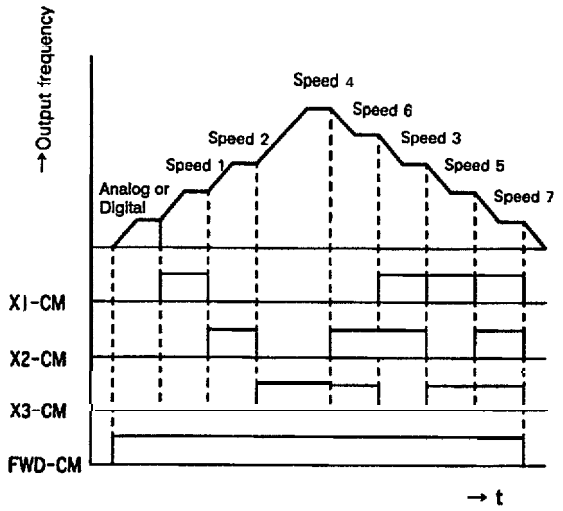
Each step speed can be set between 0.5 and 400 HZ.

The rate of change of the output frequency between each step is adjustable by the **accel/decel** time setting.

NOTE: Multi-step *frequency* ***operation*** *overrides* ***other*** *frequency reference signals.*

Relationship Between Terminals and Multi-Step Frequencies 1 to 7

Fun. Code	Depend - Loc/Rem Switch	23	24	25	26	27	28	29
Multi-Step Frequency	Analog or Digital	1	2	3	4	5	6	7
X1-CM	Off	On	Off	On	Off	On	Off	On
x 2 - w	off	Off	On	On	Off	Off	On	On
X3-CM	off	Off	Off	Off	On	On	On	On



Acc/Dec Time Settings Numbers 2 - 4.

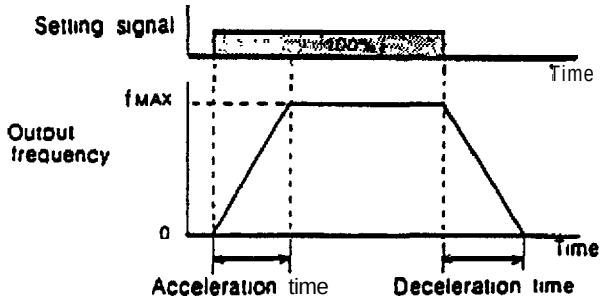
Function 2A Acceleration Time (t_{ACC2})
Function 2C Acceleration Time (t_{ACC3})
Function 2E Acceleration Time (t_{ACC4})
Function 2b Deceleration Time (t_{DEC2})
Function 2d Deceleration Time (t_{DEC3})
Function 2F Deceleration Time (t_{DEC4})

-- The time between 0 to fmax and from ***fmax*** to 0 can be set anywhere between 0.2 to 3600 seconds.

-- Accel and **Decel** time can be selected from four **different** acceleration and deceleration groups using the **RT1** and **RT2** terminals.

-- Even during acceleration and deceleration, **Accel /Decel** time can be changed using the external contact signals **RT1** and RT2.

Function Code	16/17	2A/2b	2c/2d	2E/2F
External Contact Signal	#1 Acc/Dec	#2 Acc/Dec	#3 Acc/Dec	#4 Acc/Dec
RT1-CM	Off	On	Off	On
RT2-CM	Off	Off	On	On



Function 30 Accel/Decel Pattern

ACC/DEC Pattern	Setting
Linear ACC/DEC	0
Curve ACC/DEC "A"	1
Curve ACC/DEC "B"	2

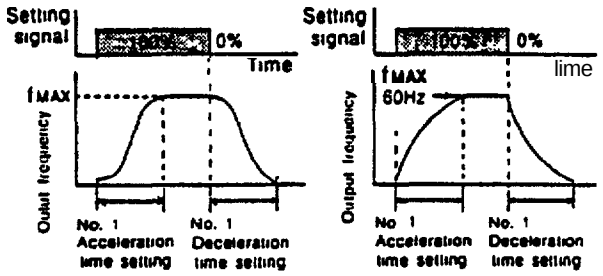
NOTE: *Linear pattern will override Pattern B if* fmax *is greater than 60 Hz.*

NOTE: *This function cannot be used in the event Multi-Step Frequency selection is made.*

Available for **selecting** two **kinds of patterns**: A **and B**

Curve A - "S" Curve

Curve B - For Use On Fans And Blowers



Function 31 Tone Control Selection

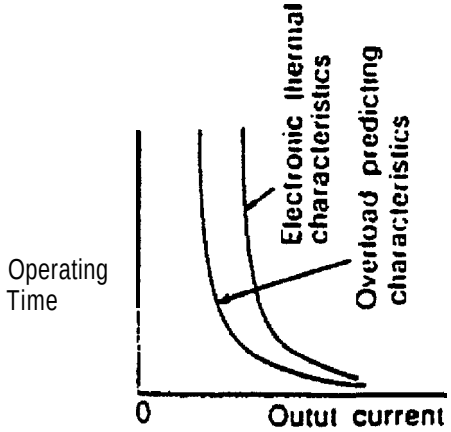
There are four different tone control selections that set the chopping frequency of the inverter output transistors. This can be used to reduce audible resonant frequencies from the motor.

Function 32 Overload Early Warning Signal

An early warning signal is given if the inverter output current exceeds the overload alarm level for a certain period of time.

This is an open collector output. If the optional relay unit is used, this signal can be used as a dry contact output.

If **"0" is** set, this function is inactive.



Torque Limiter
(Motoring and Braking)

Function 33 Motoring Torque Limit (T_{ML})
Function 34 Braking Torque Limit (T_{BL})

-- Torque is calculated from the output voltage and current of the inverter.

-- This function can improve impact load restoration characteristics during ACC/DEC and constant **speed** operations and improve instantaneous restart ability.

Regeneration Avoidance Control

Set braking torque limit to **0%,** and power regeneration to the inverter will be limited to obtain deceleration characteristics close to coasting.

Automatic ACC/DEC

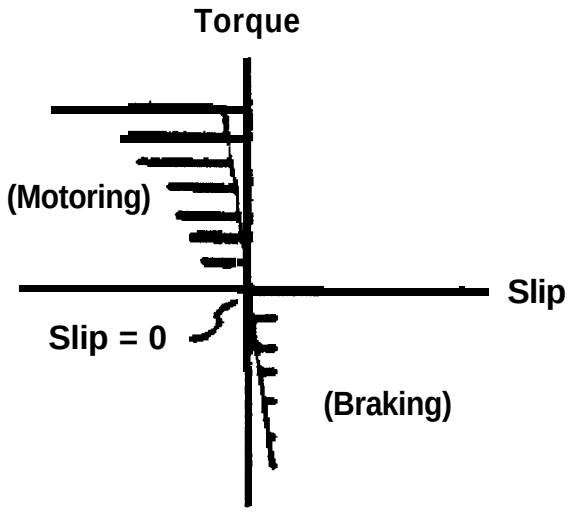
Even if the acceleration and deceleration time is reduced to less than the time fixed by load inertia, automatic acceleration and deceleration can be maintained by maintaining the torque limit level.

NOTE: *To obtain more torque on deceleration, optional brake unit and brake resistor* will *be* **required.**

Torque Limit Setting Range (Motoring and Braking)

-- Set to **"0"** to render inactive.
-- Adjustable from 20 to 180%.

NOTE: The *data setting for* ***Function Code 34 is*** *100% (assuming that a braking resistor is connected). If no braking resistor is connected, setting should be changed to 15% or less. If left at* ***100%,*** *an overvoltage trip will occur during deceleration if the* ***decel*** *time setting is too shortforthe motor/load inertia.*

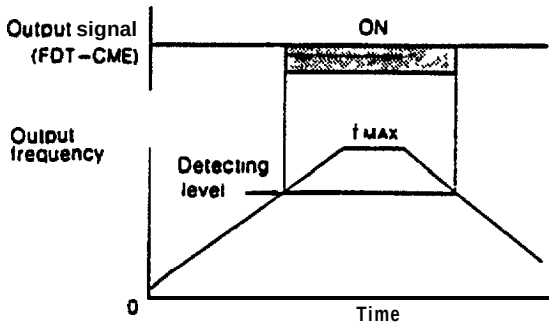


Function 35 Frequency Detection Threshold

This signal (FDT) is active (On) when the output frequency exceeds the detection level.

This is an open collector output.

If the optional relay unit is used, this signal can be used as a dry contact output.



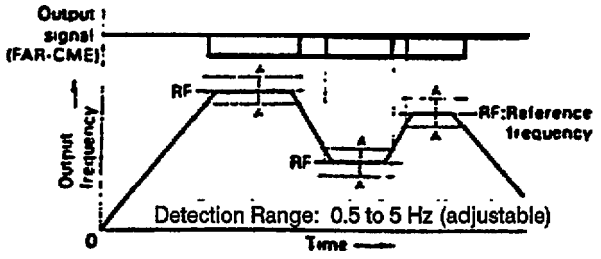
Function 36 Frequency At Reference (FAR)

This signal is active (On) when the output frequency reaches the reference frequency.

This is an open collector output.

If the optional relay unit is used, this signal can be used as a contact output.

FAR: 0.5 to 5.0 Hertz



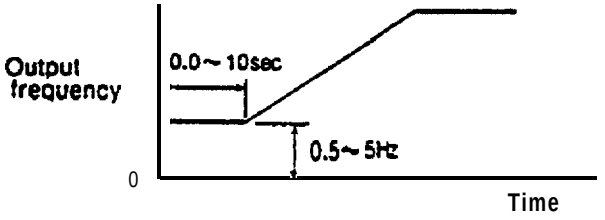
Starting Actions

Function 37 Starting Frequency (F_{Sta})
Function 36 Start Frequency Hold Time (T_{Hold})

-- Starting frequency set range is 0.5 to 5 Hz.

-- Starting frequency holding time setting range is 0.0 to 10 sec.

A starting frequency and holding time can be set that is suitable for the starting torque characteristics of the load. (Setting is ignored during deceleration and reverse operation.)



Frequency Jump

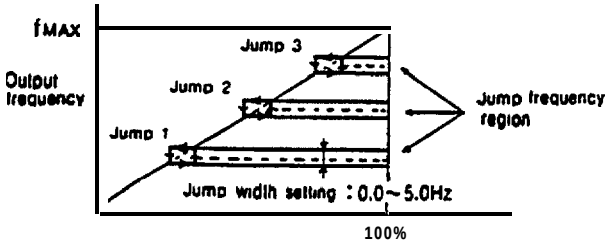
Function 39 Jump Frequency 1 (Jump 1)
Function 3A Jump Frequency 2 (Jump 2)
Function 3b Jump Frequency 3 (Jump 3)
Function 3C Jump Frequency Range.

These functions are used to avoid the natural mechanical vibration of the load.

Three jump frequencies can be set. Jump frequencies are not active during acceleration and deceleration or if the multi-step speed settings are used.

The jump frequency range is adjustable between 0 to 5 Hz. (When the jump frequency is not used, set it to '0.0'.

Jump 1, Jump 2, &Jump 3: **0.0, 0.5** to 400 Hz



Number of Electric Motor Poles
Speed Conversion Coefficient.

Function 3d Number of Motor Poles
Function 3E Machine Speed Conversion Coefficient.

These Function Codes are set to monitor the synchronous speed of the motor and the machine speed.

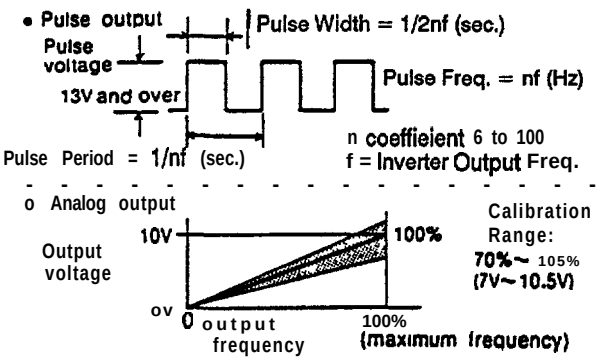
Function	Setting Ranae
3d	**2, 4, 6, 8, IO, 12**
3E	0.1 - 10.0

Machine Speed = (Motor Synchronous Speed) X (Machine Speed Conversion Coefficient).

Frequency Monitor

Function 40 Pulse Freq. Meter Coefficient
Function 50 Analog Freq. Meter Calibration

The inverter output frequency can be monitored from terminals FM1 (+) and FM2 **(-).** Set Jumper **SW2 on** the Main Control Card in the **1-2** position for an analog meter or in the **2-3** position for a pulse type meter.



Function 41 FWD/REV Command Hold (3-wire)

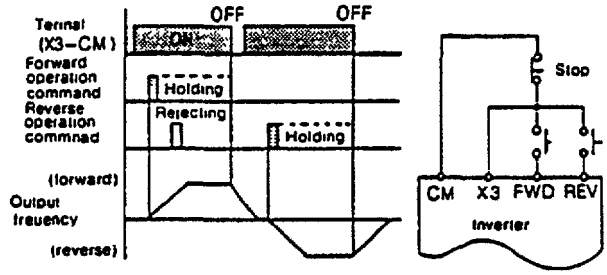
This function enables control by a one-shot (50ms minimum) RUN/STOP command (FWD and REV command), External self-sealing circuits can be omitted.

When this function is selected, multi-step speed settings from 1 to 3 can be used, but those from 4 to 7 cannot.

NOTE: ***After*** **commend** *hold, no other commands will be received* ***unless*** *a STOP command is given.*

Active: **"0" (3-wire** control, momentary contacts)
Inactive: **"-" (2-wire** control, maintained contacts)

Pulse **Signal Width must** be 60 msec. or longer

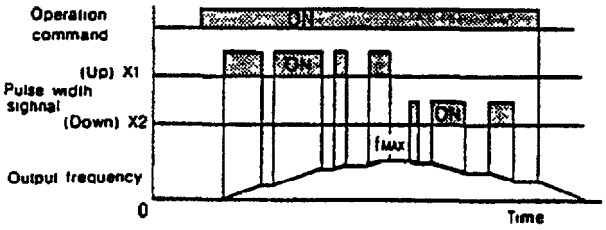


Function 42 Up-Down Control Selection

The frequency can be increased and decreased using terminals **X1** -CM and X2-CM. This function simulates a motor driven potentiometer.

In addition, if power is shut down, set value is stored. When operation is started again, inverter will come up to the stored frequency.

NOTE: *The functions of multi-step* **frequency,** ***self-****sealing selection, and* **up-down** *control use the same terminals. Thus, there are four possible combinations of operation. All three functionsca nor be used* ***simultaneously.***



FC41	FC42	Term. X1	Term. X2	Term. X3
Inact.	Inact.	Multi-Step Frequency		
Inact.	Act.	Up-Down control		Multi-Step 4 set
Act.	Inact.	Multi-Step 0-3 set		Comm-hold sel.
Act.	Act.	Up-down control		Comm-hold sel.

NOTES: ***When this control function is selected, multistep*** *speed setting 4 (Function Code 26) can be used, but the other six settings cannot.*

Multi-Step Speed Setting, ***FWD/REV*** *Command Hold, and Up-Down* ***Control*** *cannot be usedsimultaneously because the same terminals are used for these functions.*

If the **Up-Down** ***function is selected, external frequency*** **setting** ***signals (voltage and current)*** **are** ***ingored.***

If Up-Down buttons are pressed at the same time, *down will take precedence.*

Function 43 Auto-Restart For Momentary Power Failure

This function specifies whether the inverter is to be restarted automatically when power is recovered after a momentary power failure.

If automatic restart is selected, the inverter is restarted **after** power recovery under the following conditions:

1. The power failure duration is within the allowable time.
2. A RUN command is present when the power is restored.
3. Must be in "Remote" Mode.

Setting	Auto-Restart Active **Status**
0	**YES**
–	**NO**

Function44 Undervoltage Alarm

When main circuit **DC** voltage decreases to the undervoltage level, inverter will cease to give an **output.** Atthattime, it will self-hold the failure signal to indicate alarm and transmit an alarm signal ***or*** indicate that it has reached the undervoltage level without self-holding (can select either one).

Setting	Operation	Self-Hold	Alarm **Ind**	m **Sig.**
0	stop	Yes	**Yes**	Transmit
-	Stop	No	Yes	No Trans.

Function 45 Open Collector Output Selection

The **RUN, FAR, FDT,** OL, and LV terminals are used to output faults and operation monitoring signals. The output mode can be selected with this function. Three modes, 0, **1,** and 2, can be selected. If mode **1** or 2 is selected, signals are output in a bit pattern (see following charts).

This function is used for external indication of Operating Mode and/or Fault Condition.

NOTE: ***Individual*** *fault signal will output only one* ***fault,*** *even if more than one fault* ***occurs*** *simultaneous/y.*

Output Signal Selection

Setting / **Terminal Symbol**	**0**	**1**	**2**
RUN	Inverter Running	Individual Fault Output	Comb. of Operation Monitor and Individual Fault Signs
FAR	Frequency At Reference		
FDT	Frequency Level Detect		
OL	Overload Early Warning	Overload Early Warning	
LV	Undervoltage Alarm	Undervoltage Alarm	

Combination of Individual Fault Output When Setting Is "1"

Individual **Fault** / Terminal	No **Fault**	oc Overcurrent	ou Over Voltaae	**LU** Under Voltaae	**OL1** Inverter Overload	**OL2** Elec. Motor Overload	**OH1** Inverter Overheat	**OH2** External Alarm
RUN (MSB)	Off	Off	Off	Off	On	On	On	On
FAR	Off	Off	On	On	Off	Off	On	On
FDT (LSB)	Off	On	Off	On	Off	On	Off	On

Function 45 Open Collector Output Selection (continued)

Operation Monitor Output When Setting Is **"2"** (When inverter is tripped)

Individual Fault		Terminal Symbol					Decimal
		LV (LSB)	OL	FDT	FAR	RUN (MSB)	Eaulv.
	No Fault (Not Tripped)	Off	Off	Off	Off	Off	0
OC1	Overcurrent at Acceleration	On	Off	Off	Off	Off	1
OC2	Overcurrent at Deceleration	off	On	Off	Off	Off	*2*
OC3	Overcurrent at Fixed Speed	On	On	Off	Off	*Off*	*3*
OU	Overvoltage	off	Off	On	Off	*Off*	*4*
LU	Undervoltage	On	Off	On	Off	*Off*	*5*
OL1	Inverter Overload	Off	On	On	Off	*off*	*6*
OL2	Electric Motor Overload	On	On	On	Off	*off*	*7*
OH1	Inverter Overheat	Off	Off	Off	On	*off*	*6*
OH2	External Alarm	On	Off	Off	On	*Off*	*9*

Operation Monitor Output When Setting Is **"2"** (While inverter is not tripped)

Operation Monitor				Terminal Symbol					
Inverter Running	Frequency At Refer.	Frequency Level Detection	Overload Early Warn.	LV (LSB)	OL	FDT	FAR	RUN (MSB)	Decimal Equlv.
A	A	A	A	Off	Off	Off	Off	Off	0
P	A	A	A	Off	On	Off	On	Off	10
P	A	P	A	On	On	Off	On	Off	11
P	P	A	A	Off	Off	On	On	Off	12
P	P	P	A	On	Off	On	On	Off	*13*
P	A	A	P	Off	On	On	On	Off	14
P	A	P	P	On	On	*On*	*On*	*Off*	*15*
P	P	A	P	Off	Off	Off	Off	On	16
P	P	P	P	On	Off	Off	Off	On	17

NOTE: P *indicates that monitor signal is present* (True), A *indicates that monitor signal is absent (False).*

Function 46 Slip Frequency Compensation

Used for compensation of speed errors caused by load fluctuation. A slip frequency corresponding to the load current can be added; compensation range is 0.0 to 2.5 Hz.

Function 47 Operation By Signal Polarity

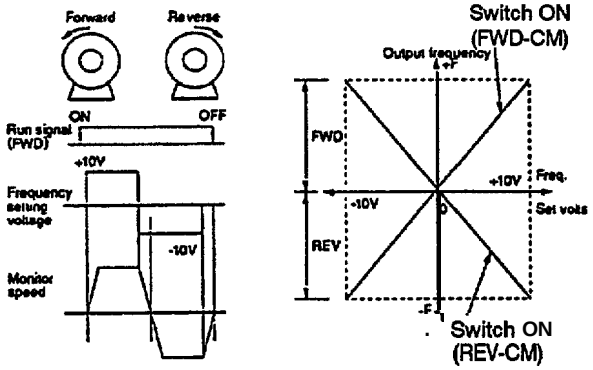
This function enables the direction of motor rotation to be changed according to the polarity (+, -) of the frequency setting voltage.

To operate the inverter with an external frequency setter, turn on the switch between terminals FWD and CM. Apply a positive voltage (+ 10 VDC) to terminal V1 to turn the motor in the forward direction, or apply a negative voltage to the terminal to turn it in the reverse dirction.

If the switch between terminals REV and CM is on, a positive voltage applied to terminal V1 turns the motor in the reverse direction and a negative voltage to the terminal turns it in the forward direction.

This means that the motor rotation direction can be changed simply by changing the polarity of the setting voltage.

"0" (zero): Active
"-" (dash): Inactive



CAUTION: If a Multi-Step Freauencv selection is made, the analoa signal is ignored. When this happens, the rotation direction will be set according to terminals FWD and REV.

DATA PROTECTION

Function 99 Data Protection

Function Code settings can be protected so that they cannot be changed or lost accidentally. Protected data cannot be changed without first cancelling the data protection function. The Data Protection function can be operated in the following two ways:

1. Select Function Code 99 using the SHIFT and UP ARROW or DOWN ARROW keys. Use the SHIFT key to set the unit to Data Setting Mode. If the data is protected, "0" is displayed. If it is not protected, "--" is displayed. Pressing the UP key displays "0", then pressing the SET Key stores this value in memory and disables a data change. Pressingthe DOWN key displays"--", then pressing the SET Key stores this value in memory and enables a data change.

2. While holding down the FAST key, press the SET key to protect the data or the RESET key to release data protection. The display shows "0" when the data cannot be changed (protected) or "--" when the data can be changed (not protected).

   The current display is held for five seconds after the keys are released and then returns to the previous display.

NOTE: *In the Fault Monitoring Mode* or *while the data display is blinking, the Data Protection function cannot be selected.*

FAULT INDICATION AND DETECTION

fault *Indication*

Functions FO - F7

The fault indication function is composed of the following three pans:

1. Indication of the present fault.
2. Indication of operating conditions at the time that the fault took place.
3. Indication of previous faults.

When a fault occurs, the mode will automatically be switched to the Fault Display Mode.

-- If the fault is a complex one (two or more simultaneous faults), use the UP or DOWN ARROW keys to change the 4th data digit and the faults will be displayed in the **1st, 2nd,** and 3rd digits.

-- Inaddition, the operating conditions at thetimeof the **present fault only** can be indicated by Functions **F1** through F4.

-- Also, the first fault for the previous three fault trips can be indicated by **F5** through F7.

NOTE: *First fault means the first detected fault when multiple faults occur at the same time.*

A more detailed explanation on obtaining fault information can be found in Section 5 of this Instruction Book. If the fault will be investigated, obtain as much fault information as possible before resetting the inverter to resume operation. Also record the date, time, and any information about the machine operation prior to the fault. This will help in determining the cause more quickly.

Function Code	Indication Data (by digit) 4th	3rd	2nd	1st	Name of Fault
	1 - 7 in order	0	c	1	Overcurrent At Acceleration
	of occurence.	0	c	2	Overcurrent At Deceleration
		0	c	3	Overcurrent At Fixed Speed
FO			0	U	Overvoltage
			L	U	Undervoltage
		0	H	1	Inverter Overheat
		0	H	2	External Alarm
		0	L	1	Inverter Overload
		0	L	2	Motor Overload
----	E	r	r	0	CPU Error
----	E	r	r	d	Memory Error
F1		6	0.	0	Output Frequency
F2		6	0.	0	Setting Frequency
F3	1	0	0.	0	Output Current
F4	1 - 4 in order		F	0	Normal Operation
	of indication		r	E	Reverse Operation
			C	L	Current Restriction
			u	L	Voltage Restriction
			u	u	Undervoltage Restriction
F5					1st Prev. Fault (Pres. Fault - 1)
F6					2nd Prev. Fault (Pres. Fault - 2)
F7					3rd Prev. Fault (Pres. Fault - 3)

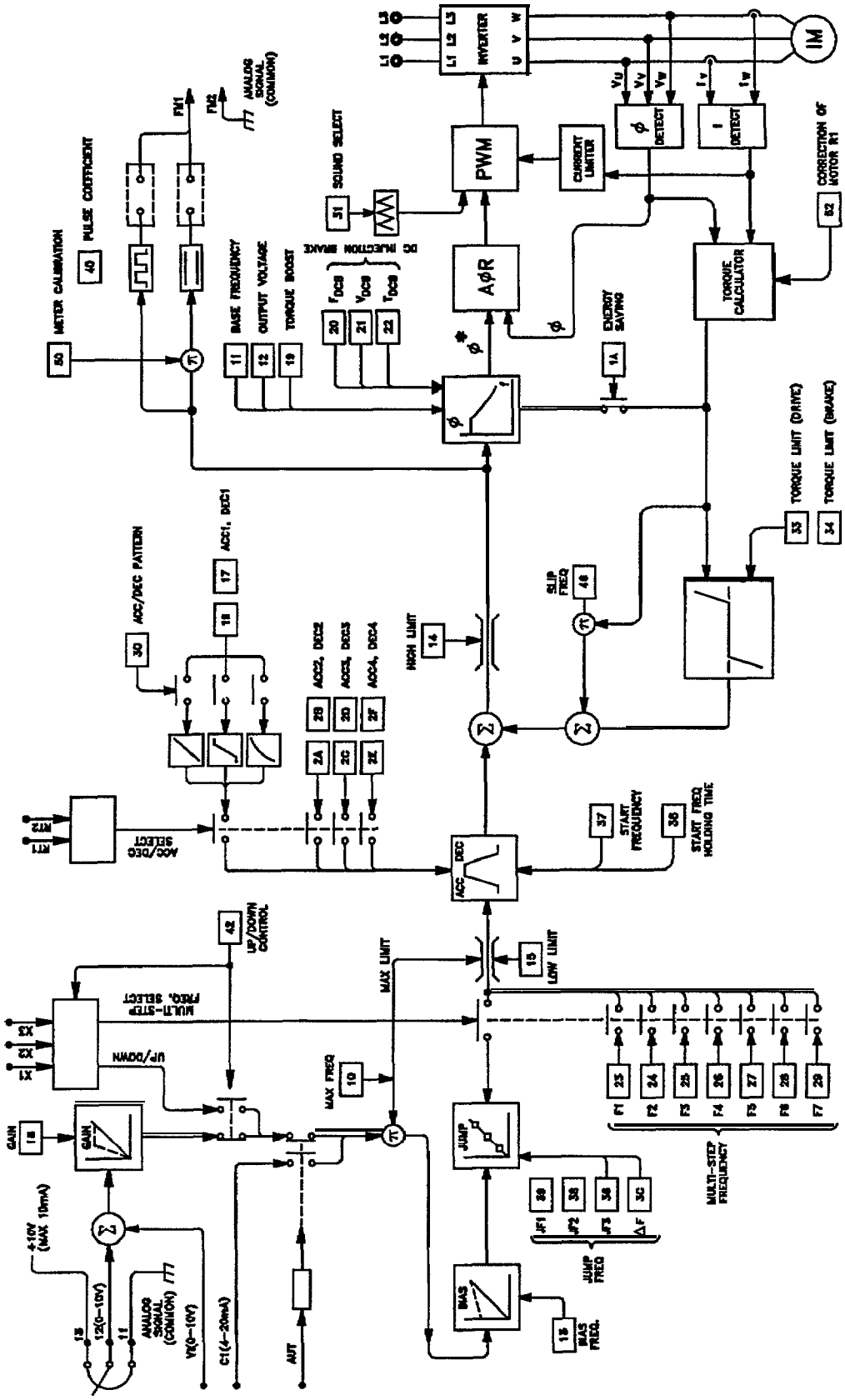


Figure 6-1. AF-300C FUNCTION CODE MAP

NOTES:

7. MAINTENANCE AND INSPECTION

So that the AF-300C inverter may give extended periods of trouble-free operation and to prevent potential problems from occuring, the following items should be periodically inspected.

WARNING - BURN AND SHOCK HAZARD: Do not conduct any Inspectlons until after disconnecting the power supply and after the "CHARGE" light on the inverter has gone out.

CAUTION: When removina or replacing wires, handle them by the connector onlv and not bv the wire itself. Be careful not to mix wires when reconnecting them.

MEGGER TEST

NOTE: Perform *this* test *only* if *absolutely* necessary.

1. Disconnect all inverter terminals and never apply test voltage to the inverter when conducting an external circuit megger test.
2. Perform the test only on the main circuit as shown in Figure 7-l when conducting a megger test on the inverter itself.
   -- Do not conduct a megger test on the control circuits.

CAUTION: Short all terminals as shown in Fiaure 7-1 and megger them all at one time. Do not megger each terminal individually.

3. Use a multi-meter set on a high resistance range to conduct a continuity test on the control circuits and not a megger or a buzzer.

PERIODIC PARTS REPLACEMENT

The life of the inverter will vary according to the installation environment and the amount of running time. However, if continuous operation is within the allowable limits, the life of the DC Link capacitor is approximately five years and the life of the cooling fan is approximately three years. It is recommended that these parts be replaced before failure occurs.

INSPECTION ITEMS

Refer to the Inspection Items Chart in this section for inspection points and corrective action.

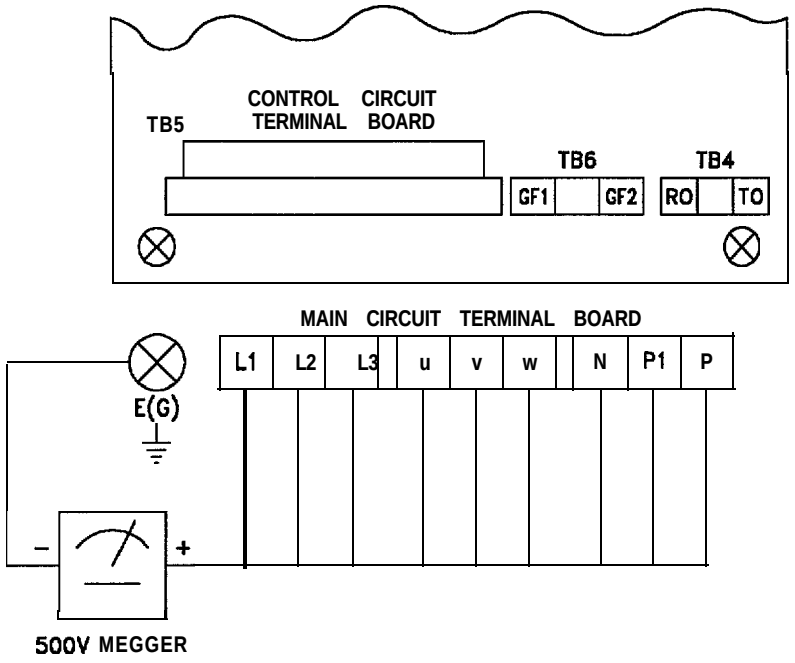


Figure 7-1. MAIN CIRCUIT MEGGER TEST CONNECTIONS

INSPECTION ITEMS CHART		
Item	**Reason of Inspection**	**Corrective Action**
Power Source Voltage	Wiihin permissible limits (323 - 506 VAC) for 380/400-460 VAC @ 50/60 Hz.	Adjust the power supply voltage
Ambient Temperature	Within permissible limits (-10° to +50° C) (+14° to +122°F)	Investigate cause and make corrections until environment is within permissible limits
Ambient Humidity	Wiihin permissible limits (20 - 90% RH) No dew condensation or freezing	Investigate cause and make corrections until environment is within permissible limits
Vibration	Within permissible limit (0.6G or less)	Investigate cause and make adjustments until within permissible limits.
Noise	Abnormal Audio Noise from cooling fan, etc.	Contact the supplier where the inverter was purchased.
Odor	Smell or burning	Contact the supplier where the inverter was purchased.
Dust	Dust accumulation on cooling fins, cooling fan, or on the control board.	Clean and blow out with compressed air. (Follow appropriate safety precautions.)
Screws/ Connectors*	Check for any loosening	Retighten as needed.

**Check internal screws/connectors* only *during periodic inspections or when cover is removed.*

Daily Inspections

For daily inspections, it is not necessary to remove the inverter cover. Check the inverter from the outside for any abnormal items per the above chart.

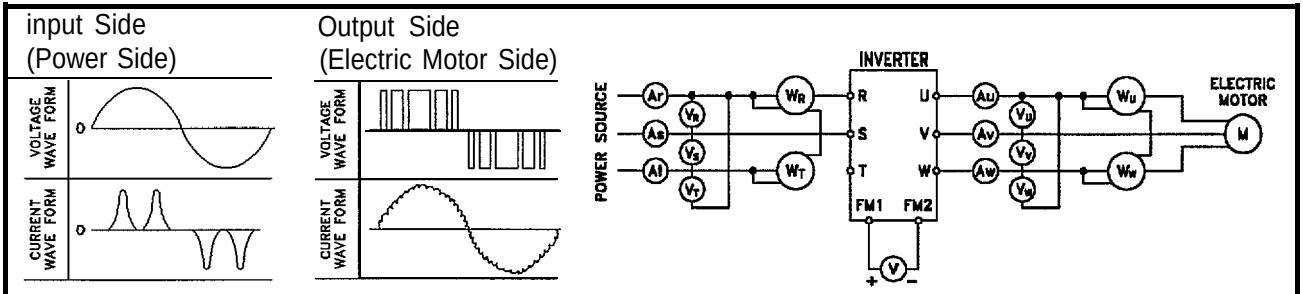
-- If anything is found, take corrective action as soon as possible.
-- If the inverter is to be kept in operation before corrections can be made, record details of the problem for future reference.

Periodic Inspections

Periodic inspection intervals will vary per the inverter application (bi-monthly, semi-annualy, annually, etc.). Pick an interval that best suits the particular application, but do not ignore the inspections entirely.

For periodic inspections, remove the inverter cover and check the inverter for any abnormal items as per the daily inspections, plus the following:

-- Check all internal screws/connections.
-- Check the DC Link capacitor for any leakage or swelling.
-- Check relay contacts for roughness.
-- Check all wires and conductors for any deformation or heat damage/deterioration.
-- Check resistor insulation for cracks.
-- Verify output voltage of each phase.



Item	Input Side Measuring Inst. (Power Side)			Output Side Measuring Inst (Electric Motor Side)		
Name of Instrument	Amperemeter AR, S, T	Voltmeter VR, S, T	Wattmeter WR, T	Amperemeter AU, V, W	Voltmeter VU, V, W	Wattmeter wu, w
Kind of Instrument	Movable Core Type	Rectifier Type or Movable Core Type	Electrodyna-mometer Type	Movable Core Type	Rectifier Type	Electro-dynamo-meter Tvoe
Symbol of Instrument						

Item	Measurement of Output Freq. Terminals FM1 and FM2
Name of Instrument	DC Voltmeter V
Kind of Inetrument	Movable Coil Type
Symbol of Instrument	

Figure 7-2. MEASUREMENT LOCATIONS AND DEVICES EXAMPLE

MEASUREMENT POINTS AND METERS

Since the inverter input/output voltage and current contains high frequencies, selection of the wrong measuring device can lead to gross miscalculations.

When using a CT (current-detection transformer) to measure the current, the amount of error will be large if the frequency is low. Because of this, always use a CT with as large a capacity as possible. See the following chart and Figure 7-2 for recommended measurement devices.

RECOMMENDED MEASUREMENT DEVICE CHART

Item	Simple Measurement	Precision Measurement
Input Voltage	Tester	Moving-Iron type voltmeter
Input Current	Clamp Meter	Moving-Iron type ammeter
Input Power	--	Electrodynamometer type wattmeter
Output Voltane	Tester	Rectifier type voltmeter
Output Current	Clamp Meter	Moving-Iron type ammeter
Output Power	--	Electrodynamometer type wattmeter

NOTES:

8. TROUBLESHOOTING

The TABLES and Figures within this Section arefurnished to aid in troubleshooting the more common problems that may be encountered with the AF-300C Inverter. If after performing all suggested procedures and the problem still is present, it may be necessary to contact the distributor that the inverter was purchased from. For parts ordering procedures (In Warranty and Out of Warranty), refer to Section 9; for replacement parts listing, refer to Section 10. For more expediant troubleshooting of Faults, it is recommended that as much information as possible be recorded from the inverter before resetting and continuing operation. See Sections 5 and 6 of this Instruction Book for additional Fault information

TABLE 7: FAULT DESCRIPTION AND OPERATION

Dis-play	Functlon	Description
CL	Current Limit	Provides restriction of the output current when the inverter reaches the current limit level.
LU	Undervoltage Trip	-- Operation will be kept on for 15 msec at full load in case of power failure. (At light load, operating time may be slightly longer.) -- When the DC Link goes below the under-voltage level, the inverter will stop output and hold the fault trip (motor coasts to stop).
OC1 oc2 oc3	Overcurrent Trip	When the output current of the inverter reaches the overcurrent protection level, the inverter will stop output and hold the fault trip (motor coasts to stop). OC1: Overcurrent during acceleration. OC2: Overcurrent during deceleration. OC3: Overcurrent during constant speed. Ground faults are detected by the ground fault detection unit (standard equipment). **NOTE:** *The ground fault detection unit is to protect the inverfer on/y. For prevention of personnel injury, efc, a leak-proof* relay, *should be connected separately.*
ou	Overvoltage Detection	When the DC Link voltage reaches the overvoltage protection level, the inverter will stop output and hold the fault trip (motor coasts to stop).
OL1	Inverter Overload	When overload operation exceeds the overload limit (inverse time characteristic) of the invetter, the inverter will stop output and hold the fault trip (motor coasts to stop).
OH1	Inverter Overheat	If the temperature of the inverter unit is too high, the inverter will stop output and hold the fault trip (motor coasts to stop).
OL2	Motor Overload (Electronic Thermal Overload)	When only one motor is driven, overload can be detected without an external thermal device. Since reduction in the cooling effect at low speed is taken into account, protection over a wide operation range can be provided. When overload operation exceeds the electronic thermal overload setting (inverse time characteristic) the inverter will stop output and hold the fault trip (motor coasts to stop). **CAUTION:** When drivina more than one motor, provide protection with a thermal relay in line with each motor.

TABLE 7: FAULT DESCRIPTION AND OPERATION (continued)

Display	Function	Description
OH2	External Fault	When the overtemp detection device of the brake unit and brake resistor (optional) or the user's external thermal relay is activated, the inverter will stop output and hold the fault trip (motor coasts to stop).
Err0 Err9	CPU Abnormal	If the inverter CPU operation is abnormal, the inverter will stop output and hold the fault trip (motor coasts to stop).
Err1	Setting Error	When low limit frequency setting exceeds that of the high limit frequency.
Err2	Setting Error	When a Function Code that cannot be set while inverter is running has been attempted to be set.
Errd	NV-RAM Error	Stored setting data is abnormal. This error message cannot be reset. Replace the Main Control Card. **NOTE:** *It* **may be possible** *to c/ear this condition by removing and reinserting any socketted integrated circuit chips on the Main Conttrol Card (total of three).*
---	Stall Prevention	Automatically extends acceleration time to restrict overcurrent during rapid acceleration (to prevent overcurrent trip). Automatically extends deceleration time to restrict overcurrent and over-voltage during rapid deceleration (to prevent overcurrent and overvoltage trip).
---	input Surge	The inverter can be protected from 5KV standard impact wave voltage which will invade from the main circuit power.

TABLE 8: FAULT CONDITION DISPLAY AND CORRECTIVE ACTION

Display	Check For:	Corrective Action
OC1	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Proper torque boost	Adjust as needed (reduce torque boost).
	Proper acceleration time	Lengthen the acceleration time.
	Loose Connections	Check on operation of external output contactor If used. Check Main Circuit wiring connections
	Other than above	Check transistors Increase inverter capacity (use larger HP inverter).
oc2	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Proper deceleration time	Lengthen the deceleration time.
	Loose Connections	Check on operation of external output contactor If used. Check Main Circuit wiring connections
	Other than above	Connect DB braking unit resistor (optional) or check operation of if already present. (Check polarity of thermal interlock connections to DB unit.) Increase inverter capacity (use larger HP inverter).
oc3	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Sudden change in load	Eliminate sudden load changes.
	Loose Connections	Check on operation of external output contactor if used. Check Main Circuit wiring connections
	Other than above	Check for electrical noise. Check on operation of external output contactor If used. Increase inverter capacity (use larger HP inverter).
ou	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Proper deceleration time	Lengthen the deceleration time.
	Dynamic Braking not functioning properly	Connect DB braking unit resistor (optional) or check operation of if already present. Check polarity of thermal interlock connections to DB unit. Check Function Code 34; increase setting value if necessary.

NOTE: Motor *will coast-to-stop when fault condition is displayed. After obtaining Fault information and correcting the cause of the problem, reset by pressing the RESET key on the Keypad Panel, or momentarily close contact between the RST-CM control circuit terminals after the motor has stopped.*

NOTE: *if a fault cannot be reset, check to be sure that a RUN command* is *not present.*

NOTE: *"LU" may be displayed when the AC supply is switched OFF and does not indicate any abnormality.*

TABLE 8: FAULT CONDITION DISPLAY AND CORRECTIVE ACTION (continued)

Display	Check For:	Corrective Action
OH1	Inverter ambient temperature not within permissible limits	Correct to proper temperature range.
	Cooling fan(s) not operating	Replace cooling fan(s).
	Load is over permissibte limits	Reduce load. Increase inverter capacity (use larger HP inverter).
OH2	Proper wiring between THR-CM	Rewire as needed.
	External thermal overload relay activated	Check relay.
	(Continuity check between external DB braking unit terminal 1-2)	Reduce load.
	Inverter ambient temperature not within permissible limits	Correct to proper temperature range. Correct ambient temp./lower braking duty cycle.
LU	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Circuit breaker is open	Close circuit breaker.
	Open AC supply phase	Correct input wiring and check AC supply fuses.
	Other than above	Check customer power supply capacity.
OL2	Electronic thermal overload set incorrectly	Check Function Code 1 b.
	Load is over permissible limits	Reduce load.
Err1	Correct Function Code selection (Low limit frequency has been set higher than high limit frequency)	Check Function Codes 14 and 15.
Err0 Err9	Electrical noise source close to inverter	Take noise prevention measures. Check to make sure noise is not coming into terminals "11" or "CM".
Err2	A Function Code that cannot be set while running has been attempted to be set	Attempt to change only those Function Codes in TABLE 5 marked with "*2" while inverter is running.
Errd	Normal stored data settings (Stored setting data has been determined to be abnormal.)	Replace Main Control card. **NOTE:** *It may be possible to clear this condition by removing and reinserting any socketted integrated circuit chips* **on** *the Main Control Card (total of three).*

NOTE: *Motor will coast-to-stop when fault condition is displayed. After obtaining fault information and* **correcting the** *cause of the problem, reset by pressing the RESET key on the Keypad Panel or momentarily close contact between the RST-CM control circuit* **terminals** *(after the motor has stopped}.*

NOTE: *If a fault cannot be reset, check to be* **sure that** *a RUN command is notpresent.*

NOTE: *"LU" may be displayed when the AC supply is switched OFF and does not indicate any abnormality.*

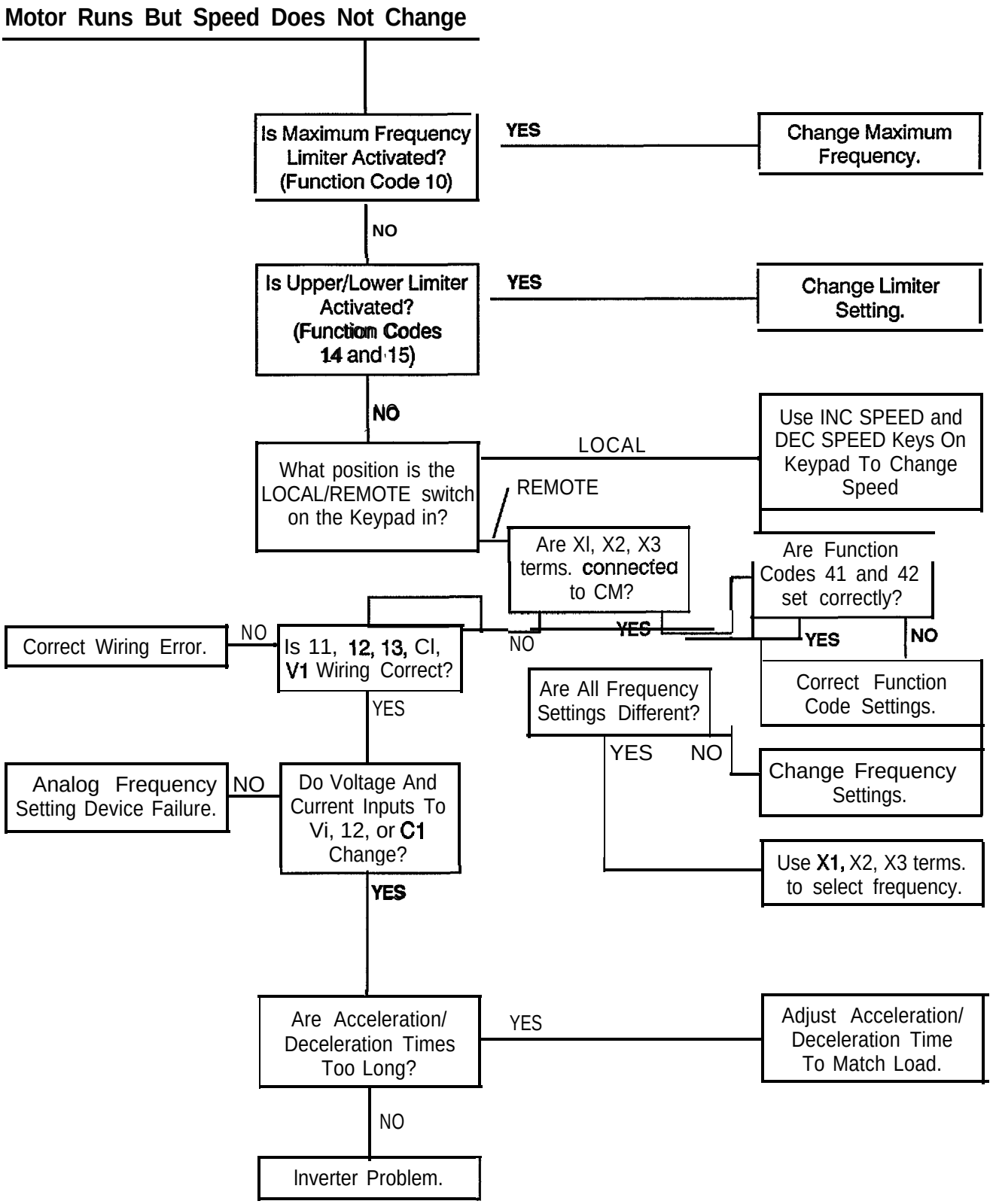


Figure 8-1. "MOTOR RUNS BUT SPEED DOES NOT CHANGE" TROUBLESHOOTING CHART

Motor Rotation Is Not Smooth

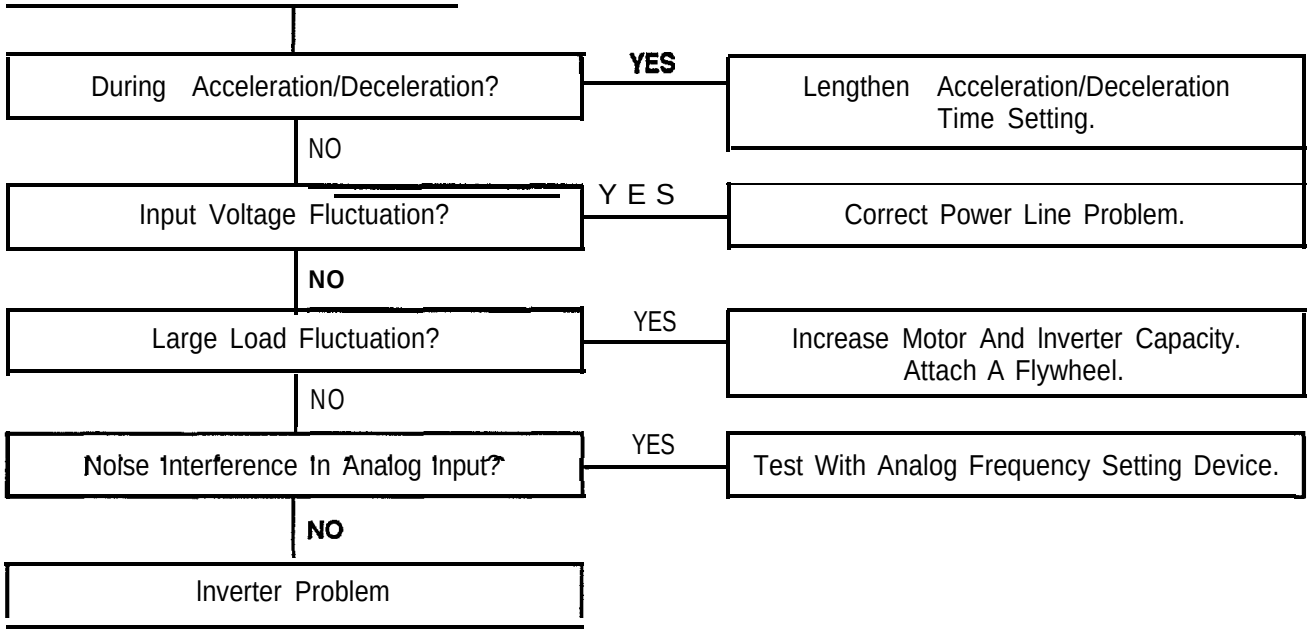


Figure 8-2. "MOTOR ROTATION IS NOT SMOOTH" TROUBLESHOOTING CHART

Motor Overheats

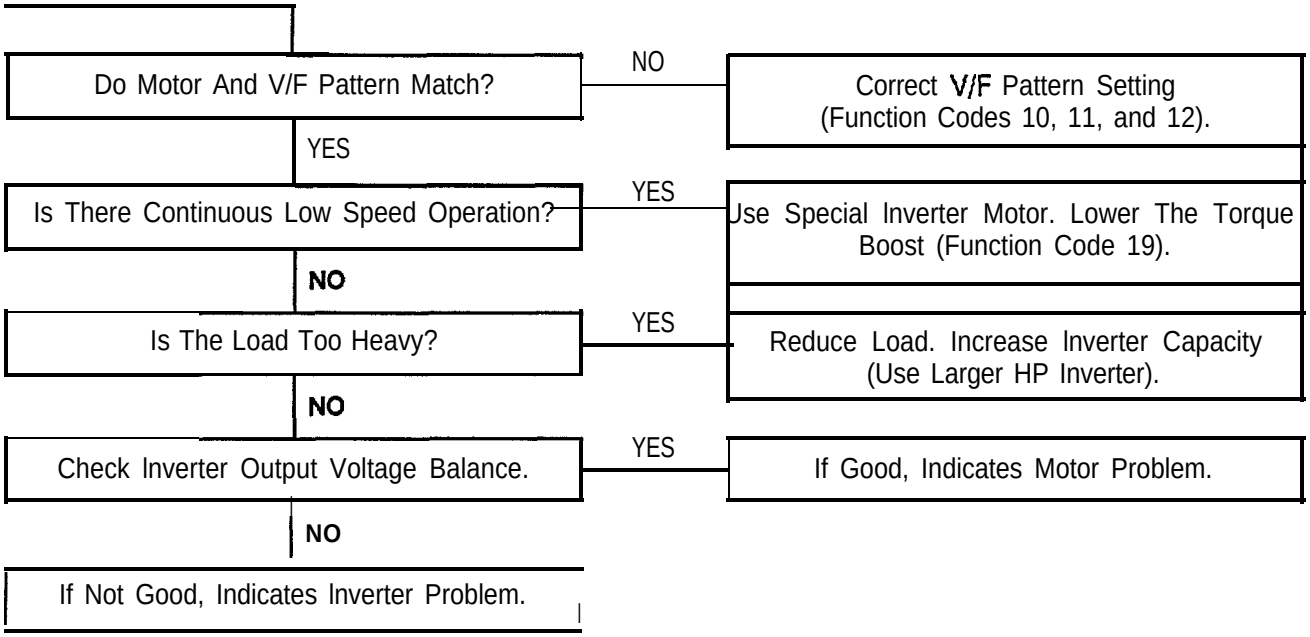


Figure 8-3. "MOTOR OVERHEATS" TROUBLESHOOTING CHART

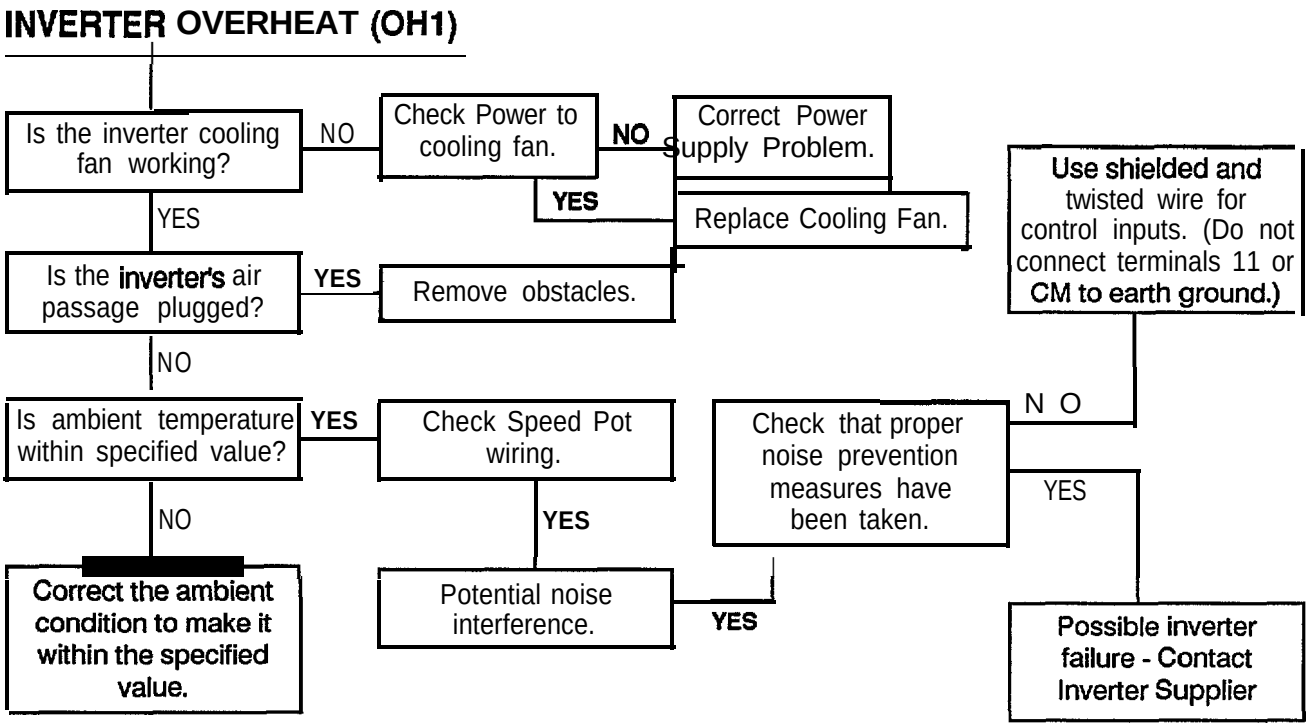


Figure 8-4. "INVERTER OVERHEATS" TROUBLESHOOTING CHART

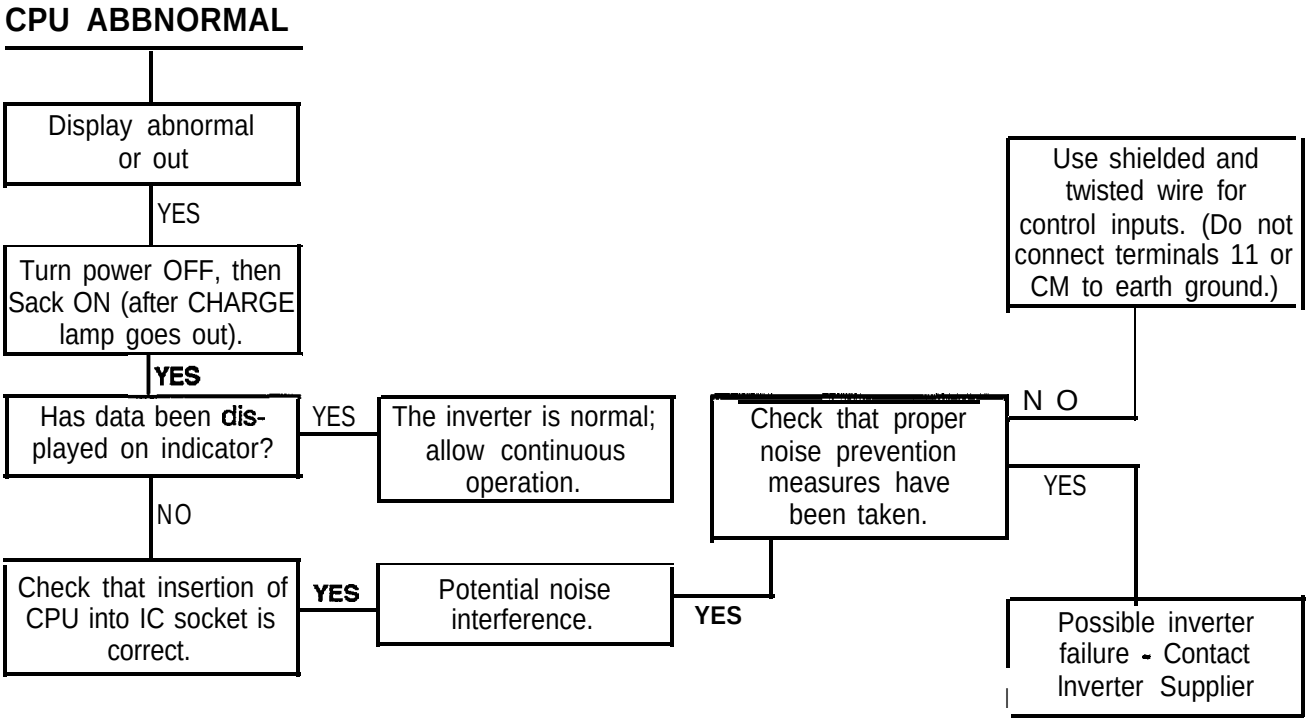


Figure 8-5. "CPU ABNORMAL" TROUBLESHOOTING CHART

EXTERNAL FAILURE (OH2)

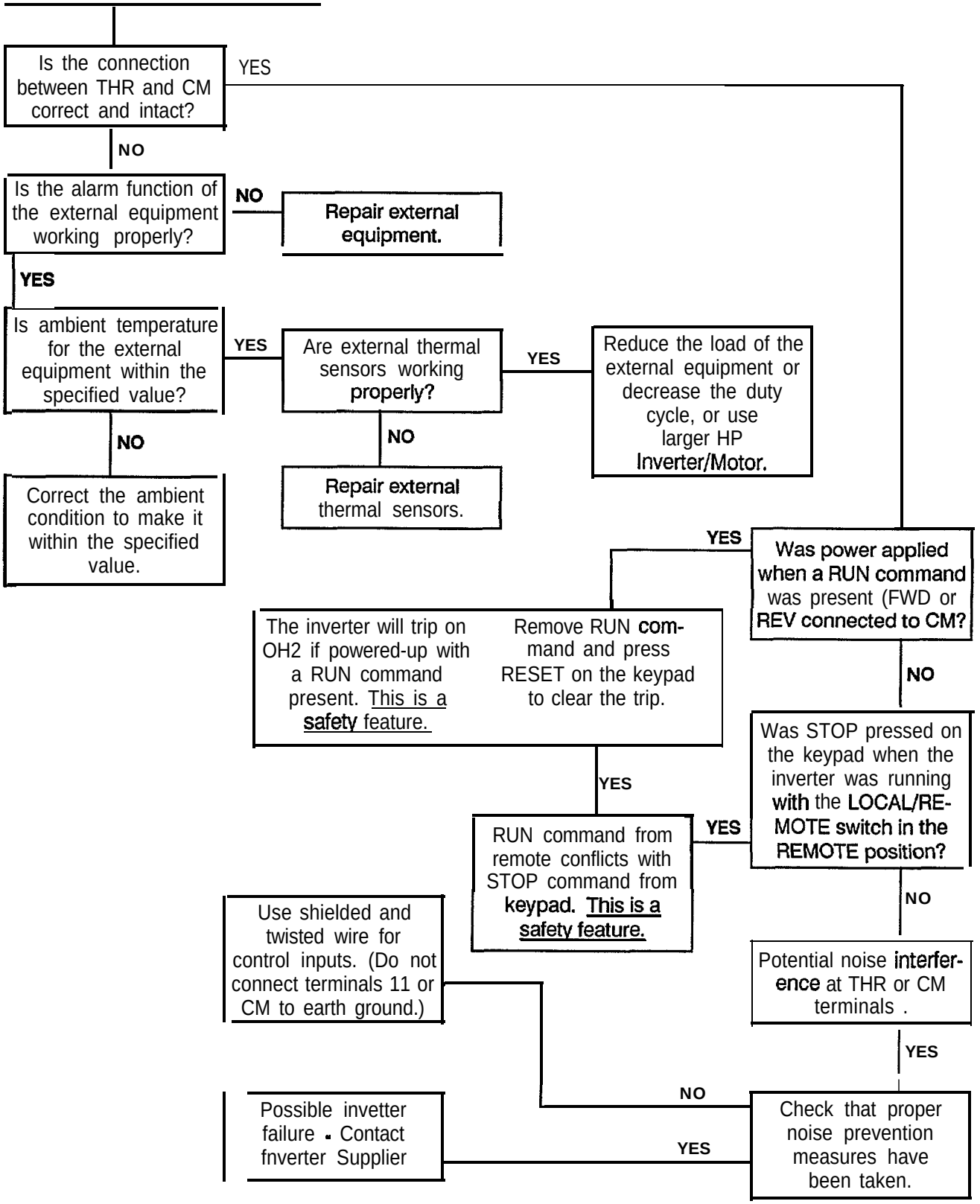


Figure 8-8. "EXTERNAL FAILURE" TROUBLESHOOTING CHART

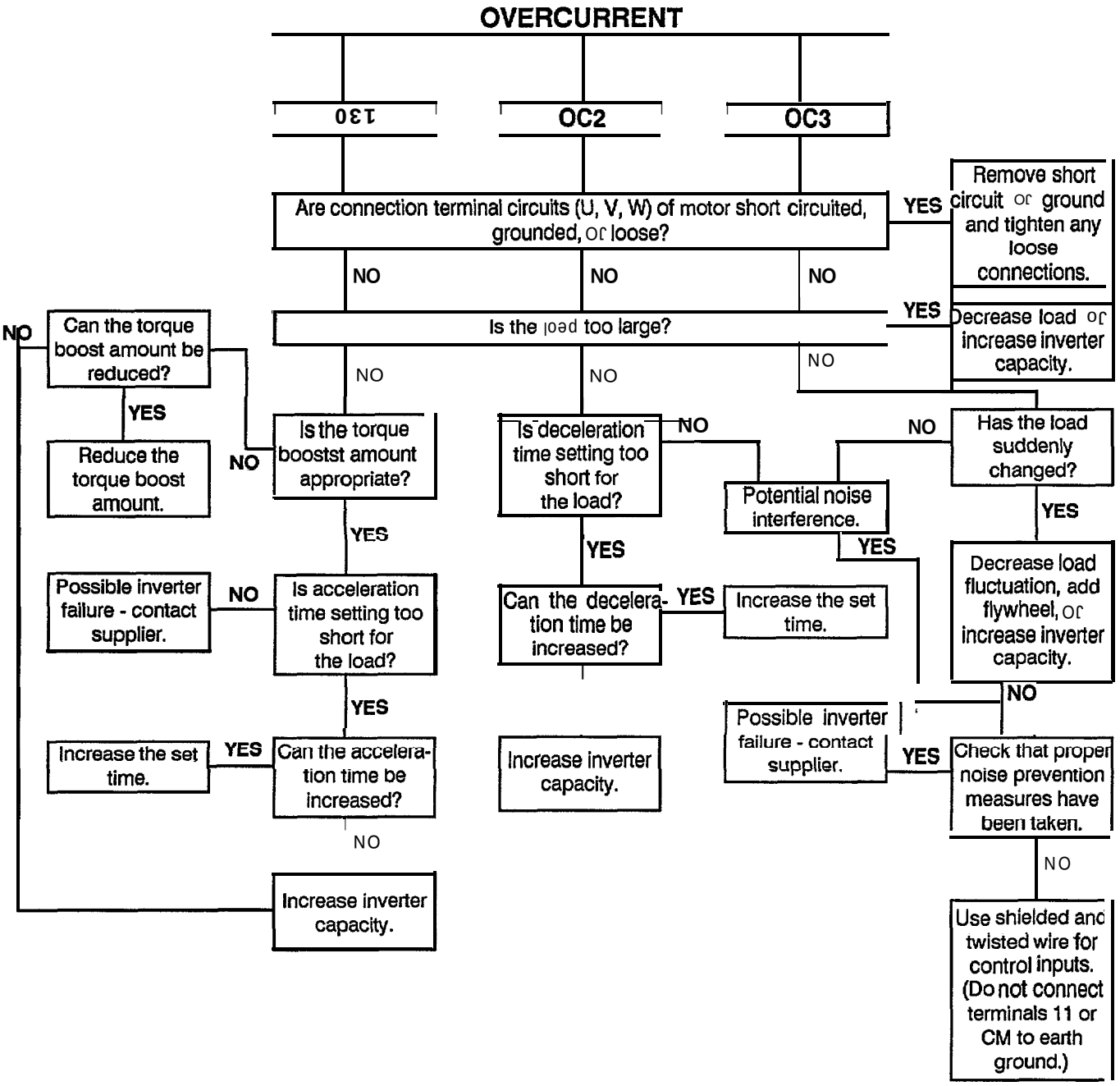
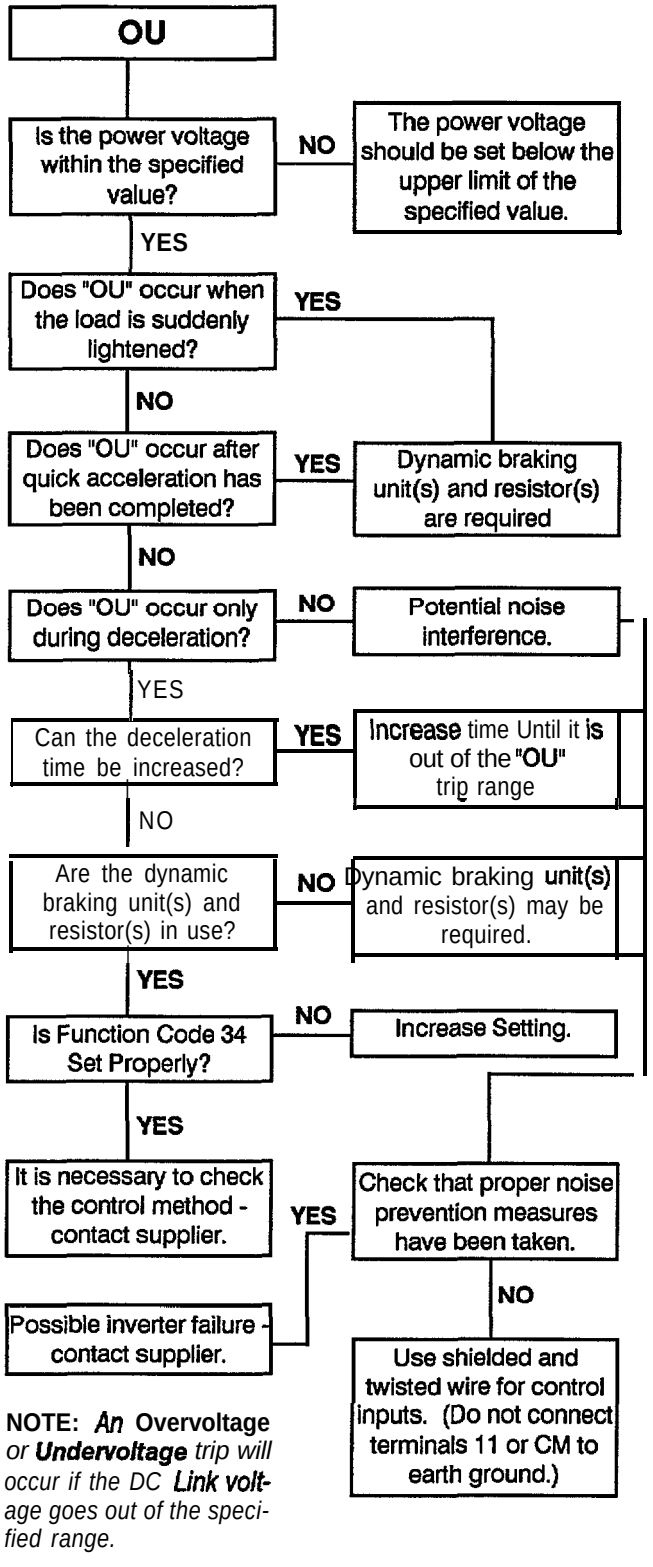


Figure 8-7. "OVERCURRENT" TROUBLESHOOTING CHART

OVERVOLTAGE



NOTE: *An* **Overvoltage** *or* ***Undervoltage*** *trip will occur if the DC* ***Link voltage*** *goes out of the specified range.*

Figure 8-8. "OVERVOLTAGE" TROUBLESHOOTING CHART

UNDERVOLTAGE

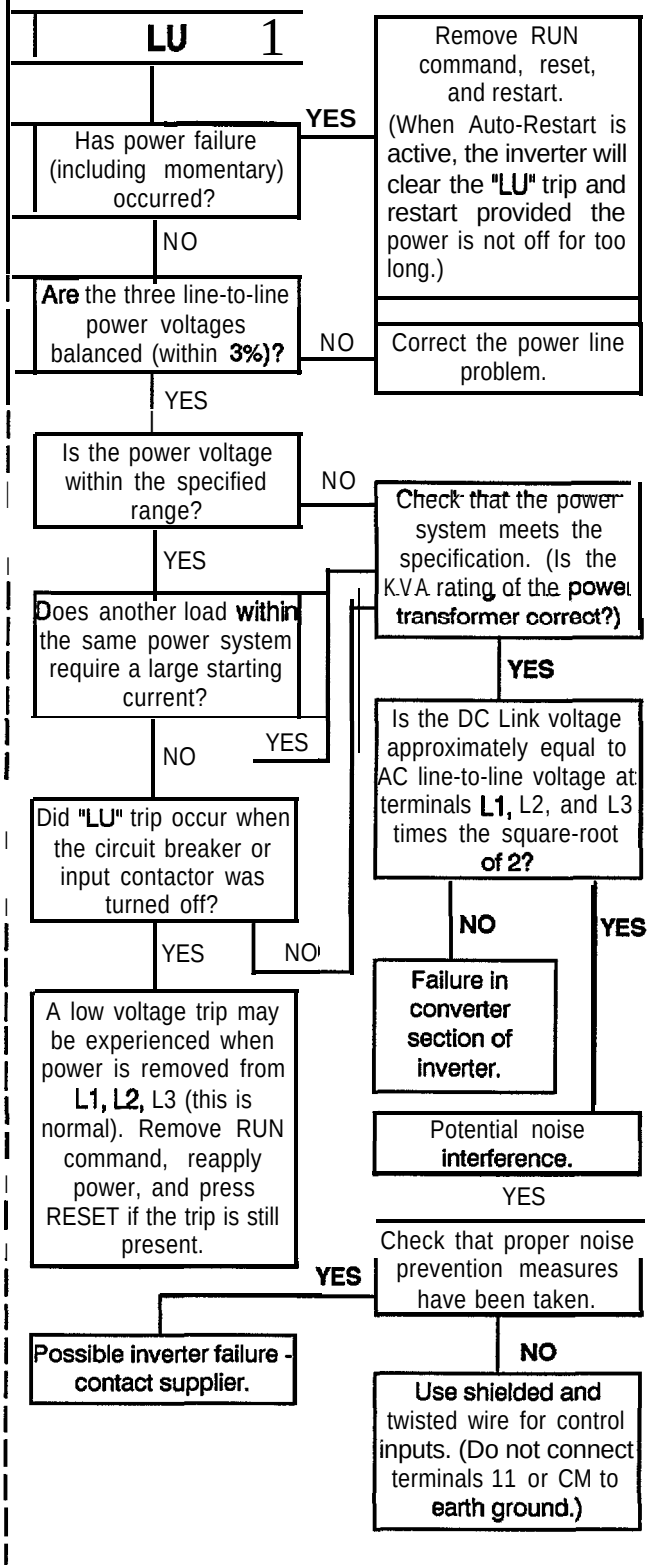


Figure 8-9. "UNDERVOLTAGE TROUBLESHOOTING CHART

OVERLOAD

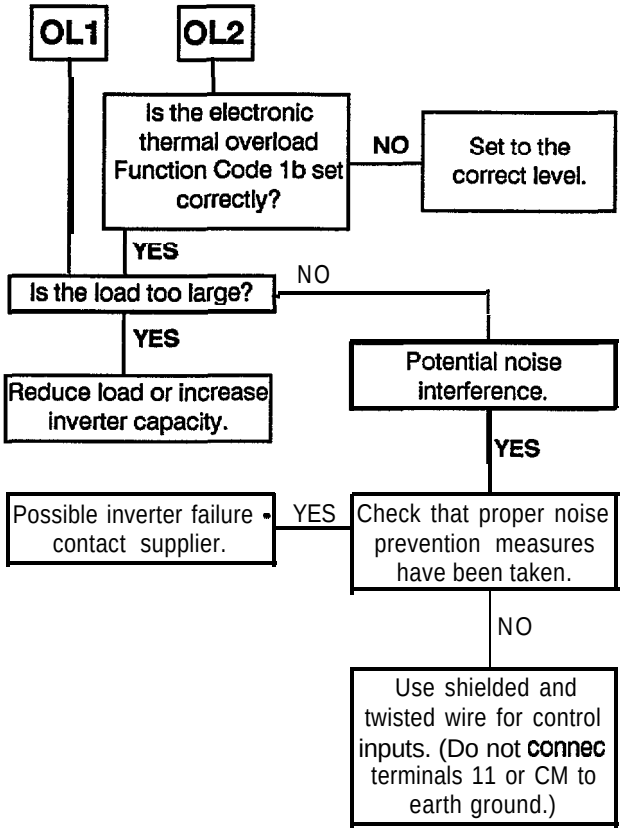


Figure 8-10. 'OVERLOAD" TROUBLESHOOTING CHART

ELECTRIC MOTOR STALLS DURING ACCELERATION

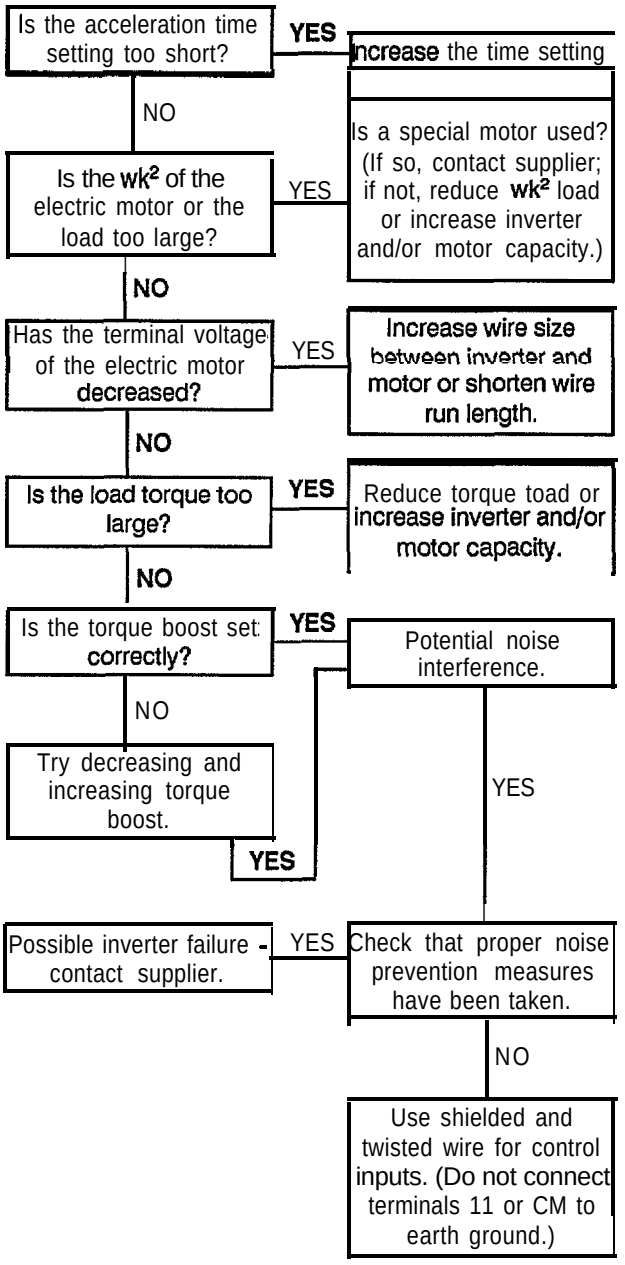


Figure 8-11. "ELECTRIC MOTOR **STALLS'** TROUBLESHOOTING CHART

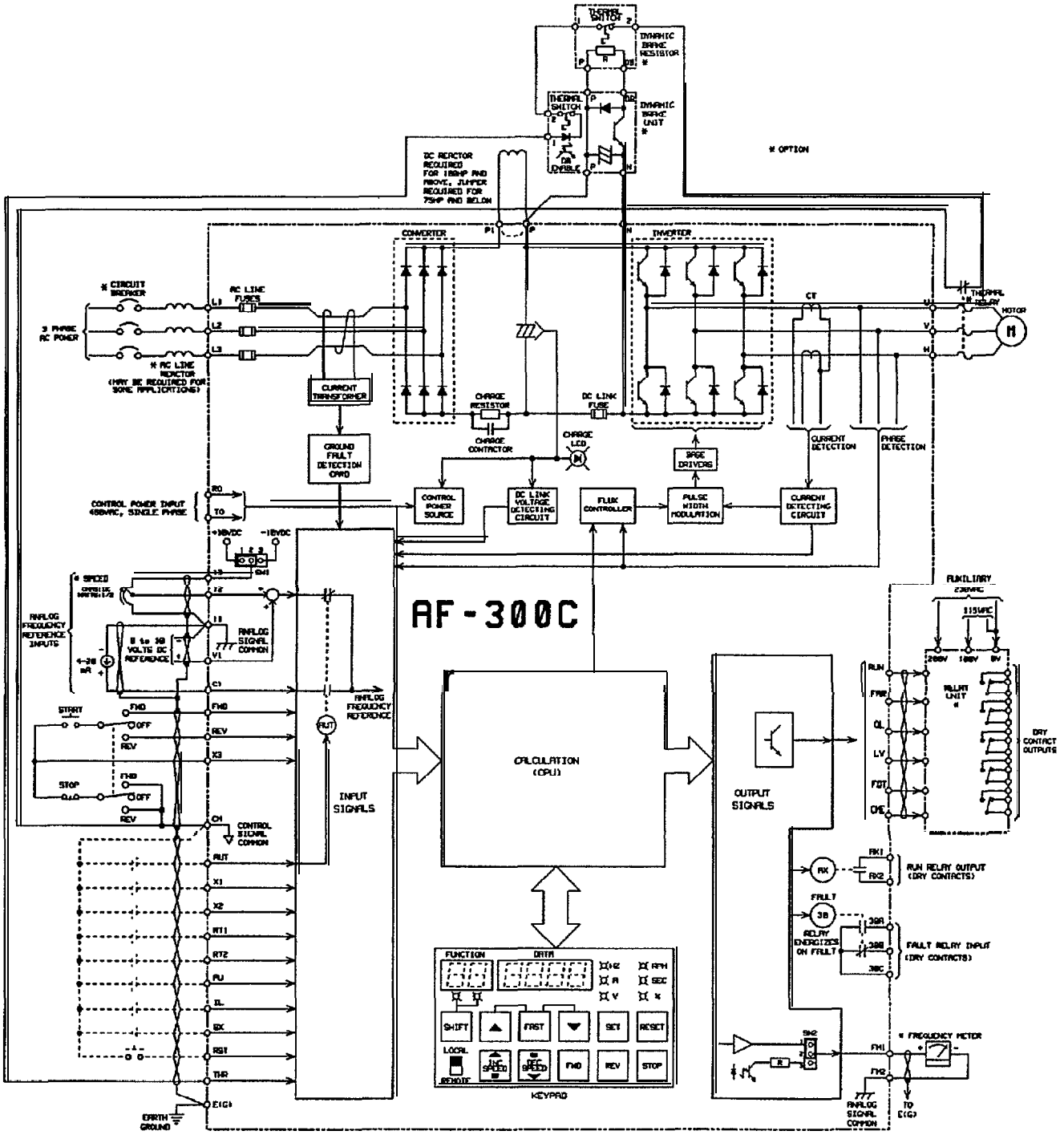


Figure 8-12. CONTROL BLOCK DIAGRAM

NOTES:

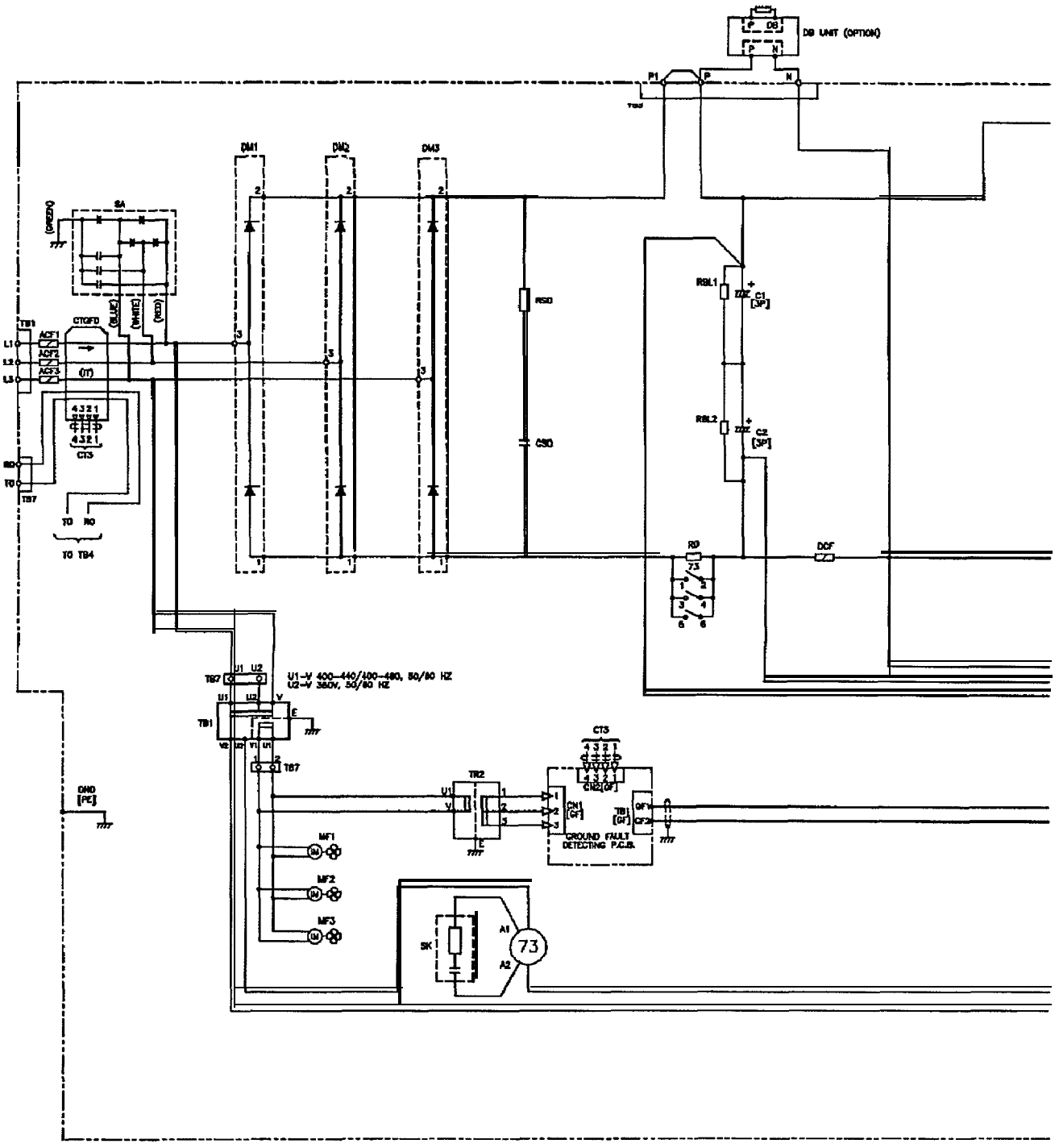


Figure 8-13 (Part 1 of 2). INTERCONNECTION DIAGRAM - 40 HP INVERTERS

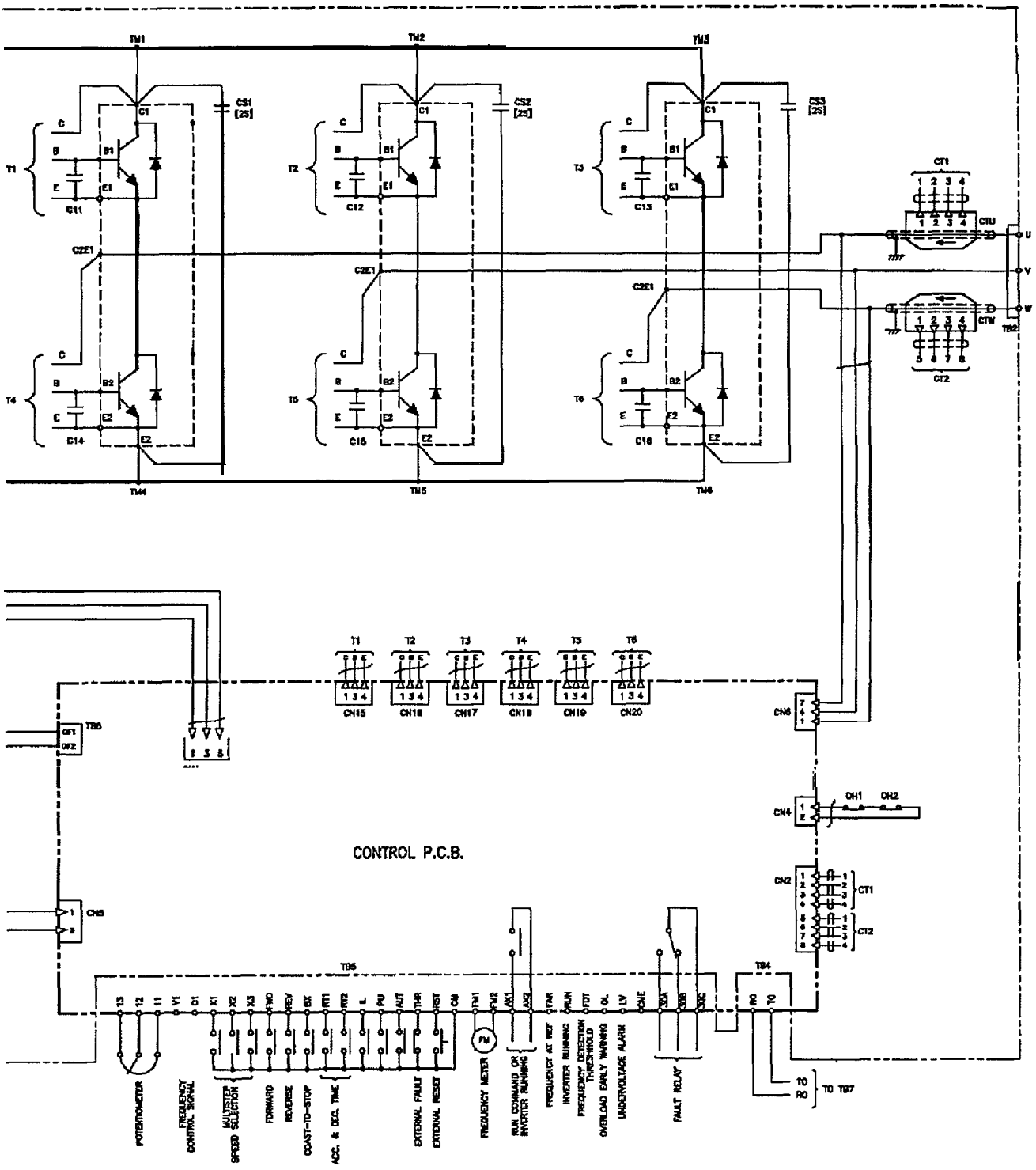


Figure 8-13 (Part 2 of 2). INTERCONNECTION DIAGRAM - 40 HP INVERTERS

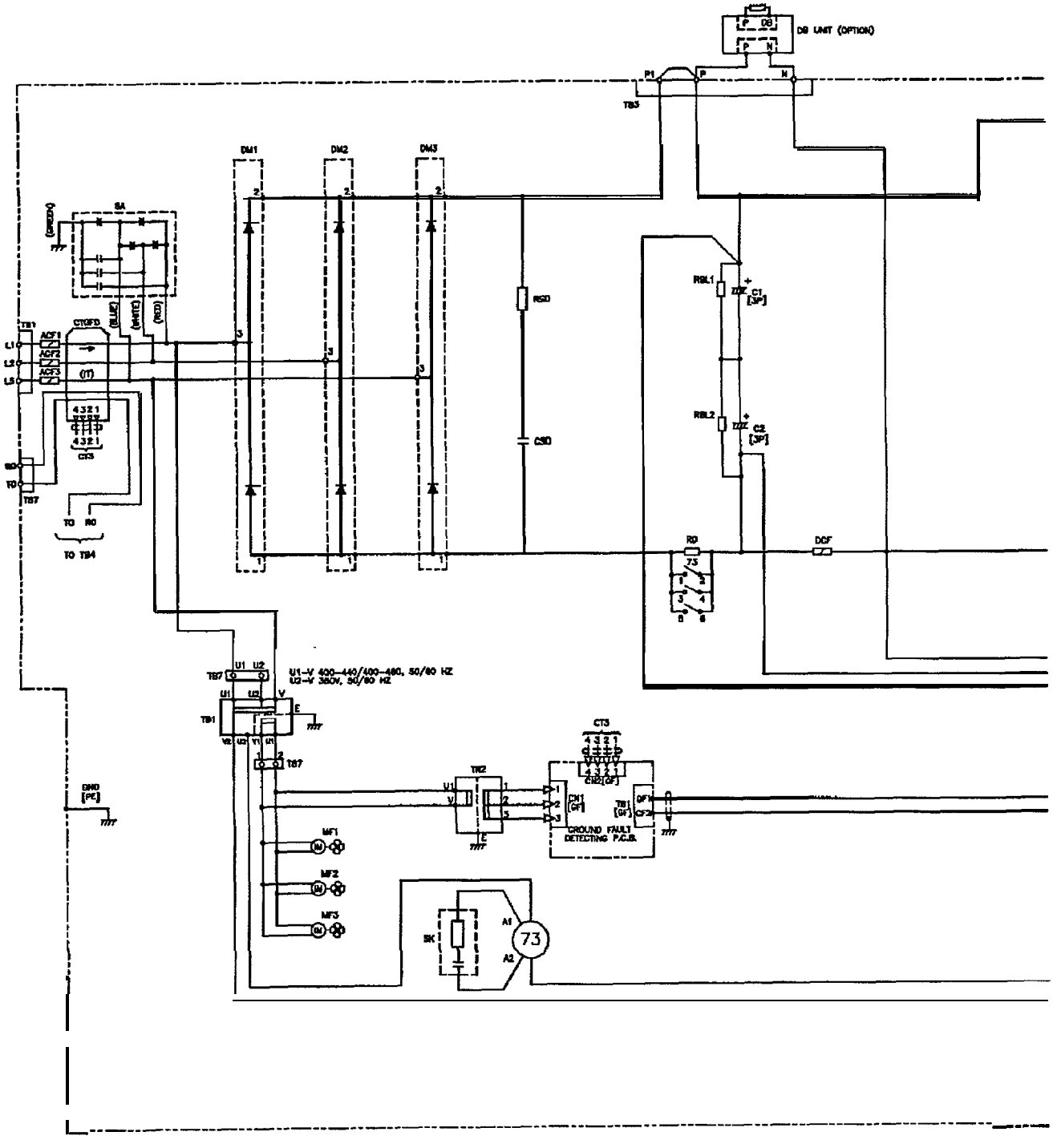


Figure 8-14 (Part 1 of 2). INTERCONNECTION DIAGRAM - 50 AND 60 HP INVERTERS

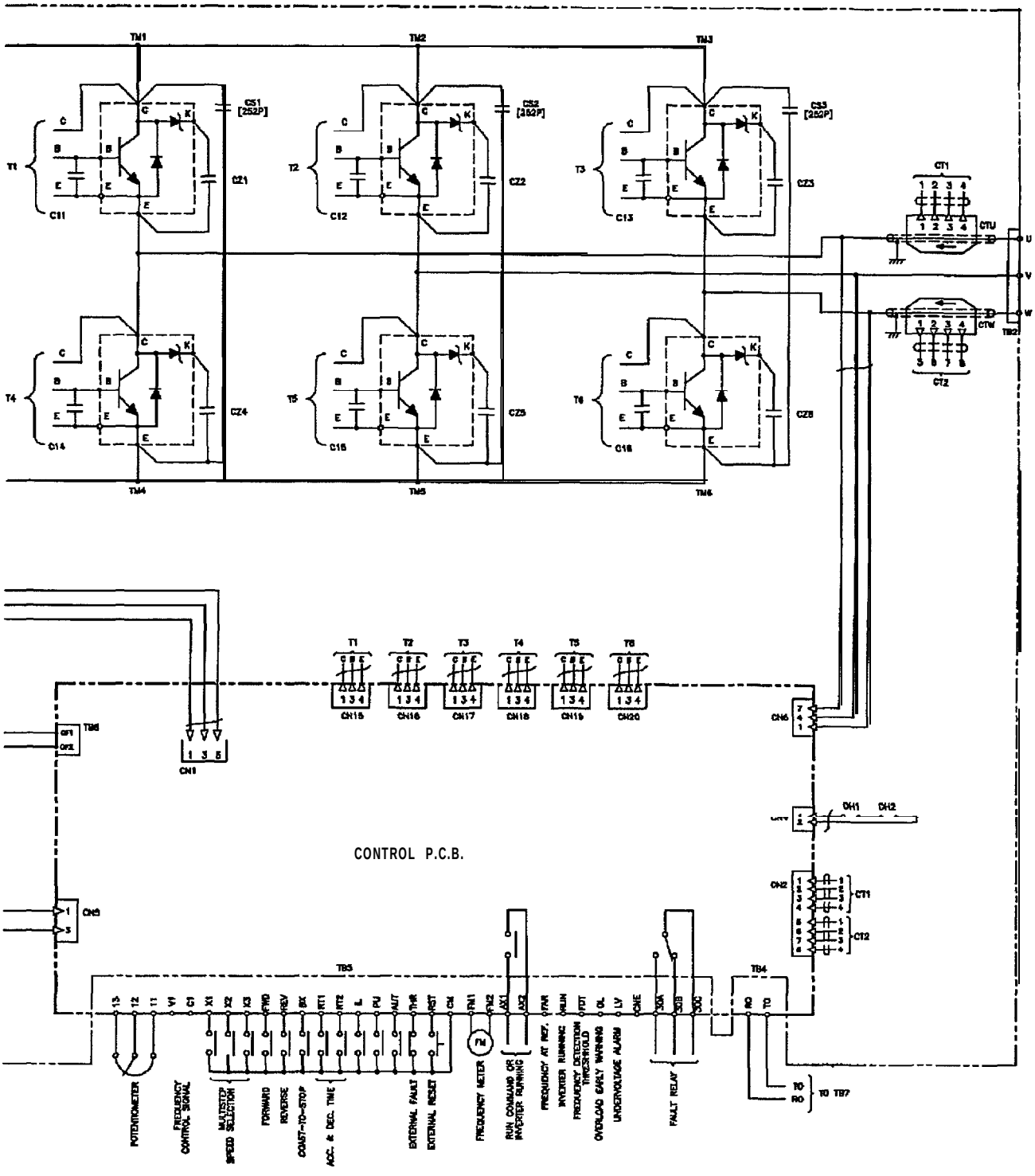


Figure 8-14 (Part 2 of 2). INTERCONNECTION DIAGRAM - 50 AND 60 HP INVERTERS

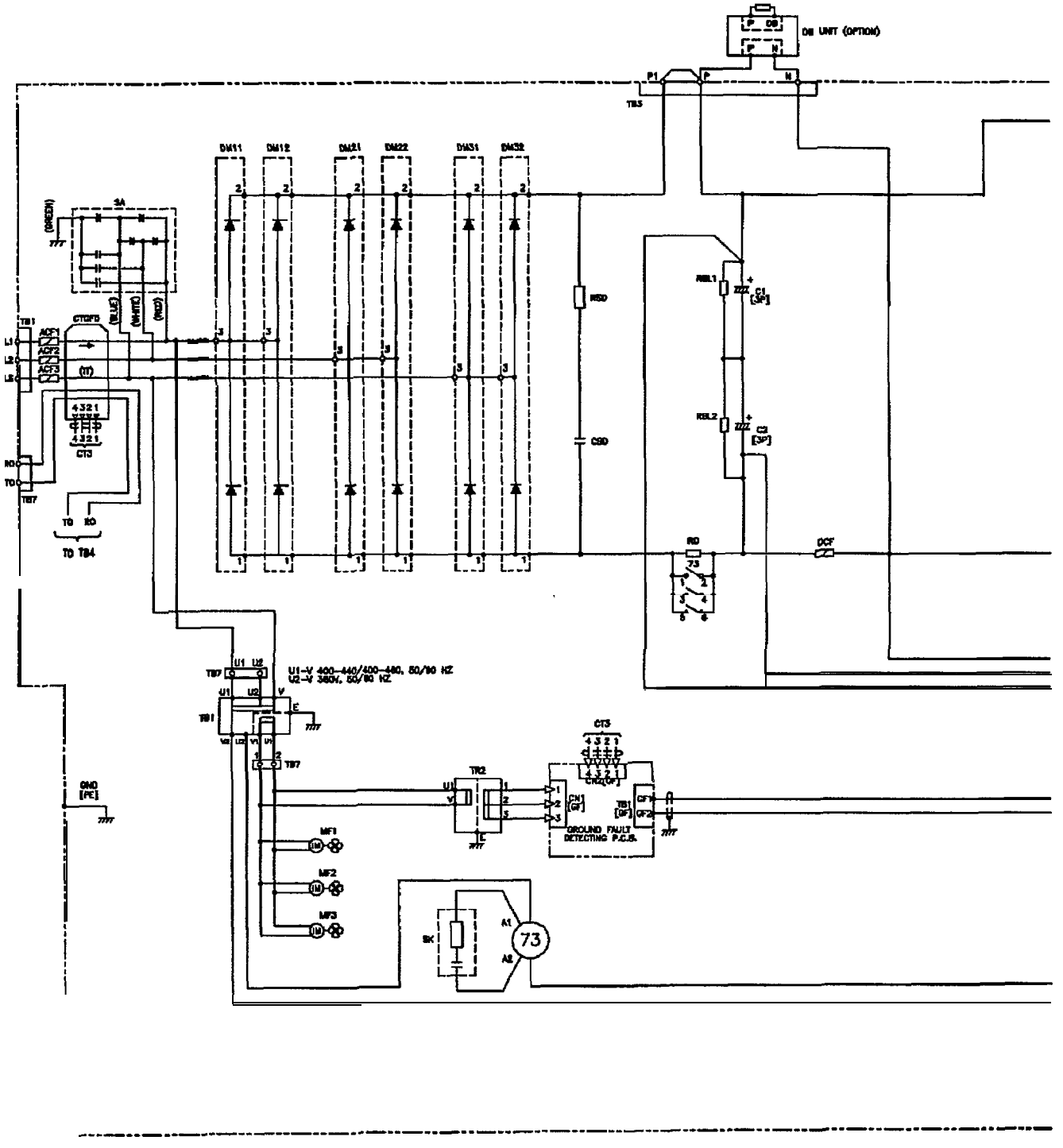


Figure 8-15 {Part 1 of 2). INTERCONNECTION DIAGRAM - 75 HP **INVERTERS**

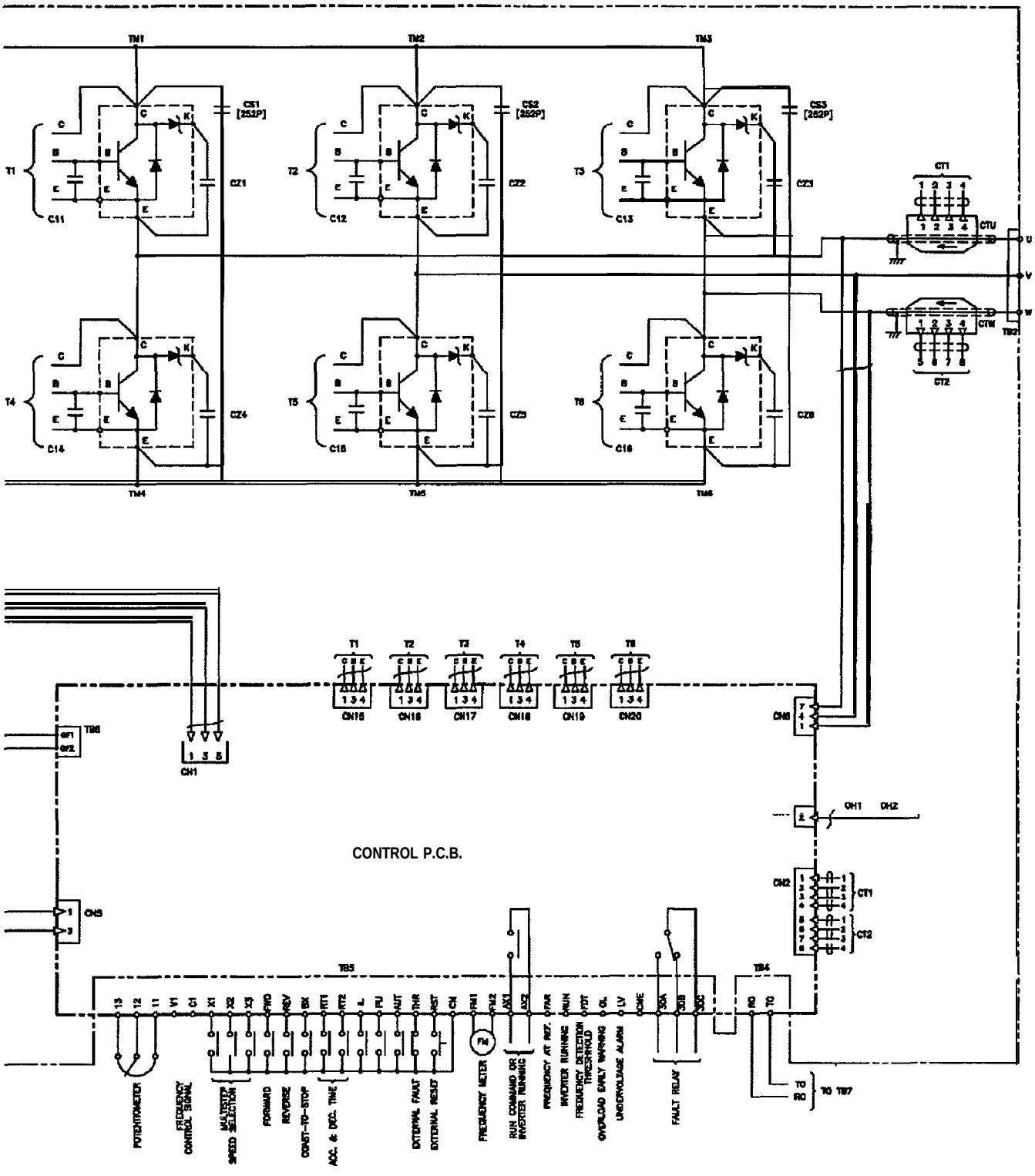


Figure 8-15 (Part 2 of 2). INTERCONNECTION DIAGRAM - 75 HP INVERTERS

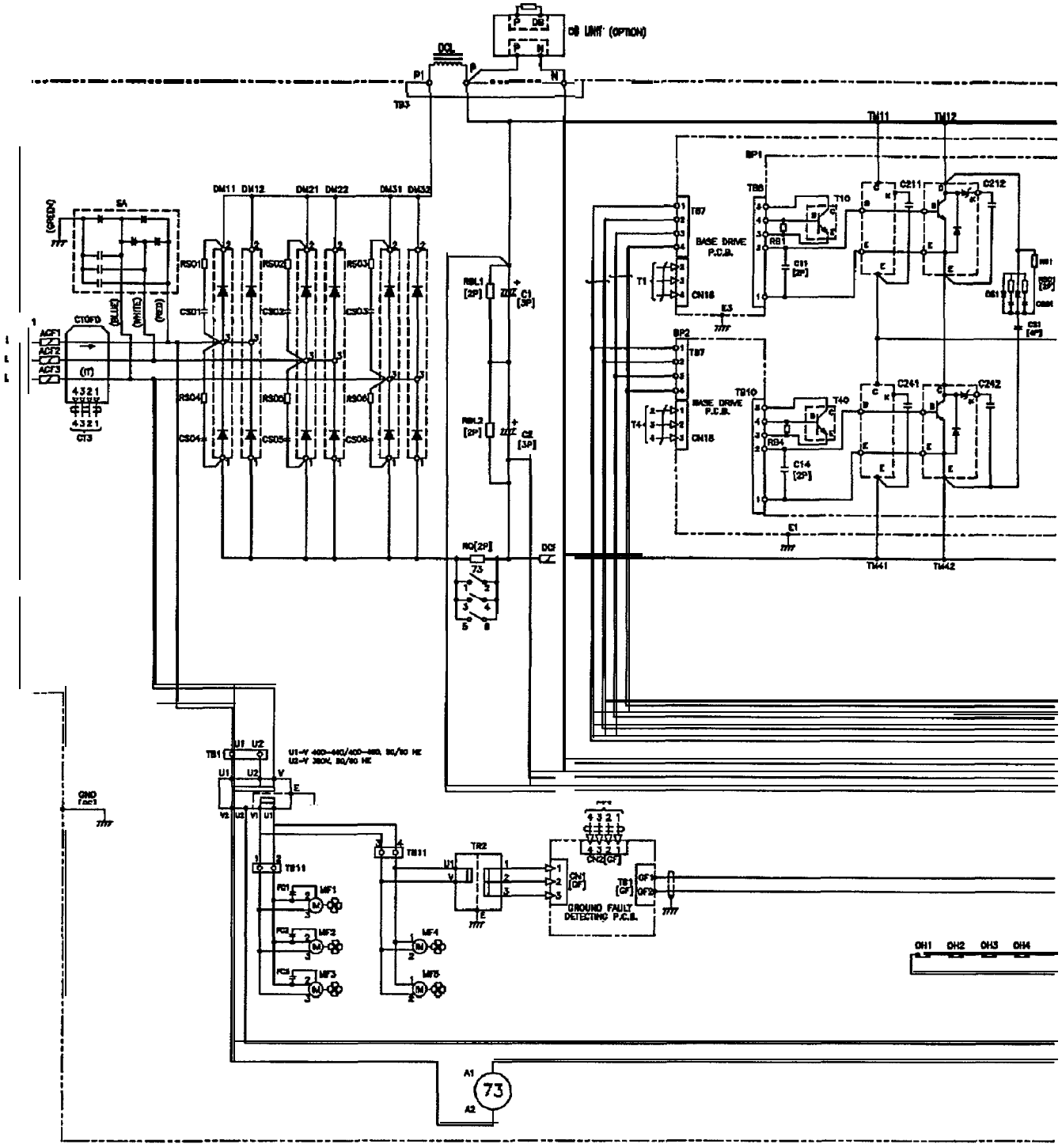


Figure 8-16 (Part 1 of 2). INTERCONNECTION DIAGRAM - 100 and 125 HP INVERTERS

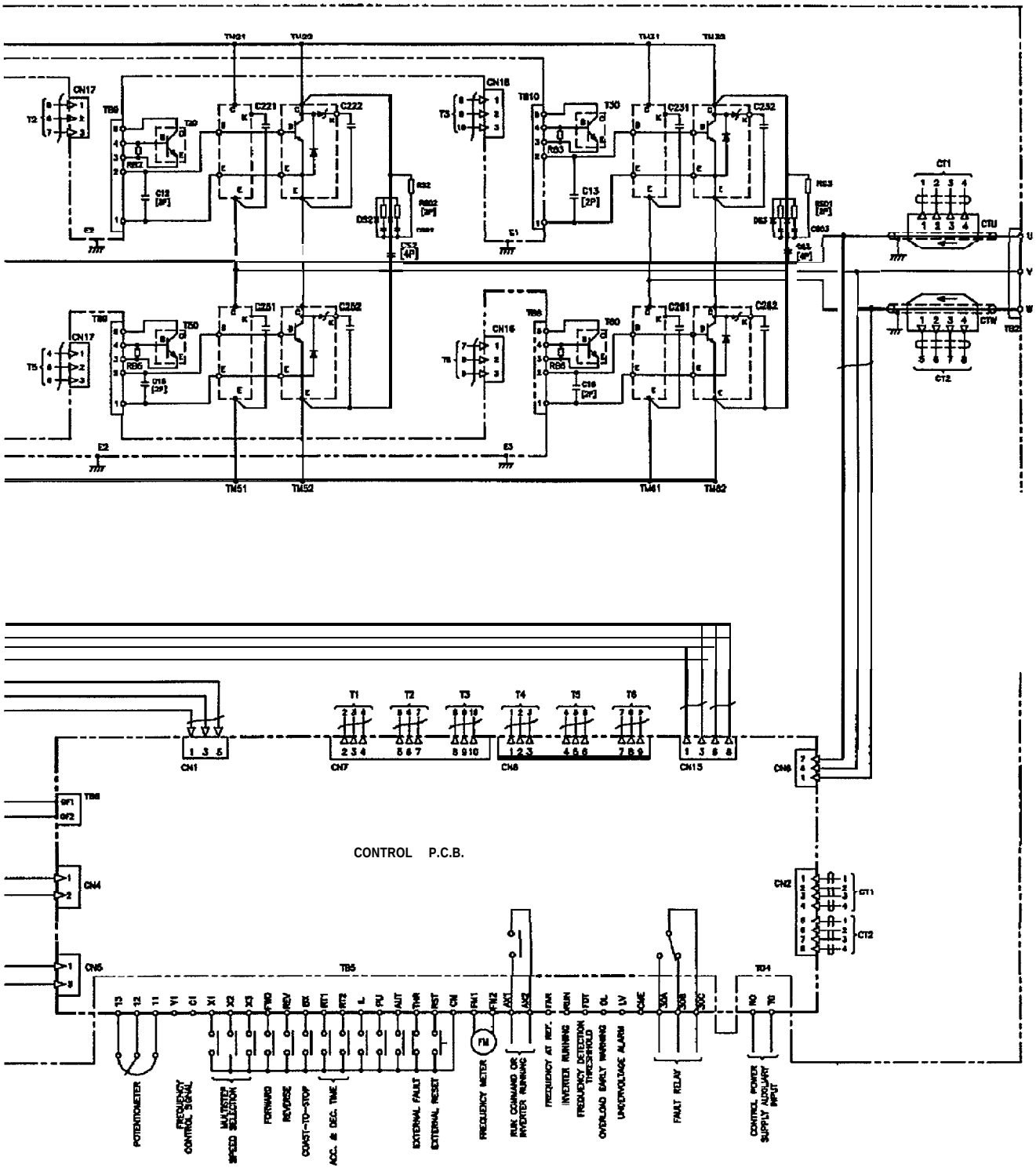


Figure 8-18 (Part 2 of 2). INTERCONNECTION DIAGRAM - 100 and 125 HP INVERTERS

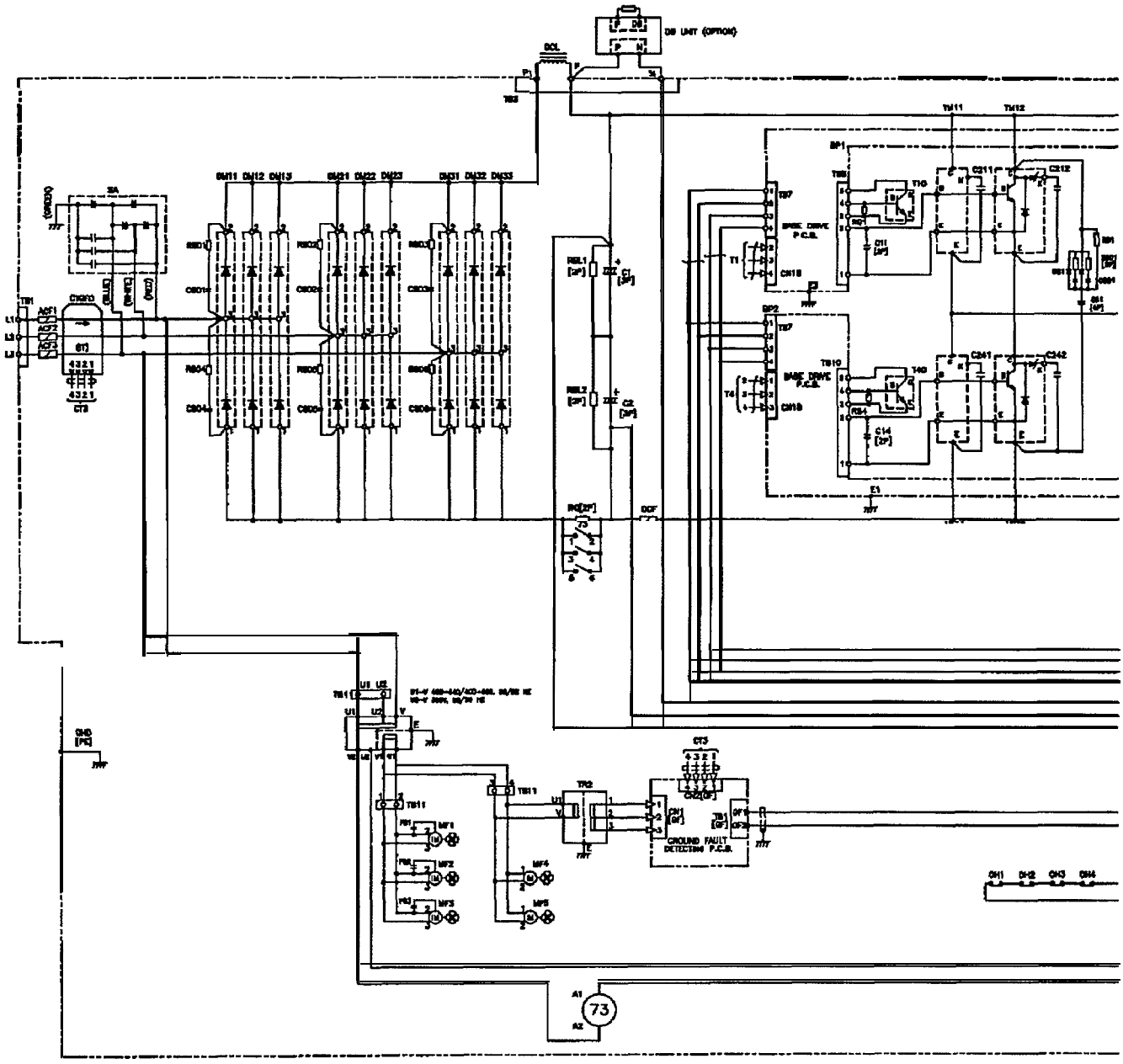


Figure 8-17 (Part 1 of 2). INTERCONNECTION DIAGRAM - 150 HP INVERTERS

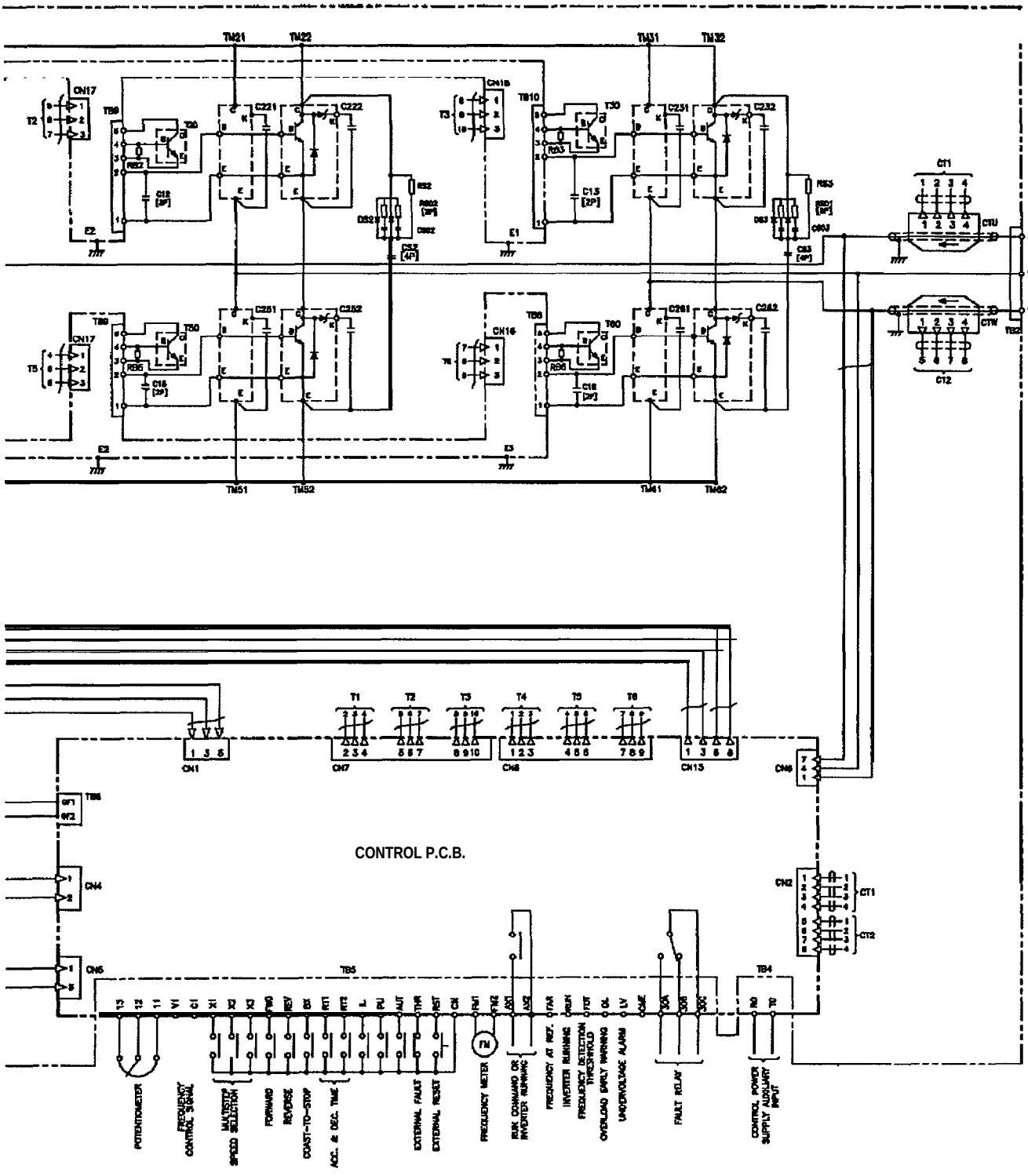


Figure 8-17 (Part 2 of 2). INTERCONNECTION DIAGRAM - 150 HP INVERTERS

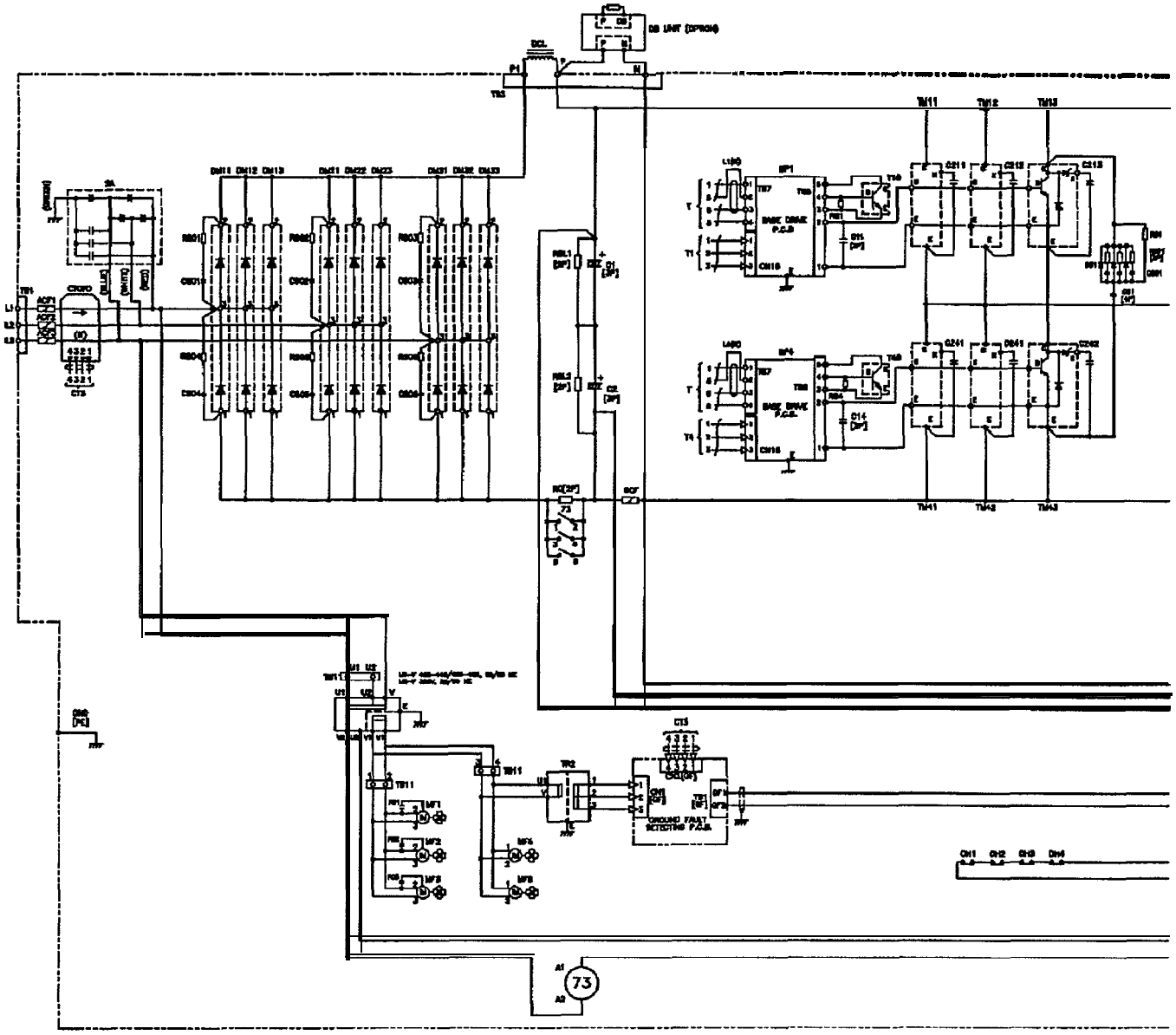


Figure 8-18 (Part 1 of 2). INTERCONNECTION DIAGRAM - 175 to 300 HP INVERTERS

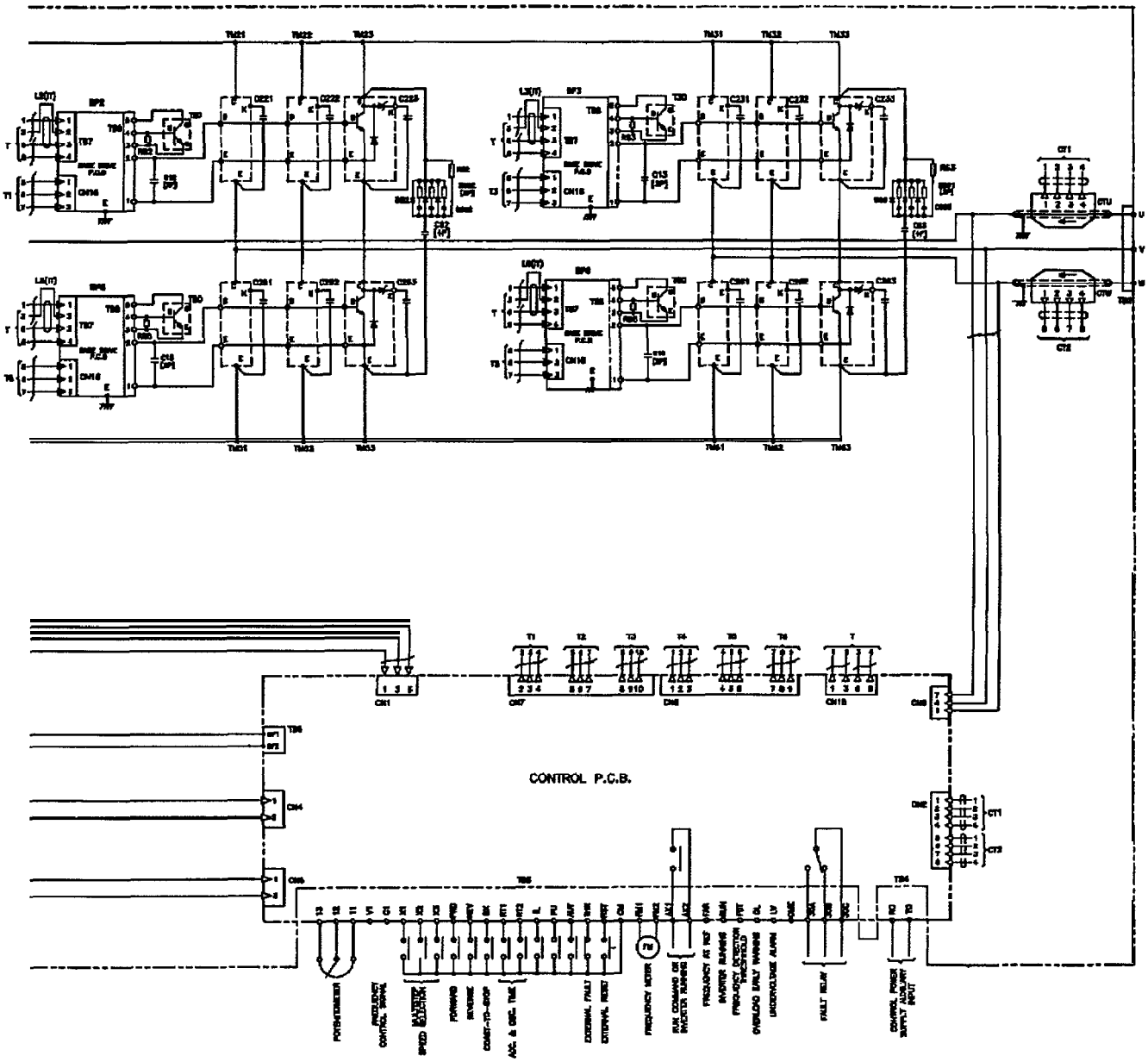


Figure 8-18 (Part 2 of 2). INTERCONNECTION DIAGRAM - 175 to 300 HP INVERTERS

NOTES:

9. WARRANTY PARTS AND SERVICE

The purpose of the following is to provide specific instructions to the user of the AF300C inverter regarding Warranty administration and how to obtain assistance on both in-warranty and out-of-warranty equipment.

For all warranty procedures, refer to Section 10 of this Instruction Book to identify the pan or assembly.

If assistance is required to either determine warranty status, identify defective pans, or obtain name of local distributor, call:

GE Drive Systems
Salem, VA
(703) 387-7595
(8 AM to 5PM, Mon. thru Fri.)

WARRANTY COVERAGE

The Warranty covers all major parts of the inverter such as the main printed circuit boards, transistor modules, etc. The warranty does not cover replacement of fuses or of the entire controller.

When the inverter is under warranty, information per the "In-Warranty-Failure Checklist" in this Instruction Book is required.

OUT-OF-WARRANTY PROCEDURES

When the defective part has been identified, contact your local Authorized AF-300C Inverter Distributor to order replacement parts.

Before calling the above number to determine warranty status, the inverter serial number will be required. This is located on the inverter nameplate. (If the inverter is still under warranty, further information will be required per the "In-Warranty Failure Checklist".)

MOTORS

Motor reoairs on GE motors are aenerallv handled bv GE Small Motor Service Centers or GE Apparatus Service Shops. For specific instructions on your motor, call GE Drive Products Operation at the phone number listed previously and furnish complete nameplate data.

IN-WARRANTY FAILURE CHECKLIST

To assist with warranty troubleshooting, the following information is required. This data is needed to evaluate the cause in an effort to eliminate any further failures.

Model No.: _____

Serial No.: _____

Start-Up Date: _____

Failure Date: _____

Status When Failure Occurred (check one):

Power-Up _____ Running Accel _____ Decel _____

Application Information (check Yes or No)

Input Transformer: **Yes** _____ **No** _____

If Yes: KVA _____

L1 Volts _____ L2 Volts _____ L3 Volts _____

Power Factor Correction Capacitors: Yes _____ **No** _____

If Yes: Microfarrad _____

IN-WARRANTY FAILURE CHECKLIST (continued)

Application Information [continued]! (check Yes or No)

Other Equipment on Same Power **Yes** _____ **No** _____

if Yes, what?

Line Reactor on input **Yes** _____ **No** _____

DC Reactor **Yes** _____ **No** _____

Input Starter **Yes** _____ **No** _____

Output Starter **Yes** _____ **No** _____

Motor Overloads **Yes** _____ **No** _____

Control Terminals Used (circle if used)

LV	OL	FAR	RUN	FDT	RT1	RT2	THR	X1	X2
x3	30A	30B	3oc	RST	BX	FWD	REV	C1	11
12	13	vi	FM1	FM2	RO	TO	AUT	IL	PU

Function Codes Different From Factory Settings

Function Code	Setting	Function Code	Setting

Failure Message (see Section 5)

Latest Fault _____ **Previous Faults:** **No Message** _____

Hz _____ **1.** _____

A _____ **2.** _____

V _____ **3.** _____

After ail of the Checklist information is acquired, contact the following number for assistance:

(703) 387-7595

When returning failed parts, reference the C- - e - m - - that the replacement parts were received on and ship failed parts to:

GE Drive Systems
1501 Roanoke Boulevard
Salem, VA 24153
(Marked C_ _ _ _ _ _ _)

10. REPLACEMENT PARTS

Refer to the Following Tables when ordering parts per the procedures outlined in Section 9 of this Instruction Book. Table is arranged per Inverter Horsepower (TABLE 9: Inverter Renewal Parts). First portion of table lists parts for 40 to 100 HP Inverters and second portion of table lists parts for 125 to 300 HP Inverters.

TABLE 9: AF-300C INVERTER RENEWAL PARTS (40 - 100 HP)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter				
			40	50	60	75	100
Power Transistor	150 Amp, 1200 PIV	2DI150Z-120	3				
Power Transistor	200 Amo. 1200 PIV	1DI200ZP-120-1		3			6
Power Transistor	200 Amp, 1200 PIV	1DI200ZN-120-1		3			6
Power Transistor	300 Amp, 1200 PIV	1DI300ZP-120-02			3	3	
Power Transistor	300 Amo. 1200 PIV	1DI300ZN-120-02			3	3	
Drive Transistor	30 Amp, 600 PIV	ETG81-050					6
Rectifier Module	60 Amp, 1600 PIV	DD60HB-160	3				
Rectifier Module	100 Amp, 1600 PIV	DD100HB-160		3	3	6	6
Snubber Diode	60 Amp, 1200 PIV	1F60A-129F					6
Snubber Capacitor	1 uF, 600 VDC	CP711A	1	1	1	1	6
Snubber Resistor	60 watt, 20 ohm	GHZ60AP	1				
Snubber Resistor	60 watt, 20 ohm	GZG60AP		1	1	1	
Snubber Resistor	20 watt, 20 ohm	GHZ20AP					6
Link Capacitor	3900 uF, 400 VDC	HF574458C-3900		6			
Link Capacitor	4700 uF, 400 VDC	HF574458C-4700	4		6		
Link Capacitor	5600 uF, 400 VDC	LNR2G562MSMAFF				6	
Link Capacitor	6800 uF, 400 VDC	HF581597					6
Charge Resistor	120 watt, 20 ohm	GZG120	1	1			2
Charge Resistor	200 watt, 20 ohm	GZG200			1	1	
Magnetic Cont.		SC-2N/UL 2A2B	1	1			
Magnetic Cont.		SC-3N/UL 2A2B			1	1	
Magnetic Cont.		SC-4N/UL 2A2B					1
DC Link Fuse	150 Amp, 600 VDC	CR6L-150/UL	1	1			
DC Link Fuse	200 Amp, 600 VDC	CR6L-200/UL			1		
DC Link Fuse	300 Amp, 600 VDC	CR6L-300/UL				1	
DC Link Fuse	300 Amo. 700 VDC	A70P-300					1
AC Line Fuse	150 Amp, 500 VAC	A50P150-4	3				

TABLE 9: AF-300C INVERTER RENEWAL PARTS (40 - 100 HP) (cont.)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter				
			40	50	60	75	100
AC Line Fuse	175Amp, 500 VAC	A50P175-4		3			
AC Line Fuse	225 Amp, 500 VAC	A50P225-4			3		
AC Line Fuse	300 Amp, 500 VAC	A50P300-4				3	3
Cont.Power Trans.	1.55 VA	HF569477	1	1	1	1	1
Cont. Power Trans.	147 VA	HF587479	1	1	1	1	
Cont. Power Trans.	185 VA	HF586561					1
Balance Resistor	150 Watt, 2.2K ohm	GZG150	2				4
Balance Resistor	150 Watt, 2.2K ohm	GHZ150			4	4	
Balance Resistor	200 Watt, 2.2K ohm	GZG200		2			
Cooling Fan	Non-UL Listed	U12CM	3	3	1	1	2
Cooling Fan	Non-UL Listed	2750M			2	2	
Cooling Fan	UL Listed	6250MG1					3
Cooling Fan	UL Listed	4715PS-22T-B30	3	3	1	1	2
Cooling Fan	UL Listed	2750MTP-15			2	2	
Surge Absorber	500 V	VCR2521	1	1	1	1	1
Main Control Card		EP3038_ -C-1040	1				
Main Control Card		EP3038_ -C-2050		1			
Main Control Card		EP3038_ -C-3060			1		
Main Control Card		EP3038_ -C-3075				1	
Main Control Card		EP3090_ -C-1100					1
Ground Fault Card		EP3077_ -C-2	1	1	1	1	1
Base Drive Card		EP3091_ -C-1					2
Keypad PC Card		6VKC3SK	1	1	1	1	1

NOTE: *All parts not specifically shown as Non-UL Listed are UL Listed.*

TABLE 9: AF-300C INVERTER RENEWAL PARTS (125 - 300 HP) (cont.)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter					
			125	150	175	200	250	300
Power Transistor	300 Amp, 1200 PIV	1DI300ZP-120-02	6	6	9	9	12	12
Power Transistor	300 Amp, 1200 PIV	1DI300ZN-120-02	6	6	9	9	12	12
Drive Transistor	30 Amp, 600 V	ETG81-050	6	6				
Drive Transistor	50 Amp, 600 V	ETK81-050			6	6	6	6
Rectifier Module	100 Amp, 1690 V	DD100HB-160	6	9	9	12	12	15
Snubber Diode	60 Amp, 1200 V	1F60A-129F	6	6	9	9	12	12
Snubber Capacitor	1 uF, 1600 V	CP711A	6	6	6	6	6	6
Snubber Resistor	20 Watt, 20 Ohm	GHZ20AP	6	6	6	6	6	6
Link Capacitor	6800 uF, 400 VDC	HF581597	6		8			
Link Capacitor	10000 uF, 400 VDC	HF574458C-1000		6		10	14	14
Charge Resistor	120 Watt, 20 Ohm	GZG120	2					
Charge Resistor	200 Watt, 20 Ohm	GZG200		2	2			
Charge Resistor	600 Watt, 5 Ohm	GRZG400				1	1	1
Magnetic Cont.		SC-4N/UL 2A2B	1	1				
Magnetic Cont.		SC-5N/UL 2A2B			1			
Magnetic Cont.		SC-8N/UL 2A2B				1	1	1
Interposing Relay		SRC50-2F/X				1	1	1
DC Link Fuse	300 Amp, 600 V	CR6L-300/UL				3		
DC Link Fuse*	350 Amp, 600 V	CR6L-350					3	3
DC Link Fuse*	400 Amp, 600 V	CR6L-400	1	1				
DC Link Fuse*	600 Amp, 600 V	CR6L-600			1			
DC Link Fuse**	400 Amp, 700 V	A70P-400	1	1				
DC Link Fuse**	500 Amp, 700 V	A70P-500			1			
AC Line Fuse	300 Amp. 500 V	A50P300-4	3					
AC Line Fuse	400 Amp, 500 V	A50P400-4		3				
AC Line Fuse	500 Amp, 500 V	A50P500-4			3			
AC Line Fuse	600 Amp, 500 V	A50P600-4				3		
AC Line Fuse	700 Amp, 500 V	A50P700-4					3	3
Cont. Power Trans.	1.55 VA	HF569477	1	1	1	1	1	1
Cont. Power Trans.	185 VA	HF586561	1	1	1			
Cont. Power Trans.	300 VA	HF587480				1	1	1

* *These Fuses are* ***not UL*** *Listed.*
** *These fuses are* ***UL*** *Listed.*

TABLE 9: AF-300C INVERTER RENEWAL PARTS (125 - 300 HP) (cont.)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter					
			125	150	175	200	250	300
Balance Resistor	150 Watt, 1.5K ohm	GZG150	4					
Balance Resistor	200 Watt, 1.5K ohm	GZG200		4	14			
Balance Resistor	400 Watt, 98 Ohm	GHZ400-96				8		
Balance Resistor	400 Watt, 192 Ohm	GHZ400					12	12
Cooling Fan	Non-UL Listed	U12CM	2	2	2	3	4	4
Cooling Fan	UL-Listed	4715PS-22T-B30	2	2	2	3	4	4
Cooling Fan	Non-UL Listed	6250MG1	3	3	3	4	5	5
Surge Absorber	500 v	VCR2521	1	1	1	1	1	1
Main Control Card		EP3090_ -C-1125	1					
Main Control Card		EP3090_C-1150		1				
Main Control Card		EP3090_ -C-2175			1			
Main Control Card		EP3090_ -C-2200				1		
Main Control Card		EP3090_ -C-2250					1	
Main Control Card		EP3090_ -C-2300						1
Ground Fault Card		EP3077_ -C-2	1	1	1	1	1	1
Base Drive Card		EP3091_ -C-2	2	2				
Base Drive Card		EP3097_ - C - I			6	6		
Base Drive Card		EP3097_ C-2					6	6
Keypad PC Card		6VKC3SK	1	1	1	1	1	1

NOTE: *All parts not specifically shown as Non-UL Listed are UL Listed.*

FUNCTION CODE SETTINGS: Factory and Field

Mode	Func. Code	Name	Setting/Indication Range	Data Unit	Incre-ment	Factory Setting	Field Setting
OPERATION MONITOR	00	Output Frequency	Indicates During Oper.	Hz	0.1 Hz	----	
	01	Reference Freauency	" " "	Hz	0.1 Hz	----	
	02(*1)	Synchronous Speed	" " "	RPM	1 RPM	----	
	03	Output Current	" " "	Amps	1 Amp	----	
	04	Output Voltage	" " "	Volts	2V	----	
	05(*1)	Machine Speed	" " "	RPM	1 RPM	----	
	06	Input Signal Status	" " "	----	----	----	
	07	Output Signal Status	" " "	----	----	----	
	08	Motoring Torque Limit	" " "	----	----	****	
	09	Braking Torque Limit	" " "	----	----	****	
	0A	Calculated Torque	" " "	----	----	****	
BASIC PARAMETER	10	Maximum Frequency	50.0 - 400.0 (*5)	Hz	0.1 Hz	60.0 Hz	
	11	Base Frequency	50 - 400 (*5)	Hz	1 Hz	60 Hz	
	12	Rated Output Voltage	320 - 460	V	1 V	460 V	
	13	Frequency Bias	0 - 400	Hz	1 Hz	0 Hz	
	14	High Limiter	O-400	Hz	1 Hz	60 Hz	
	15	Low Limiter	O-400	Hz	1 Hz	1 Hz	
	16(*3)	Acceleration Time 1	0.2 -3600	sec	0.1 sec	20.0 sec	
	1 7(*3)	Deceleration Time 1	0.2 -3600	sec	0.1 sec	20.0 sec	
	18(*2)	Frequency Gain	O-200	%	0.1%	105.0%	
	19	Torque Boost	O-20	---	1	3	
	1A	Auto Energy Saving Op.	Yes - (0), No - (-)	---	---	No (-)	
	Ib	Elec. Thermal Overload	0 (not used) 50 -105	%	1 %	105 %	
AUXILIARY PARAMETER	20	DC Brake Starting Freq.	0.0, 0.5 - 60.0	Hz	0.1 Hz	0.0 Hz	
	21	DC Brake Voltage	0.0 - 10.0	%	0.1 %	10.0 %	
	22	DC Braking Time	0.0 - 10.0	sec	0.1 sec	0.5 sec	
	23(*2)	Multi-Steo Frea. Setting 1	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	24(*2)	Multi-Step Freq. Setting 2	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	25(*2)	Multi-Step Freq. Setting 3	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	26(*2)	Multi-Step Freq. Setting 4	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	27(*2)	Multi-Step Freq. Setting 5	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	28(*2)	Multi-Step Freq. Setting 6	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	29(*2)	Multi-Step Freq. Setting 7	0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	2A	Acceleration Time 2	0.2 - 3600	sec	0.1 sec	100 sec	
	2b	Deceleration Time 2	0.2 - 3600	sec	0.1 sec	100 sec	
	2C	Acceleration Time 3	0.2 - 3600	sec	0.1 sec	100 sec	
	2d	Deceleration Time 3	0.2 - 3600	sec	0.1 sec	100 sec	
	2E	Acceleration Time 4	0.2 - 3600	sec	0.1 sec	100 sec	
	2F	Deceleration Time 4	0.2 - 3600	sec	0.1 sec	100 sec	
	30	Accel / Decel Pattern	0, 1, 2.	---	---	0	
	31	Tone Control Selection	1, 2, 3, 4.	---	---	1	
	32	Overload Early Warning	50 - 105	%	1%	105%	

FUNCTION CODE SETTINGS: Factory and Field

Mode	Func. Code	Name	Setting/Indication Ranne	Data Unit	Incre-ment	Factory Setting	Field Setting
	33(*2)	Motoring Torque Limit	0, 20 - 180	%	1%	150%	
AUXILIARY	34(*2)	Braking Torque Limit	0, 20 - 180	%	1%	100%	
PARAMETER	35(*2)	Freq. Detection Threshold	1 - 400	Hz	1 Hz	30 Hz	
(continued)	36(*2)	Freq. At Reference	0.5 - 5.0	Hz	0.1 Hz	2.5 Hz	
	37	Start Frequency	0.5 -5.0	Hz	0.1 Hz	0.5 Hz	
	38	Start Frea. Holdina Time	0.0 - 10.0	sec	0.1 sec	0.0 sec	
	39	'Jump Frequency 1	0.0, 0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	3A	Jump Frequency 2	0.0, 0.5 - 400.0	Hz	0.1 Hz	0.0 Hz	
	3B	Jump Frequency 3	0.0, 0.5 - 400.0	HZ	0.1 Hz	0.0 Hz	
	3C	Jump Freauencv Ranae	0.0 - 5.0	HZ	0.1 Hz	2.0 Hz	
	3d(*2)	Number of Motor Poles	2. 4. 6. 8. 10. 12	---	2	4	
	3E(*2)	Coefficient Mach Spd.	0.1 - 10.0	---	0.1	1.0	
	40	'Pulse Freq. Meter Coeffic.	6 - 100	---	1	30	
	41	FWD/REV Cmd. Hold (3 wire control)	Yes (0), No (-)	---	---	Yes (0)	
	42	Up / Down Control	Yes (0), No (-)	---	--	No (-)	
	43(*4)	Restart After Power Fail	Yes (0), No (-)	---	---	No (-)	
	44	Under Voltage Alarm	Yes (0), No (-)	---	---	Yes (0)	
	45	Open Coll. Output Select	0, 1, 2.		---	0	
	46	Slip Comp. Control	0.0 -2.5	Hz	0.1 Hz	0.0 Hz	
	47	Voltage Sig. Polarity Op.	Yes (0), No (-)	- -	---	No (-)	
ADJUSTMENT PARAMETER	50	Freq. Meter Calibration	70.0 - 105.0	%	0.1%	160.0%	
PROTECTION	99	Set Data Protection	Yes (0), No (-)	---	---	No (-)	
	F0	Fault Message	Ind. Failure Condition	- -	---	----	
FAILURE	F1	Output Frequency at Fault		Hz	0.1 Hz	----	
INDICATION	F2	Setting Frequency at Fault		Hz	0.1 Hz	----	
	F3	Output Current at Fault		A	1A	---	
	F4	Operating Mode at Fault		---	---	----	
	F5	Fault Memory 1		---	---	----	
	F6	Fault Memory 2		---	---	----	
	F7	Fault Memory 3		---	---	----	

*1 *Because the* data display *is four digits, if the indication exceeds four* digits *it will be shown as 1/10 of the actual automatically. (Example: 12000 RPM would be displayed as 1200 RPM.)*

*2 *These parameter* values *can be changed during inverter operation; other* **parameters** *can only be changed when the inverter is stopped.*

*3 *If the setting of Acceleration/Deceleration time exceeds 100 seconds, the minimum setting unit resolution will become 1 second.*

*4 *Momentary power failures only (duration may vary with horsepower and load).*

*5 Consult *distributor* for *operation above 120 Hertz.*



GE Drive Systems

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Salem, Virginia 24 153